Class 8

Mathematics

Volume 2

Solutions

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10. Trigonometry

WORKSHEET – 1

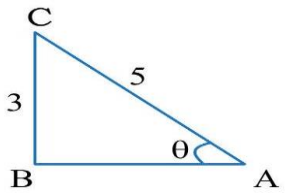
Level – I

1) B	2) C	3) D	4) A	5) D	6) C	7) D
8) D	9) B	10) A	11) B			

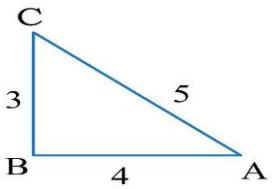
1. $135^0 = 135 \times \frac{\pi^c}{180} = \frac{3\pi}{4}$ 2. $\frac{\pi}{6} = \frac{180}{6} = 30^0$

3. $50^g = 50 \times \frac{\pi}{200} = \frac{\pi^c}{4}$

4. $5\sin\theta = 3 \Rightarrow \sin\theta = \frac{3}{5} \Rightarrow \tan\theta = \frac{3}{4}$



5. $\tan A = \frac{3}{4}$



6. $\frac{5\pi}{4} = 5 \times \frac{180}{4} = 225^0$

7. $500^g = 500 \times \frac{\pi^c}{200} = \frac{5\pi^c}{2}$

8. $1 + 1 + 1 = 3$

9. $\frac{1}{2} + 1 = \frac{3}{2}$

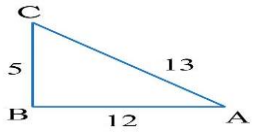
10. $2\tan^2 45 + \tan^2 60 = 2 + 3 = 5$

11. $4 + 4 = 8$

Level – II

1) B	2) D	3) B	4) B	5) A	6) A	7) C
8) A	9) A	10) C	11) A	12) D	13) C	14) B
15) A	16) D	17) A	18) A	19) C	20) A	21) B
22) C	23) C,D	24) A,C,D	25) B	26) B	27) D	28) A
29) C	30) C	31) a-s,b-p, c-q, d-q		32) 7	33) A,B,C,D	34) C
35) A	36) A	37) a-q,b-s,c-r,d-r				

1. $\sin A = \frac{5}{13}$



2. $270^0 = 270 \times \frac{\pi}{180} = \frac{3\pi^c}{2}$
3. $\sin 2(30°) = \sin 60 = \frac{\sqrt{3}}{2}$
4. $1 + 2 \times 3 = 1 + 6 = 7$
5. $\sin A = \frac{1}{\sqrt{2}} \Rightarrow A = 45 \cot 45° = 1$
6. $1 + 1 - 1 = 1$
7. $\frac{1}{2} + 1 - 1 + 1 - 2 + 2 = \frac{3}{2}$
8. $\sin 45 \cos 30 + \cos 45 \sin 30 = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} = \frac{\sqrt{3}+1}{2\sqrt{2}}$
9. $= \frac{4}{3} + \frac{4}{3} + \frac{1}{2} \Rightarrow \frac{8+8+3}{6} = \frac{19}{6}$
10. $2 \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = 1 = \sin 90°$
11. $\frac{\left(1-\frac{3}{4}\right)}{\left(\frac{1}{2}\right)} = \frac{\left(\frac{1}{4}\right)}{\left(\frac{1}{2}\right)} = \frac{1}{2} = \cos 60°$
12. $4(1)^2 - \frac{4}{3} - 2(\sqrt{3})^2 + (\sqrt{2})^2 \Rightarrow 4 - \frac{4}{3} - 6 + 2 = \frac{-4}{3}$
13. $40^0 = 40 \times \frac{\pi^c}{180} = \frac{2\pi^c}{9}$
14. Sinx is increasing function in $\theta \Rightarrow \text{Sin}\, 1^0 < \text{Sin}\, 1$
15. $\frac{1-(\sqrt{3})^2}{1+(\sqrt{3})^2} = \frac{1-3}{1+3} = \frac{-2}{4} = \frac{-1}{2} = -\cos 60$
16. $\frac{\sqrt{3}-\frac{1}{\sqrt{3}}}{1+\sqrt{3}\times\frac{1}{\sqrt{3}}} = \frac{1}{\sqrt{3}} = \tan 30°$
17. We know $\text{Sin}\, 30° = \frac{1}{2}$

    angle $\theta = 30°$

    $\therefore$ Sum of two angles $= 180° - 30° = 150°$
18. $= \frac{4}{3}\left[(\sqrt{3})^2 + 3\left(\frac{\sqrt{3}}{2}\right)^2 - 2\left(\frac{2}{\sqrt{3}}\right)^2 - \frac{3}{4}\left(\frac{1}{\sqrt{3}}\right)^2\right] = \frac{4}{3}\left(\frac{3}{1} + \frac{9}{4} - \frac{8}{3} - \frac{1}{4}\right) = \frac{4}{3}\left(\frac{36+27-32-3}{12}\right) = \frac{28}{9}$
19. $\frac{\left(\frac{1}{2}\right)}{\left(1-\frac{\sqrt{3}}{2}\right)} - \frac{\left(1-\frac{\sqrt{3}}{2}\right)}{\left(\frac{1}{2}\right)} = \frac{1}{2-\sqrt{3}} - \frac{2-\sqrt{3}}{1} = \frac{1-(2-\sqrt{3})^2}{2-\sqrt{3}}$

    $= \frac{1 - 4 - 3 + 4\sqrt{3}}{2 - \sqrt{3}} = \frac{-6 + 4\sqrt{3}}{2 - \sqrt{3}} = 2\sqrt{3}\left[\frac{-\sqrt{3} + 2}{2 - \sqrt{3}}\right] = 2\sqrt{3} = 2\cot 30°$
20. $\text{Sin}\, \theta = \frac{7}{25} \Rightarrow \text{Cot}\, \theta = \frac{24}{7} \Rightarrow \tan \theta = \frac{7}{24} \Rightarrow \frac{7\cot\theta - 24\tan\theta}{7\cot\theta + 24\tan\theta} = \frac{24-7}{24+7} = \frac{17}{31}$

21. $\sqrt{3}\sin\theta - \cos\theta = 0$

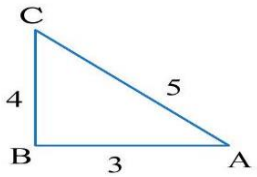
$$\Rightarrow \tan\theta = \frac{1}{\sqrt{3}} \Rightarrow \frac{2\left(\frac{1}{\sqrt{3}}\right)}{1-\left(\frac{1}{\sqrt{3}}\right)^2} = \frac{\left(\frac{2}{\sqrt{3}}\right)}{\left(\frac{2}{3}\right)} = \sqrt{3}$$

22. $\frac{3\pi}{2} = \frac{60}{360} \times \pi r^2 \Rightarrow r = 3$; perimeter $= 2r + l = 2\mathrm{r} + \mathrm{r}\theta = 2r + 3 \cdot \frac{\pi}{3} = 6 + \pi$

23. We know

$$\frac{D}{90^0} = \frac{G}{100} = \frac{c}{\frac{\pi}{2}} \Rightarrow \frac{D}{90} = \frac{G}{100} \text{ and } \frac{D}{90} = \frac{c}{\frac{\pi}{2}} \Rightarrow D = \frac{9G}{10} \text{ and } D = \frac{180c}{\pi}$$

24. $\sin A = \frac{4}{5} = \frac{\text{opp.side.to } A}{\text{hyp}}$



$$\cos C = \frac{4}{5} = \frac{\text{adj.side.to } C}{\text{hyp}} \Rightarrow \sin C = \frac{3}{5}$$

25. $l = \frac{\theta}{360} \times 2\pi r = \frac{60}{360} \times 2 \times \frac{22}{7} \times 6 = \frac{44}{7}$ cm

26. $A = \frac{\theta}{360} \times \pi r^2 = \frac{30}{360} \times \frac{22}{7} \times 10 \times 10 = \frac{550}{21}$ sqcm

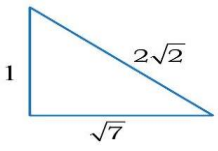
27. $l = \frac{\theta}{360} \times 2\pi r = \frac{110}{7} = \frac{\theta}{360} \times 2 \times \frac{22}{7} \times 10 \Rightarrow \theta = 90$

28. $15\cos A - 8\sin A = 0 \Rightarrow \tan A = \frac{15}{8} \Rightarrow \cot A = \frac{8}{15} \sin A = \frac{15}{17} \sec A = \frac{17}{8} \Rightarrow \frac{8+15}{15+34} = \frac{23}{49}$

29. $\frac{\sin A + \cos A}{2\cos A - \sin A} = \frac{\frac{15}{17} + \frac{8}{17}}{\frac{16}{17} - \frac{15}{17}} = \frac{23}{1} = 23$

30. $\frac{\left(\frac{17}{15} - \frac{17}{8}\right)}{\frac{17}{15} + \frac{17}{8}} = \frac{17 \times -7}{17 \times 23} = \frac{-119}{391}$

31. a) $\tan\theta = \frac{1}{\sqrt{7}}$



$$\frac{\left(\frac{8}{1} - \frac{8}{7}\right)}{\left(\frac{8}{1} + \frac{8}{7}\right)} = \frac{8 \times 6}{8 \times 8} = \frac{3}{4}$$

b) $\tan B = \frac{5}{12} \Rightarrow \cos B = \frac{12}{13}$

c) $\sec A \cdot \cos A = \cos A \cdot \frac{1}{\cos A} = 1$

d) $\tan A = \frac{3}{4}, \sin^2 A + \cos^2 A = \left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2 = 1$

32. $3\tan A = 4 \Rightarrow \frac{1+\cot A}{1-\cot A} = \frac{1+\frac{3}{4}}{1-\frac{3}{4}} = \frac{\left(\frac{7}{4}\right)}{\left(\frac{1}{4}\right)} = 7$

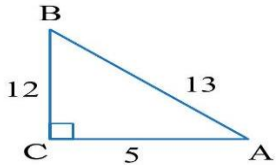
33. $\sin B = \frac{1}{2} = \cos A$, then

$B = 30°$ and $A = 60°$ and also $c = 90°$

$3\cos 30° - 4\cos^3 30° = \frac{3\sqrt{3}}{2} - 4\left(\frac{3\sqrt{3}}{8}\right) = 0$

$\tan 0° = 0, \cos c = \cos 90 = 0, \cot 90 = 0$

34.



$\sin B + \sin A = \frac{5}{13} + \frac{12}{13} = \frac{17}{13}$

35. $\tan B = \frac{5}{12}$

36. $\sin^2 A + \cos^2 A = \left(\frac{12}{13}\right)^2 + \left(\frac{5}{13}\right)^2 = \frac{144+25}{169} = 1$

37. $25x^2 - 35x + 12 = 0 \Rightarrow \sin\alpha + \sin\beta = \frac{35}{25} = \frac{7}{5}$

$\sin\alpha\sin\beta = \frac{12}{25} \Rightarrow (a-b)^2 = (a+b)^2 - 4ab$

$\sin\alpha - \sin\beta = \sqrt{\frac{49}{25} - \frac{48}{25}} = \sqrt{\frac{1}{25}} = \frac{1}{5} \Rightarrow$ since $\sin\alpha + \sin\beta, \sin\alpha - \sin\beta(+ve)$

$\sin\alpha + \sin\beta = \frac{7}{5}$

$\sin\alpha - \sin\beta = \frac{1}{5}$

$2\sin\alpha = \frac{8}{5} \Rightarrow \sin\alpha = \frac{4}{5}$

$\sin\beta = \frac{7}{5} - \frac{4}{5} \Rightarrow \sin\beta = \frac{3}{5}$

a) $\cos\alpha + \cos\beta = \frac{3}{5} + \frac{4}{5} = \frac{7}{5}$

b) $\tan\alpha + \tan\beta = \frac{4}{3} + \frac{3}{4} = \frac{25}{12}$

c) $\sec\alpha + \operatorname{cosec}\alpha = \frac{5}{3} + \frac{5}{4} = \frac{35}{12}$

d) $\sec\beta + \operatorname{cosec}\beta = \frac{5}{3} + \frac{5}{4} = \frac{35}{12}$

WORKSHEET – 2

Level – I

1) 2	4) 1	7) 1	10) 1	13) 3
2) 3	5) 4	8) 1	11) 1	14) 2
3) 1	6) 3	9) 4	12) 4	15) 3

1. $\sin(180-\theta)=\sin\theta$
2. $=\tan(90+\theta)=-\cot\theta$
3. $=-\sin 15^\circ+\sin 15^\circ=0$
4. $\sin(180+\theta)=-\sin\theta$
5. $\tan(90+30)=-\cot 30^\circ=-\sqrt{3}$
6. $\operatorname{cosec}(90-\theta)=\sec\theta$
7. $\tan(180+\theta)=\tan\theta$
8. Concept $\operatorname{cosec}(270-\theta)=-\sec\theta$
9. $\cos 585^\circ=\cos[6\times 90+45]=-\cos 45^\circ=-\frac{1}{\sqrt{2}}$
10. $\cot 1140^\circ=\cot[13\times 90^\circ-30^\circ]=\tan 30^\circ=\frac{1}{\sqrt{3}}$
11. $\tan 17\frac{\pi}{4}=\tan\left(\frac{16\pi}{4}+\frac{\pi}{4}\right)=\tan\left(4\pi+\frac{\pi}{4}\right)\Rightarrow\tan\frac{\pi}{4}=1$
12. $\cos\left(-9\frac{\pi}{4}\right)=\cos 9\frac{\pi}{4}=\cos\left[2\pi+\frac{\pi}{4}\right]=\cos\frac{\pi}{4}$
13. $\sec 15\frac{\pi}{4}=\sec\left(4\pi-\frac{\pi}{4}\right)=\sec\frac{\pi}{4}=\sqrt{2}$
14. $\tan 27^\circ\tan 63^\circ=\tan 27^\circ\cot 27^\circ=1\ \therefore \sin ce\tan 63^\circ=\tan(90^\circ-27^\circ)=\cot 27^\circ$
15. $A+B=90,\frac{\cot 54^\circ}{\cot 54^\circ}+\frac{\tan(90-70)}{\cot 70^\circ}=1+1=2$

Level - II

1) B	2) B	3) A	4) C	5) A	6) 3	7) 2
8) A	9) A	10) C	11) C	12) B	13)2	14) 2
15) B	16) C	17)D	18) C	19) B	20)1	21) 4
22) C	23) A	24) A,C,D	25) a-r,b-p,c-q,d-r		26)-3	27) 4
28) C	29) a-p,q;b-p,q;c-r,t;d-s		30) – 2	31) B	32) a-r,b-p,c-q,d-t	
33) A	34) C	35) B				

1. $\sin 10^\circ+\sin 20^\circ+\sin 30^\circ+\cdots\ldots+\sin 360=0\Rightarrow \text{since}-\sin\theta=\sin(180+\theta)$
2. $\tan 1^\circ\tan 2^\circ\tan 3^\circ\ldots\ldots\tan 89^\circ=1$

   since $A+B=90;\tan A\tan B=1$

   $\log 1=0$

3. we know $\cos A = \sin B$

   $\cos A = \cos(90 - B) \Rightarrow A + B = 90 \Rightarrow A = 90 - B \sin 2\theta = \cos 3\theta \Rightarrow 2\theta + 3\theta = 90$

   $5\theta = 90 \Rightarrow \cot 5\theta = \cot 90^\circ = 0$

4. since $\cos 90^\circ = 0 \ \therefore \cos 1^\circ \cdot \cos 2^\circ \cos 3^\circ \ldots \ldots \cos 90^\circ \ldots \ldots \cos 179^\circ = 0$

5. $A + B = 90 \Rightarrow \sin A = \cos B \Rightarrow 31 + 59 = 90^\circ \Rightarrow \sin 31^\circ = \cos 59^\circ \Rightarrow \frac{\sin 31^\circ}{\cos 59^\circ} = 1$

6. $\sin 72 = \cos 18^\circ$ and $\sec 32^\circ = \operatorname{cosec} 58^\circ \Rightarrow \frac{\sin 72}{\cos 18} = 1$ and $\frac{\sec 32}{\operatorname{cosec} 58} = 1$

7. $\sin 65^\circ = \cos 25$ and $\cos 38^\circ = \sin 52^\circ \Rightarrow \sin 15^\circ \cos 38^\circ = \cos 25^\circ \sin 52^\circ$

8. $\frac{\cos\theta}{\sin\theta} \cdot \tan\theta = 1$

9. $A + B = 90^\circ \Rightarrow \cot A \cot B = 1, \cot 9^\circ \times 1 \times \cot(90 - 9) = \cot 9^\circ \times \tan 9^\circ = 1$

10. $\frac{\cot\theta}{\cot\theta} + \frac{\sin\theta \tan\theta \operatorname{cosec}\theta}{\cos\theta \tan\theta \sec\theta} = 1 + 1 = 2$

11. $\frac{\sin 30 - 5\cos 60 + 7\tan 45}{-\cot 45^\circ - 3\sin 30^\circ} = \frac{\frac{1}{2} - \frac{5}{2} + 7}{-1 - \frac{3}{2}} = \frac{15-5}{-5} = -2$

12. $A + B = 180^\circ \Rightarrow \cos A + \cos B = 0$

    $\cos A + \cos C + \cos B + \cos D = 0 + 0 = 0$

13. by simplification $\frac{2}{\cos\theta} = 2\sec\theta$ since θ is $Q_2 \sec\theta$-ve

14. $\sin\theta + \cos\theta = \sqrt{2}$ exists at $\theta = 45^\circ$

15. $\cos\frac{[2(A+B+C)+(A-B)]}{2} + \cos\left(\frac{A-B}{2}\right)$

    $\cos\left[180 + \left(\frac{A-B}{2}\right)\right] + \cos\left(\frac{A-B}{2}\right)$

    $-\cos\left(\frac{A-B}{2}\right) + \cos\left(\frac{A-B}{2}\right) = 0$

16. $\tan(A + B) = \sqrt{3} \Rightarrow A + B = 60$

    $\tan(A - B) = 1 \Rightarrow \frac{A - B = 45}{2\ B = 15}$

    $\therefore \tan 6\ B = \tan 45^\circ = 1$

17. Exp: $= \sin^2\left(\pi - \frac{\pi}{3}\right) + \cos^2\left(\pi - \frac{\pi}{6}\right) - \tan^2\left(\pi - \frac{\pi}{4}\right) = \frac{3}{4} + \frac{3}{4} - 1 = \frac{6}{4} - 1 = \frac{1}{2}$

18. $A + B + C + D = 360$

    $$\frac{A + B}{2} = 180 - \left(\frac{C + D}{2}\right) \Rightarrow \tan\left(\frac{A + B}{2}\right) = -\tan\left(\frac{C + D}{2}\right)$$

19. Exp: $= \frac{(-\sin\theta)\cdot\sec\theta(-\tan\theta)}{\sec\theta(-\sin\theta)(\tan\theta)} = -1$

20. $\sqrt{1 - \cos^2 215} \cdot \operatorname{cosec} 215 = -\sin 215^\circ \operatorname{cosec} 215^\circ = -1$

21. $\frac{\tan(270-20)+\tan(360-20)}{\tan(180+20)-\tan(90+20)} = \frac{\cot 20 - \tan 20}{\tan 20 + \cot 20} = \frac{1-k^2}{1+k^2}$

22. $\frac{p\tan\theta - q}{p\tan\theta + q} = \frac{p^2 - q^2}{p^2 + q^2}$

23. $\cos\frac{\pi}{7} + \cos\frac{2\pi}{7} + \cos\frac{3\pi}{7} + \cos\left(\pi - \frac{3\pi}{7}\right) + \cos\left(\pi - \frac{2\pi}{7}\right) + \cos\left(\pi - \frac{\pi}{7}\right) = 0$

24. $A + B = 90°$, put $A = 30° B = 60° \Rightarrow \sin^2 30 + \sin^2 60 = \frac{1}{4} + \frac{3}{4} = 1$

25. a) $\sec\frac{13\pi}{3} = \sec\left(4\pi + \frac{\pi}{3}\right) = \sec\frac{\pi}{3} = 2$ b) $\tan\left(\pi + \frac{\pi}{3}\right) = \tan\frac{\pi}{3} = \sqrt{3}$

c) $\sec(270 - \theta) = -\operatorname{cosec}\theta$

d) $\sqrt{3}\sec(8 \times 90 + 30°) = \sqrt{3}\sec 30 = \sqrt{3} \times \frac{2}{\sqrt{3}} = 2$

26. $2(1)^2 - (3\sqrt{2})^2 + (1)^2 = 2 - 6 + 1 = -3$

27. $-\cos A - \cos B - \cos C - \cos D = -(\cos A + \cos B + \cos C + \cos D) = 0$

28. $A + B \Rightarrow \tan A + \tan B = 0$

29. Conceptual

30. $\frac{\sin 23\theta + \sin 7\theta}{\sin 2\theta - \sin 15\theta} = \frac{-1-1}{0-(-1)} = -2$

31. In $\triangle ABC$, put $A = 30\ \ B = 60\ C = 90 \Rightarrow \sin^2 A + \sin^2 B + \sin^2 C = \frac{1}{4} + \frac{3}{4} + 1 = 2$

32. $a \rightarrow x(\sqrt{3})^2 + 4\left(\frac{\sqrt{3}}{2}\right)^2 = 9 \Rightarrow k = 2$

$b \rightarrow 8\sin^2 x + 3(1 - \sin^2 x) = 5; 5\sin^2 x = 2 \Rightarrow \sin x = \sqrt{\frac{2}{5}} \Rightarrow \cot x = \sqrt{\frac{3}{2}}$

$c \rightarrow \cos 72 = \sin 18$ and $\cos 54 = \sin 36$(complimentary angle) $\Rightarrow \frac{\cos 72 \cos 54}{\sin 18 \sin 36} = 1$

$d \rightarrow \sin\alpha = \frac{3}{5} \cos\alpha = \frac{4}{5}$; $\alpha \& \beta$ complimentary $\Rightarrow \sin\beta = \frac{4}{5}\ \cos\beta = \frac{3}{5}$

$\Rightarrow \sin\alpha\cos\beta - \cos\alpha\sin\beta = \frac{9}{25} - \frac{16}{25} = \frac{-7}{25}$

comprehension type:

$\tan(A + B) = \sqrt{3}$ $\tan(A - B) = \frac{1}{\sqrt{3}}$

$A + B = 60$ $A - B = 30$

$A = 45°$ $= B = 15°$

33. $\sin^2 60 + \cos^2 30 = \frac{3}{4} + \frac{3}{4} = \frac{6}{4} = \frac{3}{2}$

34. $\cot^2 45° + \tan^2 30 + \tan 45° = 1 + \frac{1}{3} + 1 = \frac{7}{3}$

35. $\sin^2 45° + \sin^2 60 + \sin^2 45° = \frac{1}{2} + \frac{3}{4} + \frac{1}{2} = \frac{7}{4}$

WORKSHEET – 3

Level – I

1) B	2) C	3) A	4) C	5) D	6) C	7) A
8) A	9) B	10) D	11) C	12) D	13) B	

1. $\cot\theta = \frac{3}{4}$ $\operatorname{cosec}\theta = +\sqrt{1+\cot^2\theta}$ ($\therefore \theta$ is acute)

   $\sqrt{1+\frac{9}{16}} = \frac{5}{4}$

2. $4\sin^2\theta + \tan^2\theta \Rightarrow 4(1-\cos^2\theta) + (\sec^2\theta - 1)$

   $4\left(1-\frac{3}{4}\right) + \left(\frac{4}{3}-1\right) = 1 + \frac{1}{3} = \frac{4}{3}$

3. $(1+\tan^2\theta)(1-\sin^2\theta) = \sec^2\theta \cdot \cos^2\theta = 1$

4. $(\cos^2\theta - 1)\operatorname{cosec}^2\theta = -\sin^2\theta \operatorname{cosec}^2\theta = -1$

5. $\cos^2\theta + \frac{1}{\operatorname{cosec} 2}\theta = \cos^2\theta + \sin^2\theta = 1$

6. $\frac{1}{\sec^2\theta} + \frac{1}{\operatorname{cosec} e^2\theta} = \cos^2\theta + \sin^2\theta = 1$

7. $\sin^2\theta + \tan^2\theta = 1 - \cos^2\theta + \sec^2\theta - 1 = \frac{4}{3} - \frac{3}{4} = \frac{16-9}{12} = \frac{7}{12}$

8. $\frac{\cos\theta}{1+\sin\theta} = \frac{\cos\theta(1-\sin\theta)}{1-\sin^2\theta} = \frac{1-\sin\theta}{\cos\theta}$

9. we know $\sec^2\theta - \tan^2\theta = 1 \Rightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

10. $xy = (\sec\theta + \tan\theta)(\sec\theta - \tan\theta) \Rightarrow \sec^2 - \tan^2\theta = 1$

11. $(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$

    $(\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2 = 2(\sin^2\theta + \cos^2\theta) = 2$

12. $\sin\theta = \frac{12}{13} \Rightarrow \tan\theta = \frac{12}{5}$

13. $\frac{(1-\cos^2 A)}{\cos^2 A} = \frac{\sin^2 A}{\cos^2 A} = \tan^2 A$

Level – II

1) C	2) C	3) D	4) C	5) A	6) B	7) D
8) C	9) D	10) C	11) A	12) D	13) C	14) A
15) C	16) B	17) C	18) D	19) C	20) B	21) C
22) B	23) B,C	24) A	25) D	26) C	27) B	28) A
29) A	30) a-r,s;b-q,c-t,d-p		31) B	32) A,D	33) A,B	
34) a-s,b-q,c-t,d-p		35) B	36) C	37) B		

1. $\cot\theta - \tan\theta = \frac{\cos\theta}{\sin\theta} - \frac{\sin\theta}{\cos\theta} = \frac{\cos^2\theta - \sin^2\theta}{\sin\theta\cdot\cos\theta} = \frac{2\cos^2\theta - 1}{\sin\theta\cdot\cos\theta}$

2. $\tan\theta = 3 \Rightarrow \sin\theta = \frac{-3}{\sqrt{10}}$ $\therefore$ ($'\theta'$ is in Q_3)

3. $$\frac{\tan^2\theta}{1+\tan^2\theta} + \frac{\cot^2\theta}{1+\cot^2\theta} = \frac{\tan^2\theta}{\sec^2\theta} + \frac{\cot^2\theta}{\operatorname{cos}ec^2\theta} = \sin^2\theta + \cos^2\theta = 1$$

4. $\frac{1+\sin\theta}{\cos\theta} + \frac{1-\sin\theta}{\cos\theta} = 2\sec\theta$

5. $\sec^2 A \cdot \operatorname{cosec}^2 A - \tan^2 A - \cot^2 A$

$(\sec^2 A - \tan^2 A) + (\operatorname{cosec}^2 A - \cot^2 A) = 1 + 1 = 2$

6. $\sec\theta(1-\sin\theta)(\sec\theta+\tan\theta) = (\sec\theta - \sec\theta\cdot\sin\theta)(\sec\theta+\tan\theta)$

$= (\sec\theta - \tan\theta)(\sec\theta+\tan\theta) = 1$

7. $\frac{\sec\theta-1}{\sec\theta+1} = \frac{\frac{1}{\cos\theta}-1}{\frac{1}{\cos\theta}+1} = \frac{1-\cos\theta}{1+\cos\theta}$

8. wer know $\operatorname{cosec} e^2\theta - \cot^2\theta = 1 \Rightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

9. $\operatorname{cosec}\theta = \sqrt{n+1} \Rightarrow \sin\theta = \frac{1}{\sqrt{n+1}}$

$$\cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{1-\frac{1}{n+1}} = \sqrt{\frac{n}{n+1}}$$

10. $\cos\theta\left(\frac{2}{1-\sin^2\theta}\right) = 4 \Rightarrow \frac{2}{\cos\theta} = 4 \Rightarrow \cos\theta = \frac{1}{2}$

11. $x^2 = \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta$

$x^2 = 1 + 2y$

12. $xy = ab((\sec\theta+\tan\theta)^2(\sec\theta-\tan\theta))^2 = ab$

$x^2y^2 = a^2b^2$

13. $\frac{\tan\theta}{\sec^4\theta} + \frac{\cot\theta}{\operatorname{cosec}^4\theta}$

$= \frac{\sin\theta}{\sec^3\theta} + \frac{\cos\theta}{\operatorname{cosec} e^3\theta} = \sin\theta\cos^3\theta + \cos\theta\cdot\sin^3\theta = \sin\theta\cos\theta(\cos^2\theta+\sin^2\theta)\ \sin\theta\cos\theta$

14. $\sin\theta = \frac{-5}{13}, \tan\theta = \frac{5}{12}(+ve\sin ce'\theta' \text{ is in } Q_3)$

15. $\sqrt{\frac{x}{a}} + \sqrt{\frac{y}{b}} = \cos^2\theta + \sin^2\theta = 1$ 16. $x^2 + y^2 + z^2 = r^2$

16. $\sqrt{\frac{1+\sin A}{1-\sin A}} = \frac{1+\sin A}{\cos A} +$ ve in I and II because $1 \le 1 + \sin A \le 2$ and $\cos A +$ ve in Q_1Q_4

18.. $m^2 = \tan^2\theta + \sin^2\theta + 2\tan\theta\cdot\sin\theta$

$\underline{n^2 = \tan^2\theta + \sin^2\theta - 2\tan\theta\cdot\sin\theta}$

$m^2 - n^2 = 4\tan\theta\cdot\sin\theta = 4\sqrt{mn}$

and $mn = \tan^2\theta - \sin^2\theta = \sin^2\theta\left(\frac{1-\cos^2\theta}{\cos^2\theta}\right) = \sin^2\theta\frac{\sin^2\theta}{\cos^2\theta}$

19. $\frac{x}{a} = \operatorname{cosec}\alpha + \cot\alpha$ and $\frac{y}{b} = \operatorname{cosec}\alpha - \cot\alpha \Rightarrow xy = ab$

20. We know

$\sec A + \tan A = 3$

$\underline{\sec A - \tan A = \frac{1}{3}}$

$2\sec A = 3 + \frac{1}{3}$

$\sec A = \frac{5}{3}$

21. $\sqrt{\frac{1-\cos\theta}{1+\cos\theta}} + \sqrt{\frac{1+\cos\theta}{1-\cos\theta}} = \frac{1-\cos\theta}{\sin\theta} + \frac{1+\cos\theta}{\sin\theta} = \frac{2}{\sin\theta}$

22. $\left(\frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta\right)^2 = 1^2$

$\underline{\left(\frac{x}{a}\sin\theta - \frac{y}{b}\cos\theta\right)^2 = 1^2}$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$

23. $\frac{x^2}{4} = \text{cosec}^{\,2}\theta$ and $\frac{y^2}{4} = \cot^2\theta \Rightarrow \text{cosec}^{\,2}\theta - \cot^2\theta = 1 \Rightarrow \frac{x^2}{4} - \frac{y2}{4} = 1$

24. $a^2 = x^2\cos^2\theta + y^2\sin^2\theta + 2xy\cos\theta\sin\theta$

$\underline{b^2 = x^2\sin^2\theta + y^2\cos^2\theta - 2xy\cos\theta\sin\theta}$

$a^2 + b^2 = x^2 + y^2$

25. $a^2 + b^2 = 1^2 + 1^2 = 2$

26. $x^2 + y^2 = 4 + 2 = 6$

27. $\frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta = 1$ and

$\frac{x}{a}\sin\theta - \frac{y}{b}\cos\theta = 1 \Rightarrow\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$

28. $(x+y)^3 = x^3 + y^3 + 3xy(x+y)$ and $(x+y)^2 = x^2 + y^2 + 2xy$

$(\sin x + \cos x)^2 = 1 + 2\sin x\cos x$

$\sin x\cos x = \frac{p^2-1}{2}$

$\therefore x^3 + y^3 = (x+y)^3 - 3xy(x+y)$

$\sin^3 x + \cos^3 x = (\sin x + \cos x)^3 - 3\sin x\cos x(\sin x) + \cos x$

$q = p^3 - 3\left(\frac{p^2-1}{2}\right)\cdot p$

$2q = 2p^3 - 3p^3 + 3p$

$p^3 + 2q = 3p$

29. $\sec^2\theta - \tan^2\theta = 1 \Rightarrow xy = 1$

30. a-r,s b-q c-t d-p

(a) $\sin^4 A - \cos^4 A = (\cos^2 A + \sin^2 A)(\sin^2 - \cos^2 A) = 1 - 2\cos^2 A$

(b) $1 - \frac{1-\sin^2 A}{1+\sin^2 A} == 1 - (1 - \sin A) = \sin A$

(C) $1 - \frac{1-\cos^2 \text{A}}{1+\cos \text{A}} = 1 - (1 - \cos A) = 1 - 1 + \cos A = \cos A$

(d) $\tan^2 A(1 + \tan^2 A)\big((\sec^2 A - 1)\cdot \sec^2 A = \sec^4 A - \sec^2 A$

31. $\sin\theta + \operatorname{cosec}\theta = 2$

squaring on both sides

$(\sin\theta + \operatorname{cosec}\theta)^2 = 4$

$\sin^2\theta + \operatorname{cosec}^2\theta + 2\sin\theta \times \operatorname{cosec}\theta = 4$

$\sin^2\theta + \operatorname{cosec}^2\theta + 2 = 4 \ \therefore \sin\theta \cdot \operatorname{cosec}\theta = 1$

$\sin^2\theta + \operatorname{cosec}^2\theta = 4 - 2$

$\therefore \sin^2\theta + \operatorname{cosec}^2\theta = 2$

32. $a\sin^2 x + b\cos^2 x = c$

dividing $\cos^2 x$ on both sides

$a\tan^2 x + b = c\sec^2 x$

$a\tan^2 x + b = c(1 + \tan^2 x)$

$(a - c)\tan^2 x = c - b$

$\tan^2 x = \left(\frac{c-b}{a-c}\right)$ similarly $\tan^2 y = \left(\frac{d-a}{b-d}\right)$

but $a\tan x = b\tan y$

$$\frac{a^2}{b^2} = \frac{\tan'^2 y}{\tan^2 x} = \frac{(d-a)}{(b-d)} \times \frac{(a-c)}{(c-b)}$$

33. $\sin x + \sin^2 x + \sin^3 x = 1 \Rightarrow \cos^6 x - 4\cos^4 x + 8\cos^2 x$

$\sin x[1 + \sin^2 x] = 1 - \sin^2 x$

$\sin x[1 + \sin^2 x] = \cos^2 x$

squaring on both sides

$\sin^2 x[1 + \sin^2 x]^2 = \cos^4 x \Rightarrow (1 - \cos^2 x)[(1 + 1 - \cos^2 x)]^2 = \cos^4 x$

$(1 - \cos^2 x)(4 + \cos^4 x - 4\cos^2 x) = \cos^4 x$

$4 + \cos^4 x - 4\cos^2 x - 4\cos^2 x - \cos^6 x + 4\cos^4 x = \cos^4 \mathrm{x}$

$= \cos^6 \cos^6 x - 4\cos^4 x + 8\cos^2 x = 4$

34. Conceptual

35. $\sec\theta + \tan\theta = 5$ then $\sec\theta - \tan\theta =$

$\sec^2\theta - \tan^2\theta = 1 \Rightarrow (\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 1$

$5(\sec\theta - \tan\theta) = 1$

$$\sec\theta - \tan\theta = \frac{1}{5}$$

36. $\sec\theta + \tan\theta = 4$ then $\sin\theta = \frac{k^2-1}{k^2+1}$

$\sec\theta + \tan\theta = k$

$$\therefore k = 4\ \sin\theta = \frac{16-1}{16+1} = \frac{15}{17}$$

37. $\operatorname{cosec}\theta + \cot\theta = 3$ then $\sec\theta = \cos\theta = \frac{k^2-1}{k^2-1}, \sec\theta = \frac{k^2+1}{k^2-1}, \sec\theta = \frac{3^2+1}{3^2-1} = \frac{9+1}{9-1} = \frac{10}{8} = \frac{5}{4}$

WORKSHEET – 4

Level – I

1.	C	3.	B	5.	B	7.	A
2.	C	4.	B	6.	C	8.	C

1. $\sin 60^\circ = \frac{\sqrt{3}}{2}$
2. $\tan(22^\circ + 23^\circ) = 1$
3. $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{\frac{1}{2}+\frac{1}{3}}{1-\frac{1}{2}\times\frac{1}{3}} = \frac{\left(\frac{5}{6}\right)}{\left(\frac{5}{6}\right)} = 1$
4. $\sin 75^\circ + \cos 75^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}} + \frac{\sqrt{3}-1}{2\sqrt{2}} = \sqrt{\frac{3}{2}}$
5. $\cos(28+32) = \cos 60^\circ = \frac{1}{2}$
6. $\tan(\alpha+\beta) = \frac{\frac{1}{2}+\frac{1}{3}}{1-\frac{1}{2}\times\frac{1}{3}} = 1 = \tan 45^\circ \Rightarrow \alpha + \beta = 45^\circ$
7. $\tan 75 - \cot 75^\circ = 2 + \sqrt{3} - 2 + \sqrt{3} = 2\sqrt{3}$
8. $\sin 15^\circ \cdot \cos 15^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}}, \frac{\sqrt{3}+1}{2\sqrt{2}} = \frac{3-1}{4\times 2} = \frac{1}{4}$

Level - II

1) A	2) D	3) C	4) A	5) D	6) A	7) C
8) B	9) B	10) C	11) C	12) B	13) A	14) A
15) A	16) B	17) A	18) D	19) A	20) C	21) D
22) B	23) B	24) C	25) C	26) B	27) C	
28) a-q,b-s,p;c-q,d-r		29) 0	30) A, C	31) A	32) A	33) B
34) C	35) a-p,b-q,c-q,d-r					

1. Given $A + B + C = \frac{\pi}{2} \Rightarrow A + B = \frac{\pi}{2} - c \Rightarrow \tan(A+B) = \tan\left(\frac{\pi}{2} - c\right)$

$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{1}{\tan C} \Rightarrow \tan A \tan B + \tan B \tan C + \tan C \tan A = 1$$

2. $\cos^2 45^\circ - \sin^2 15^\circ = \left(\frac{1}{\sqrt{2}}\right)^2 - \left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)^2$

$$= \frac{1}{2} - \left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)^2 = \frac{1}{2} - \frac{4-2\sqrt{3}}{4\times 2} = \frac{1}{2} - \frac{2-\sqrt{3}}{4} = \frac{2-2+\sqrt{3}}{4} = \frac{\sqrt{3}}{4}$$

3. $\sin 75 = \sin(30+45) = \sin 30 \cos 45 + \cos 30 \sin 45$

$$= \frac{1}{2}\cdot\frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2}\cdot\frac{1}{\sqrt{2}} = \frac{1+\sqrt{3}}{2\sqrt{2}}$$

4. $\cos A = \frac{3}{5} \Rightarrow \tan A = \frac{4}{3} \Rightarrow \sin B = \frac{5}{13} \Rightarrow \tan B = \frac{5}{12}$

$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{\frac{4}{3} + \frac{5}{12}}{1 - \frac{4}{3} \times \frac{5}{12}} = \frac{48 + 15}{36 - 20} = \frac{63}{16}$$

5. $\sin 20^\circ + \sin(60^\circ - 20^\circ) - \sin(60^\circ + 20^\circ)$

$= \sin 20^\circ - 2\cos 60^\circ \sin 20^\circ = \sin 20^\circ - 2 \cdot \frac{1}{2} \sin 20^\circ = \sin 20^\circ - \sin 20^\circ = 0$

6. $\cos(30 + \theta) = \cos 30 \cos\theta - \sin 30 \sin\theta = \frac{\sqrt{3}}{2}\cos\theta - \frac{1}{2}\sin\theta$

$\sin(30 - \theta) = \sin 30^\circ \cos\theta - \cos 30 \sin\theta = \frac{1}{2}\cos\theta - \frac{\sqrt{3}}{2}\sin\theta$

$$\cos^2(30 + \theta) - \cos^2(30 - \theta) = \left(\frac{\sqrt{3}\cos\theta - \sin\theta}{2}\right)^2 - \left(\frac{\cos\theta - \sqrt{3}\sin\theta}{2}\right)^2$$

$$= \frac{1}{4}\left[3\cos^2\theta + \sin^2\theta - 2\sqrt{3}\sin\theta\cos\theta - \cos^2\theta - 3\sin^2\theta + 2\sqrt{3}\sin\theta\cos\theta\right] = \frac{1}{2}\cos 2\theta$$

7. We $A + B = 90^\circ \rightarrow \tan A \tan B = 1 \Rightarrow \tan(\theta + x)\tan(\theta - x) = 1 \Rightarrow \theta + \theta = 90 \Rightarrow \theta = 45^\circ$

8. $9x = 4x + 5x \Rightarrow \tan 9x = \frac{\tan 4x + \tan 5x}{1 - \tan 4x \cdot \tan 5x}$

$\rightarrow \tan 4x + \tan 5x - \tan 9x = -\tan 9x \tan 5x \tan 4x \Rightarrow k = -1$

9. $A + B = 45^\circ \Rightarrow \frac{\tan A + \tan B}{1 - \tan A \tan B} = 1$

$\tan A + \tan B + \tan A \tan B = 1$

$1 + \tan A + \tan B(1 + \tan A) = 2$

$(1 + \tan A)(1 + \tan B) = 2$

10. $A + B = 90^\circ \Rightarrow \tan B + 2\tan(A - B) = \tan A$

Consider $55 = 35 + 20$

$$\tan 55 = \frac{\tan 35 + \tan 20}{1 - \tan 35 \cdot \tan 25}$$

$\Rightarrow \tan 55(1 - \tan 35 \tan 20) = \tan 35 + \tan 20$

$\Rightarrow \tan 55 = \tan 35 + \tan 20 + \tan 55 \cdot \tan 35 \cdot \tan 20$

$\Rightarrow \tan 55 = \tan 35 + \tan 20 + \tan 20$

$\Rightarrow \tan 55 = \tan 35 + 2\tan 20 \Rightarrow x = 55^\circ$

11. $\frac{\tan 45^\circ - \cot 81^\circ \cot 69^\circ}{\cot 81^\circ + \cot 69^\circ} = -\cot 150^\circ = \sqrt{3}$

12. $A + B = 135 \Rightarrow (1 + \cot A)(1 + \cot B) = 2$

13. $A + B = 45^\circ \Rightarrow (1 + \tan A)(1 + \tan B) = 2$

$$\frac{(1 + \tan 13)(1 + \tan 32^\circ)}{(1 + \tan 12)(1 + \tan 33^\circ)} = \frac{2}{2} = 1$$

14. $\frac{1}{a}-\frac{1}{b}=\frac{1}{\tan x+\tan y}-\frac{1}{\cot y+\cot x}$

$=\dfrac{\cot x\cot y}{\cot x+\cot y}-\dfrac{1}{\cot y+\cot x}=\dfrac{\cot x\cot y-1}{\cot y+\cot x}=\cot(x+y)$

15. $A+B=360-60^\circ$

$\cot(A+B)=-\cot 60^\circ$

$\dfrac{\cot A\cot B-1}{\cot B+\cot A}=\dfrac{-1}{\sqrt{3}}\Rightarrow\dfrac{\cot A\cot B-1}{\cot B+\cot A}=\dfrac{-\sqrt{3}}{3}$

$\sqrt{3}\cot A+\sqrt{3}\cot B+3\cot A\cot B=3$

$1+\sqrt{3}\cot A+\sqrt{3}\cot B+3\cot A\cot B=3+1$

$(1+\sqrt{3}\cot A)+\sqrt{3}\cot B(1+\sqrt{3}\cot A)=4$

$(1+\sqrt{3}\cot A)(1+\sqrt{3}\cot B)=4$

16. $\frac{\tan 70-\tan 20}{\tan 50}=\frac{\cot 20-\tan 20}{\tan 50}=\frac{2\cot 40}{\tan 50}=2[\because \cot\theta-\tan\theta=2\cot 2\theta]$

17. $\sum\frac{(\sin A\cos B-\cos A\sin B)}{\cos A\cos B}=\sum(\tan A-\tan B)=0$

18. $\tan 75+\cot 75^\circ=2+\sqrt{3}+2-\sqrt{3}=4$

19. $\cot 20=p$

$$\frac{-\tan 20+\cot 20}{1+\tan 20\cdot\cot 20}=\frac{\dfrac{-1}{p}+p}{1+\dfrac{1}{p}\cdot p}=\frac{-1+\mathrm{p}^2}{1+1}=\frac{\mathrm{p}^2-1}{2\mathrm{p}}$$

20. $A+B+C=360\Rightarrow\frac{A}{2}+\frac{B}{2}+\frac{C}{2}=180\Rightarrow\frac{\mathrm{A}}{2}+\frac{\mathrm{B}}{2}=180-\frac{\mathrm{C}}{2}$

$$\tan\frac{A}{2}+\tan\frac{B}{2}+\tan\frac{C}{2}=\tan\frac{A}{2}\times\tan\frac{B}{2}\times\tan\frac{C}{2}$$

21. $A+B+C=180^\circ\Rightarrow\frac{A}{2}+\frac{B}{2}=90-\frac{C}{2}\Rightarrow\tan\left(\frac{A}{2}+\frac{B}{2}\right)=\left(90-\frac{1}{2}\right)$

$$\frac{\tan\frac{A}{2}+\tan\frac{B}{2}}{1-\tan\frac{A}{2}\tan\frac{B}{2}}=\cot\frac{C}{2}\Rightarrow\frac{\frac{5}{6}+\frac{20}{37}}{1-\frac{5}{6}\times\frac{20}{37}}=\cot\frac{C}{2}\Rightarrow\left(\frac{185+120}{222-100}\right)=\frac{305}{122}=\cot\frac{C}{2}$$

$$\tan\frac{C}{2}=\frac{122}{305}=\frac{2}{5}$$

22. $A+B+C=180^\circ$

put $A=B=C=60^\circ$

$(\cot B+\cot C)(\cot C+\cot A)(\cot A+\cot B)$

$=\left(\frac{1}{\sqrt{3}}+\frac{1}{\sqrt{3}}\right)\left(\frac{1}{\sqrt{3}}+\frac{1}{\sqrt{3}}\right)\left(\frac{1}{\sqrt{3}}+\frac{1}{\sqrt{3}}\right)=\frac{8}{3\sqrt{3}}=\operatorname{cosec}A\cdot\operatorname{cosec}B\operatorname{cosec}C$

$=\dfrac{2}{\sqrt{3}}\cdot\dfrac{2}{\sqrt{3}}\cdot\dfrac{2}{\sqrt{3}}=\dfrac{8}{3\sqrt{3}}$

23. $\tan(A-B)=\frac{\tan A-\tan B}{1+\tan A\tan B}$

$$=\frac{\frac{1}{35}-\frac{18}{17}}{1+\frac{1}{35}\times\frac{18}{17}}=\frac{17-35\times18}{35\times17+18}=\frac{-613}{613}=-1$$

24. $\frac{\tan A+\tan B}{1-\tan B\tan B}=1\Rightarrow\tan A+\tan B+\tan A\tan B=1\Rightarrow(1+\tan A)(1+\tan B)=2$

25. $\cot(A+B)=\cot 45$

$$\frac{\cos 20(1-\tan 20)}{\cos 20(1+\tan 20)}=\frac{\tan 45-\tan 20}{1+\tan 45\tan 20}=\tan(45-20)=\tan 25$$

$$\cot A\cot B-1=\cot A+\cot B\Rightarrow\cot B\cot A-\cot B-\cot A-1=0$$

$$\cot A[\cot B-1]-[\cot B-1]-2=0\Rightarrow(\cot B-1)(\cot A-1)=2$$

26. Conceptual

27. Conceptual

28. a-q; b-s,p; c-q; d-r

a) $2\cos 60\cos\theta=\cos\theta$

b) $\cos 15=\cos(45-30)=\frac{\sqrt{3}+1}{2\sqrt{2}}=\frac{(\sqrt{3}+1)(\sqrt{3}-1)}{2\sqrt{2}(\sqrt{3}-1)}=\frac{3-1}{2\sqrt{2}(\sqrt{3}-1)}=\frac{1}{\sqrt{2}(\sqrt{3}-1)}$

c) $\sin(90-\theta)=\cos\theta$

d) $-\sin(A+B)\sin(A-B)=\cos^2A-\cos^2B$

$$-\sin 60\cdot\sin 30=-\frac{\sqrt{3}}{4}$$

29. $A+B+C=180\Rightarrow\tan A+\tan B+\tan C-\tan A\tan B\tan C=0$

30. $\cos(\theta-\alpha)=a$ and $\sin(\theta-\beta)=b$

$\sin^2(\theta-\alpha)=1-\mathrm{a}^2\cos^2(\theta-\beta)=1-\mathrm{b}^2$

$\sin(\theta-\alpha)=\sqrt{1-\mathrm{a}^2}\ \cos(\theta-\beta)=\sqrt{1-\mathrm{b}^2}$

$\cos^2(\alpha-\beta)+2ab(\sin\alpha-\beta)$

$=\cos^2[(\theta-\beta)-(\theta-\alpha)]+2ab\sin[(\theta-\beta)-(\theta-\alpha)]$

$=[\cos(\theta-\beta)\cos(\theta-\alpha)+\sin(\theta-\alpha)\cdot\sin(\theta-\beta)]^2$

$+2ab[\sin(\theta-\beta)\cos(\theta-\alpha)-\cos(\theta-\beta)\sin(\theta-\alpha)]$

$=\left(a\sqrt{1-b^2}+b\sqrt{1-a^2}\right)^2+2ab\left[ab-\sqrt{1-a^2}\sqrt{1-b^2}\right]$

$=a^2(1-b^2)+b^2(1-a^2)+2ab\sqrt{1-a^2}\sqrt{1-b^2}+2a^2b^2-2ab\sqrt{1-a^2}\sqrt{1-b^2}$

$=a^2+b^2=(a+b)^2-2ab=\frac{(a+b)^2+(a-b)^2}{2}$

31. By componendo & dividendo

32. $\frac{\cos 9+\sin 9}{\cos 9-\sin 9} = \tan(45+9) = \tan 54 = \cot 36$

33. $\sin^2\left(\frac{\pi}{8}+\frac{A}{2}\right) - \sin^2\left(\frac{\pi}{8}-\frac{A}{2}\right) = \sin\frac{\pi}{4}\cdot\sin A = \frac{1}{\sqrt{2}}\sin A$

34. $\cos\left(\frac{\pi}{3}\right)\cos 2\theta = \frac{1}{2}\cos 2\theta$

35. a) $\frac{\sin(A+B)+\sin(A-B)}{\cos(A+B)+\cos(A-B)} = \frac{2\sin A\cos B}{2\cos A\cos B} = \tan A$

b) $\sum\frac{\sin A\cos B-\cot A\sin B}{\cos A\cos B} = \sum(\tan A-\tan B) = 0$

c) $\sum\frac{(A-B)}{\sin A\sin B} = \sum(\cot B-\cot A) = 0$

d) $\tan(A+B)\cdot\tan(A-B)$

$= \frac{\sin(A+B)(\sin A-B)}{\cos(A+B)\cos(A-B)} = \frac{\sin^2 A-\sin^2 B}{\cos^2 A-\sin^2 B}$

WORKSHEET – 5

Level – I

1.	B	3.	A	5.	B	7.	B	9.	C
2.	C	4.	A	6.	A	8.	C		

1. $\cos\theta+\sin\theta = a \Rightarrow \cos^2\theta+\sin^2\theta+2\sin\theta\cos\theta = a^2 \Rightarrow \sin 2\theta = a^2-1$

2. $\cos^2 72^\circ - \sin^2 54^\circ = \left(\frac{\sqrt{5}-1}{4}\right)^2 - \left(\frac{\sqrt{5}+1}{4}\right)^2 = \frac{-4\sqrt{5}\times 1}{16} = \frac{-\sqrt{5}}{4}$

3. $\sec 72^\circ - \sec 36^\circ = \frac{4}{\sqrt{5}-1} - \frac{4}{\sqrt{5}+1} = \frac{4[\sqrt{5}+1-\sqrt{5}+1]}{5-1} = 2$

4. $\frac{1-\cos\theta+\sin\theta}{1+\cos\theta+\sin\theta} = \tan\frac{\theta}{2}$

5. for $0<\theta<\frac{\pi}{4}, \sin\theta<\cos\theta \Rightarrow \sin\theta-\cos\theta<0$

6. $\sin\theta = \frac{3}{5}, \sin 3\theta = 3\sin\theta-4\sin^3\theta = \frac{9}{5}-\frac{108}{125} = \frac{225-108}{125} = \frac{117}{125}$

7. $\sin\theta\cdot\sin(60-\theta)\cdot\sin(60+\theta) = \frac{1}{4}\sin 3\theta$

$\sin 20^\circ\sin 40^\circ\sin 80^\circ = \frac{1}{4}\sin 60^\circ = \frac{\sqrt{3}}{8}$

8. $\frac{2\sin\theta\cdot\cos\theta}{2\cdot\cos^2\theta} = \mathrm{Tan}\,\theta$

9. $\tan A + \frac{1-\tan^2 A}{2\tan A} = \frac{2\tan^2 A+1-\tan^2 A}{2\tan A} = \frac{1+\tan^2 A}{2\tan A} = \mathrm{cosec}\,2A$

Level – II

1) D	2) C	3) B	4) C	5) A	6) A	7) B
8) A	9) B	10) B	11) C	12) D	13) A	14) B

15) A 16) C 17) C 18) B 19) A 20) C 21) C

22) B 23) B 24) B,D 25) C 26) D 27) D 28) B

29) a-s,b-r,c-p,d-q 30) A,B,C,D 31) a-r,b-s,c-p,d-q 32) A 33) C

34) A 35) C 36) C 37)B

1. $\cos^2\frac{3\pi}{5}+\cos^2\frac{4\pi}{5}=\cos^2\frac{2\pi}{5}+\cos^2\frac{\pi}{5}=\cos^2 72^\circ+\cos^2 36$

   $\frac{(\sqrt{5}+1)^2}{16}+\frac{(\sqrt{5}-1)^2}{16}=\frac{2(5+1)}{16}=\frac{6}{8}=\frac{3}{4}$

2. $\frac{\sin\theta+2\sin\theta\cdot\cos\theta}{\cos\theta+2\cos^2\theta}=\frac{\sin\theta}{\cos\theta}=\text{Tan}\,\theta$

3. $4\cos^3\theta-3\cos\theta=\cos 3\theta\Rightarrow\cos 45^\circ=\frac{1}{\sqrt{2}}$

4. $\cos\frac{\pi}{5}+\cos\frac{3\pi}{5}=\cos\frac{\pi}{5}-\cos\frac{2\pi}{5}\Rightarrow\frac{\sqrt{5}+1}{4}-\frac{\sqrt{5}-1}{4}=\frac{1}{2}$

5. $\cot\theta=\frac{4}{3}\Rightarrow\cos\theta=\frac{-4}{5}\ (\because 180^\circ<\theta<270^\circ)$

   $\sin\frac{\theta}{2}=+\sqrt{\frac{1-\cos\theta}{2}}\left(\because 90<\frac{\theta}{2}<135^\circ\right)$

   $=\sqrt{\frac{1+\frac{4}{5}}{2}}=\sqrt{\frac{9}{10}}=\frac{3}{\sqrt{10}}$

6. $\cos\theta=\frac{1}{2}\left(a+\frac{1}{a}\right)\Rightarrow\cos 2\theta=2\cos^2\theta-1$

   $=2\left[\frac{1}{2}\left(a+\frac{1}{a}\right)\right]^2-1=\frac{2}{4}\cdot\left[a^2+\frac{1}{a^2}+2\right]-1=\frac{1}{2}\left[a^2+\frac{1}{a^2}\right]$

7. $\frac{\text{Tan}\,3A}{\text{Tan}\,A}=a\Rightarrow\frac{3-\text{Tan}^2 A}{1-3\text{Tan}^2 A}=a\Rightarrow 3-\text{Tan}^2 A=\hat{a}-3a\text{Tan}^2 A$

   $(3a-1)\text{Tan}^2 A=a-3\Rightarrow\text{Tan}^2 A=\frac{a-3}{3a-1}\ a\notin\left(\frac{1}{3},3\right)$

   $\cot^2 A=\frac{3a-1}{a-3}\Rightarrow\text{cosec}^2-1=\frac{3a-1}{a-3}$

   $\text{cosec}^2 A=1+\frac{3a-1}{a-3}=\frac{a-3+3a-1}{a-3}\Rightarrow\sin^2 A=\frac{a-3}{4(a-1)}$

   $\frac{\sin 3A}{\sin A}=3-4\sin^2 A=3-\cancel{4}\,\frac{a-3}{\cancel{4}(a-1)}=\frac{3a-3-a+3}{a-1}=\frac{2a}{a-1}$

8. $\text{Tan}\,A=\frac{2\sin^2\frac{B}{2}}{2\sin\frac{B}{2}\cos\frac{B}{2}}=\text{Tan}\frac{B}{2}\Rightarrow\text{Tan}\,2A=\text{Tan}\,B\Rightarrow\text{Tan}\,2A-\text{Tan}\,B=0$

9. $\left(\frac{\sin 3A}{\sin A}\right)^2-\left(\frac{\cos 3A}{\cos A}\right)^2=\left(\frac{3\sin A-4\sin^3 A}{\sin A}\right)^2-\left(\frac{4\cos^3 A-3\cos A}{\cos A}\right)^2$

   $=(3-4\sin^2 A)^2-(4\cos^2 A-3)^2$

$= \emptyset - 24\sin^2 A + 16\sin^4 A - 16\cos^4 A + 24\cos^2 A - \emptyset$

$= 24[\cos^2 A - \sin^2 A] - 16[\cos^4 A - \sin^4 A] = 24\cos 2A - 16\cos 2A = 8\cos 2A$

10. $\cos 60^\circ = \frac{1}{2}$

11. $\cos^2\left(\frac{\alpha-\beta}{2}\right) = \frac{1+\cos(\alpha-\beta)}{2} = \frac{1}{2}[1 + \cos\alpha\cos\beta + \sin\alpha\sin\beta]$

$= \frac{1}{2}\left[1 + \frac{3}{5}\times\frac{5}{13} + \frac{4}{5}\times\frac{12}{13}\right] = \frac{1}{2}\left[\frac{65+15+48}{65}\right] = \frac{128}{2\times 65} = \frac{64}{65}$

12. $\frac{2\sin A\cdot\cos A}{1-\cos A}\cdot\frac{\cos A}{1+\cos 2A} = \frac{2\sin A\cos A\cdot\cos A}{(1-\cos A)\cdot 2\cos^2 A} = \frac{\sin A}{1-\cos A} = \cot A/2$

13. $\alpha + \beta = \cos^2 36^\circ + \sin^2 18^\circ = \frac{3}{4} \Rightarrow \alpha\beta = \cos^2 36^\circ \cdot \sin^2 18^\circ = \frac{1}{16}$

Quadratic Equ $x^2 - (\alpha+\beta)x + \alpha\beta = 0$

$\Rightarrow x^2 - \frac{3}{4}x + \frac{1}{16} = 0 \Rightarrow 16x^2 - 12x + 1 = 0$

14. $= \frac{2\text{Tan}\,\alpha}{1+\text{Tan}^2\,\alpha} - \frac{1-\text{Tan}^2\,\alpha}{1+\text{Tan}^2\,\alpha}\cdot\text{Tan}\,\alpha = \frac{2\text{Tan}\,\alpha - \text{Tan}\,\alpha + \text{Tan}^3\,\alpha}{1+\text{Tan}^2\,\alpha} = \text{Tan}\,\alpha$

15. $\frac{(\cos A+\sin A)^2-(\cos A-\sin A)^2}{\cos^2 A-\sin^2 A} = \frac{4\sin A\cos A}{\cos 2A} = 2T\text{an}\,2A$

16. $\cos 6^\circ\sin 24^\circ\cos 72^\circ = \frac{1}{8}$

17. $x = \sqrt{\frac{1-\cos\theta}{1+\cos\theta}} = \text{Tan}\,\theta/2 \Rightarrow \frac{2x}{1-x^2} = \frac{2\text{Tan}\,\theta/2}{1-\text{Tan}^2\,\theta/2} = \tan\theta$

18. $a^2 + b^2 = 2 + 2\cos(\alpha-\beta) \Rightarrow a^2 + b^2 = 2\left(2\cos^2\left(\frac{\alpha-\beta}{2}\right)\right)$

$a^2 + b^2 = 4\cos^2\left(\frac{\alpha-\beta}{2}\right) \Rightarrow a^2 + b^2 - 3 = 4\cos^2\theta - 3 = \frac{\cos 3\theta}{\cos\theta}$

19. $\alpha + \beta = \text{Tan}\,22\frac{1}{2} + \cot 22\frac{1}{2} = 2\text{cosec}\,45^\circ = 2\sqrt{2}$

$\alpha\beta = \text{Tan}\,22\frac{1}{2}\cdot\cot 22\frac{1}{2} = 1$

Required Equation is

$x^2 - (\alpha+\beta)x + \alpha\beta = 0 \Rightarrow x^2 - 2\sqrt{2}x + 1 = 0$

20. Let $k = \frac{3}{4} \Rightarrow a^6 + b^6 = (a^2+b^2)^3 - 3a^2b^2(a^2+b^2)$

$\sin^6 A + \cos^6 A = (\sin^2 A + \cos^2 B)^3 - \frac{3}{4}\cdot 4\sin^2 A\cos^2 A(\cos^2 A + \sin^2 A) = 1 - \frac{3}{4}\sin^2 2A$

21. $\text{Cos}\,x = \text{Tan}\,y \Rightarrow 1 + \cos^2 x = \cot^2 z = \frac{\cos^2 z}{\sin^2 z} = \frac{\cos^2 z}{1-\cos^2 z}$

$1 + \cos^2 x = \frac{\text{Tan}^2\,x}{1 - \text{Tan}^2\,x}$ $(\because \cos^2 z = \text{Tan}\,x)$

$1 + \cos^2 x = \frac{\left(\frac{\sin^2 x}{\cos^2 x}\right)}{\left(1 - \frac{\sin^2 x}{\cos^2 x}\right)} \Rightarrow \frac{1+\cos^2 x}{1} = \frac{\sin^2 x}{\cos^2 x - \sin^2 x}$

$\cos^2 x - \sin^2 x + \cos^4 x - \cos^2 x \sin^2 x = \sin^2 x$

$1 - \sin^2 x - \sin^2 x + (1 - \sin^2 x)^2 - (1 - \sin^2 x) \cdot \sin^2 x = \sin^2 x$

$2\sin^4 x - 6\sin^2 x + 2 = 0$

$\sin^2 x = \frac{6 \pm \sqrt{36 - 16}}{2 \times 2} = \frac{3 - \sqrt{5}}{2} = \left(\frac{\sqrt{5} - 1}{2}\right)^2$

$\sin x = \frac{\sqrt{5} - 1}{2} \Rightarrow \sin x = \frac{2[\sqrt{5} - 1]}{4} = 2\sin 18^\circ$

22. $8\sin\theta \cdot \cos\theta \cos 2\theta \cos 4\theta = \sin 8\theta$

23. $\sqrt{\frac{a+b}{a-b}} + \sqrt{\frac{a-b}{a+b}} = \sqrt{\frac{1+b/a}{1-b/a}} + \sqrt{\frac{1-b/a}{1+b/a}}$

$= \sqrt{\frac{1+\tan\theta}{1-\tan\theta}} + \sqrt{\frac{1-\tan\theta}{1+\tan\theta}} = \frac{1 + \mathrm{Tan}\,\theta + 1 - \mathrm{Tan}\,\theta}{\sqrt{1 - \mathrm{Tan}^2\,\theta}} = \frac{2}{\sqrt{1 - \mathrm{Tan}^2\,\theta}}$

$= \frac{\left(\frac{2}{\sqrt{1 + \mathrm{Tan}^2\,\theta}}\right)}{\sqrt{\frac{1 - \tan^2\theta}{1 + \tan^2\theta}}} = \frac{2\cos\theta}{\sqrt{\cos 2\theta}}$

24. $\sin^2 A - \sin^2 B = \cos(A + B)\cos(A - B) = \sin 54 \cdot \sin 30 = \frac{\sqrt{5}+1}{8}$

25. Statement : $\frac{\cos 10 - \sqrt{3}\sin 10}{\sin 10 \cdot \cos 10} = 4\left[\frac{\frac{1}{2}\cos 10 - \frac{\sqrt{3}}{2}\sin 10}{\sin 20}\right] = 4$

Statement : $\frac{1}{\sin 20} - \frac{1}{\sqrt{3}\cos 20} = \frac{4}{\sqrt{3}}\left[\frac{\frac{\sqrt{3}}{2}\cos 20 - \frac{1}{2}\sin 20}{\sin 40}\right] = \frac{4}{\sqrt{3}}\left[\frac{\cos 50}{\sin 40}\right] = \frac{4}{\sqrt{3}}$

26. $\mathrm{Tan}\,2A$: $\frac{2\mathrm{Tan}\,A}{1 - \mathrm{Tan}^2\,A} = \frac{2\left(\frac{4}{3}\right)}{1 - \left(\frac{4}{3}\right)^2} = \frac{\left(\frac{8}{3}\right)}{\frac{9-16}{9}} = -\frac{24}{7}$

27. $\mathrm{Tan}\,3A$: $\frac{3\mathrm{Tan}\,A - \mathrm{Tan}^3\,A}{1 - 3\mathrm{Tan}^2\,A} = \frac{3\left(\frac{4}{3}\right) - \left(\frac{4}{3}\right)^3}{1 - 3\left(\frac{4}{3}\right)^2} = \frac{(44/27)}{\left(-\frac{19}{3}\right)} = \frac{-44}{117}$

28. $\mathrm{Cos}\,2A = \frac{1 - \mathrm{Tan}^2\,A}{1 + \mathrm{Tan}^2\,A} = \frac{1 - \left(\frac{4}{3}\right)^2}{1 + \left(\frac{4}{3}\right)^2} = \frac{9-16}{9+16} = \frac{-7}{25}$

29. a-s b-r c-p d-q

1. $2\,\mathrm{A} + 3\,\mathrm{A} = 90 \Rightarrow A = 18^\circ$
2. $3\,\mathrm{A} + 7\,\mathrm{A} = 90 \Rightarrow A = 9^0$
3. $\mathrm{A} + 3\,\mathrm{A} = 90 \Rightarrow 4A = 22\frac{1}{2}$
4. $\mathrm{A} + 2\,\mathrm{A} = 90 \Rightarrow A = 30^\circ$

30. formulae

31. a) $\sin\theta \cdot \sin(60 - \theta)(\sin 60 + \theta) = \frac{1}{4}\sin 3\theta$

$\sin 10^\circ \sin 50 \sin 70^\circ = \frac{1}{4}\sin 30^\circ$

$\sin 10^\circ \sin 50 \sin 70^\circ \sin 30^\circ = \frac{1}{4} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{16}$

b) $\sin 20 \cdot \sin 40 \sin 60 \sin 80 = \frac{1}{4}\sin 60 \cdot \sin 60 = \frac{3}{16}$

c) $\tan 20 \tan 40 \tan 60 \tan 80 = \tan 20 \tan 40 \tan 60 \tan 80$

$\tan 3 \times 20 \cdot \tan 60 \Rightarrow \tan 60 \cdot \tan 60 = \sqrt{3}\sqrt{3} = 3$

d) $\sin\frac{\pi}{4} \cdot \sin\frac{3\pi}{4} \cdot \sin\frac{5\pi}{4} \cdot \sin\frac{7\pi}{14} = \frac{1}{8}$

32. By Compodendo and devodendo

$$\frac{\tan A - \tan B}{\tan A + \tan B} = \frac{x-1}{x+1}$$

$$\frac{\frac{\sin A}{\cos A} - \frac{\sin B}{\cos B}}{\frac{\sin A}{\cos A} + \frac{\sin B}{\cos B}} = \frac{x-1}{x+1}$$

$$\frac{\sin A\cos B - \cos A\sin B}{\sin A\cos B + \cos A\sin B} = \frac{x-1}{x+1} \Rightarrow \frac{\sin(A-B)}{\sin(A+B)} = \frac{x-1}{x+1}$$

33. $a\cos\theta = c - b\sin\theta$

$a^2(1 - \sin^2\theta) = c^2 + b^2\sin^2\theta - 2bc\sin\theta$

$(a^2 + b^2)\sin^2\theta - 2bc\sin\theta + c^2 - a^2 = 0$

since $\sin\alpha, \sin\beta$ are roots $\Rightarrow \sin\alpha\sin\beta = \frac{c^2-a^2}{a^2+b^2}$

34. $\cos\alpha\cos\beta = \frac{c^2-b^2}{a^2+b^2}$

35. $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta = \frac{a^2-b^2}{a^2+b^2}$

36. $\frac{\cos^3 21 + \cos^3 39}{\cos 21 + \cos 39} = \cos^2 21 + \cos^2 39 - \cos 21\cos 39$

$$A + B = 60^\circ \Rightarrow \mathrm{Cos}^2 A + \cos^2 B - \mathrm{Cos}\, A\mathrm{Cos}\, B = \frac{3}{4}$$

37. K = 2

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + 2\cos\theta}}} = 2\cos\theta$$

WORKSHEET – 6

Level – I

1) B	2)B	3)B	4)B	5)B	6)B	7)C
8)A	9)B	10)B	11)B			

Level – II

1) B	2)B	3)C	4)B	5)C	6)C	8)B
9)D	10)B	11)C	12)C	13)B	14)C	15)C
16)D	17)C	18)A	19)D	20)A	21)B	22)B
23)D	24)D	25)C	26)a-r,b-s,c-p,d-q		27)B	28)0
29)A	30)A	31)A	32)B	33)B	34) a-q; b-r; c-s;d-t	
35)A	36)B	37)B	38)12			

1. $\cos 48^\circ \cdot \cos 12^\circ$

   multiply and divided by ' 2 '

   $\frac{2}{2}\cos 48^\circ \cdot \cos 12^\circ$

   $2\cos A\cos B = \cos(A+B) + \cos(A-B)$

   $= \frac{1}{2}\left(\cos(48^\circ + 12^\circ) + \cos(48^\circ - 12^\circ)\right) = \frac{-1}{2}(\cos 60^\circ + \cos 36^\circ)$

   $= \frac{1}{2}\left(\frac{1}{2} + \frac{\sqrt{5}+1}{4}\right) = \frac{1}{2}\left[\frac{2+\sqrt{5}+1}{4}\right] = \frac{\sqrt{5}+3}{4\times 2} = \frac{\sqrt{5}+3}{8}$

2. $\sin 20^\circ \cdot \sin 40^\circ \cdot \sin 60^\circ \cdot \sin 80^\circ = 3/16$

   $\sin 60^\circ = [\sin 20^\circ \cdot \sin 40^\circ \cdot \sin 80] \Rightarrow \sqrt{3}/2[\sin 20^\circ \cdot \sin(60-20) \cdot \sin(60+20^\circ)]$

   $\Rightarrow \sqrt{3}/2\left[\frac{\sin(20^\circ)^3}{4}\right] = \frac{\sqrt{3}}{2}\left[\frac{\sin 60^\circ}{4}\right] = \frac{\sqrt{3}}{2}\left[\frac{\frac{\sqrt{3}}{2}}{4}\right] = \frac{\sqrt{3}}{2}\times\frac{\sqrt{3}}{8} = \frac{3}{16}$

3. $\cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(\frac{6\pi}{7}\right) =$

   $= \frac{2\sin\left(\frac{2\pi}{7}\right)}{2\sin\left(\frac{2\pi}{7}\right)}\left[\cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(\frac{6\pi}{7}\right)\right]$

   $= \left[2\sin(2\pi/7)\cos\left(\frac{2\pi}{7}\right) + 2\sin\left(\frac{2\pi}{7}\right)\cos\left(\frac{4\pi}{7}\right) + 2\sin\left(\frac{2\pi}{7}\right)\cos\left(\frac{6\pi}{7}\right)\right] \times \frac{1}{2\sin\left(\frac{2\pi}{7}\right)}$

   $= \left[\sin\left(\frac{4\pi}{7}\right) + 2\sin\left(\frac{2\pi}{7}\right)\cos\left(\frac{4\pi}{7}\right) + 2\sin\left(\frac{2\pi}{7}\right)\cos\left(\frac{6\pi}{7}\right)\right] \times \frac{1}{2\sin\left(\frac{2\pi}{7}\right)}$

   $= \left[\sin\left(\frac{4\pi}{7}\right) + \sin\left(\frac{6\pi}{7}\right) - \sin\left(\frac{2\pi}{7}\right) + \sin\left(\frac{8\pi}{7}\right) - \sin\left(\frac{4\pi}{7}\right)\right] \times \frac{1}{2\sin\left(\frac{2\pi}{7}\right)}$

   $= \left[\sin\left(\frac{6\pi}{7}\right) - \sin\left(\frac{2\pi}{7}\right) + \sin(8\pi/7)\right] \times \frac{1}{2\sin\left(\frac{2\pi}{7}\right)}$

$= [\sin(\pi/7) - \sin(2\pi/7)] - \sin(\pi/7)] \times \frac{1}{2\sin(2\pi/7)} = \frac{-\sin(2\pi/7)}{2\sin(2\pi/7)} = \frac{-1}{2}.$

4. $1 + \cos 2\theta + \cos 4\theta + \cos 6\theta =$

$$\begin{bmatrix}\cos 2\theta = 2\cos^2\theta - 1 \\ 2\cos^2\theta = 1 + \cos 2\theta\end{bmatrix} = 2\cos^2\theta + 2\cos\left[\frac{4\theta + 6\theta}{2}\right]\cos\left[\frac{4\theta - 6\theta}{2}\right]$$
$$= 2\cos^2\theta + 2\cos 5\theta\cos\theta$$
$$= 2\cos\theta[\cos\theta + \cos 5\theta] = 2\cos\theta \cdot 2\cos\left(\frac{\theta + 5\theta}{2}\right)$$
$$= 2\cos\theta \cdot 2\cos\left(\frac{\theta + 5\theta}{2}\right)\cos\left(\frac{\theta - 5\theta}{2}\right)$$
$$= 4\cos\theta \cdot \cos 2\theta \cdot \cos 3\theta$$

5. $\sin A + \sin A + \sin 5\ A + \sin 7\ A$

$= 2\sin\left(\frac{A + 3A}{2}\right)\cos\left(\frac{A - 3A}{2}\right) + 2\sin\left(\frac{5A + 7A}{2}\right)\cos\left(\frac{5A - 7A}{2}\right)$

$= 2\sin 2A\cos A + 2\sin 6A\cos A = 2\cos A[\sin 2A + \sin 6A]$

$= 2\cos A \cdot 2\sin\left(\frac{2\ A + 6\ A}{2}\right)\cos\left(\frac{2\ A - 6\ A}{2}\right) = 4\cos A \cdot \cos 2\ A \cdot \sin 4\ A$

6. $\sin 47^\circ + \sin 13^\circ = 2\sin\left(\frac{47+13}{2}\right)\cos\left(\frac{47^\circ - 13}{2}\right)$

$= 2\sin 30^\circ\cos 17^\circ = 2 \times \frac{1}{2} \times \cos 17^\circ = \cos 17^\circ$

7. $A + B + C = 180^\circ$

$\cos 2\ A + \cos 2\ B + \cos 2c = 2\cos(A - B) + 2\cos^2 c - 1$

$= -1 + 2\cos^2 c + 2\cos(180^\circ - c)\cos(A - B) = -1 + 2\cos^2 c - 2\cos c\cos(A - B)$

$= -1 + 2\cos c[\cos c - \cos(A - B)] = -1 + 2\cos c[\cos(180 - (A + B)) - \cos(A - B)]$

$= -1 + 2\cos c[-\cos(A + B) - \cos(A - B)]$

$= -1 - 2\cos C[2\cos A\cos B] = -1 - 4\cos A\cos B\cos C$

8. $\sin 78^\circ - \sin 18^\circ + \cos 132^\circ \Rightarrow 2\cos\left(\frac{78+18}{2}\right)\sin\left(\frac{78-18}{2}\right) + \cos(180 - 48)$

$= 2\sin 30^\circ\cos 48^\circ + \cos(180^\circ - 48^\circ) = 2 \cdot \frac{1}{2} \cdot \cos 48^\circ - \cos 48^\circ = 0$

9. $A + B + C = \pi \Rightarrow A + B = \pi - c$

$\Rightarrow \text{Tan}(A + B) = \text{Tan}(\pi - c) \Rightarrow \frac{\text{Tan}\,A + \text{Tan}\,B}{1 + \text{Tan}\,A \cdot \text{Tan}\,B} = \text{Tan}\,c$

$\Rightarrow \text{Tan A} + \text{Tan B} = -\text{Tanc}[1 + \text{Tan A} \cdot \text{Tan B}]$

$\Rightarrow \text{TanA} + \text{Tan}\,B + \text{Tan}\,C = \text{TanA.TanB.TanC}$

10. $\pi < \alpha - \beta < 3\pi, \text{Sin}\,\alpha + \text{Sin}\,\beta = \frac{-21}{65}, \text{Cos}\,\alpha + \text{Cos}\,\beta = \frac{-27}{65}$

squaring and adding them

$\Rightarrow 1 + 2\sin\alpha\sin\beta + 1 + 2\cos\alpha\cos\beta = \frac{441}{65^2} + \frac{729}{65^2} = \frac{1170}{65^2}$

$\Rightarrow 1+\cos(\alpha-\beta)=\frac{9}{65}\Rightarrow 2\cos^2\left(\frac{\alpha-\beta}{2}\right)=\frac{9}{65}\Rightarrow \cos\left(\frac{\alpha-\beta}{2}\right)=\frac{-3}{\sqrt{130}}$

Since $\frac{\alpha-\beta}{2}$ lies in II or III quadrant.

11. $\tan 15^\circ\cdot\tan 45^\circ\cdot\tan 75^\circ$

$=\tan 15^\circ\cdot\tan(60-45)\cdot\tan(60+15^\circ)=\tan(3\times 45^\circ)=\tan 45^\circ=1$

12. $A+B+C=\pi\Rightarrow \sin 2A+\sin 2B+\sin 2C=(\sin 2A+\sin 2B)+\sin 2C$

$=2(\sin(A+B)\cdot\cos(A-B))+\sin 2C=2(\sin(\pi-c)\cdot\cos(A-B))+2\sin C\cos C$

$=2\sin C[\cos(A-B)+\cos C]=2\sin c[\cos(A-B)-\cos(A+B)]=4\sin A\cdot\sin B\cdot\sin C$

13. $\mathrm{Cos}(A+B)\cdot\mathrm{Cos}(A-B)=2\sin C[\cos(A-B)+\cos(A+B)]$

$=2\sin C[2\sin A\sin B]=4\sin A\sin B\sin C$

$=\cos(A+B)\cdot\cos(A-B)=(\cos A\cos B-\sin A\sin B)(\cos A\cos B+\sin A\sin B)$

$=\cos^2 A\cos^2 B-\sin^2 A\sin^2 B$

$=\cos^2 A(1-\sin^2 B)-\sin^2 A\sin^2 B=\cos^2 A-\cos^2 A\sin^2 B-\sin^2 A\sin^2 B$

$=\cos^2 A-\sin^2 B(\cos^2 A-\sin^2 A)=\cos^2 A-\sin^2 B$

14. $\cos 2A+\cos 2B+\cos 2C=(\cos 2A+\cos 2B)+\cos 2C$

$=2\cos(A+B)\cos(A-B)+\cos^2 C=2\cos(\pi-c)\cos(A-B)+\cos 2C$

$=-1-2\cos C(\cos(A-B)-\cos C)$

$=-1-4\cos A\cos B\cos C$

$A+C=2B\Rightarrow B=\frac{A+C}{2}$

15. Put $A=B=C=60^\circ$ and verify

16. Put $A=B=30^\circ$ and verify (shortcut)

17. $\sqrt{3}\cos(20^\circ)-4\cos(20^\circ)=\frac{\sqrt{3}\cos(20^\circ)-4\sin 20^\circ\cos 20^\circ}{\sin 20^\circ}$

$=\frac{2\left[\sqrt{3}/2\cos 20^\circ-2\sin 20^\circ\cos 20^\circ\right]}{\sin 20^\circ}$

$=\frac{2\sin 60^\circ\cos 20^\circ-\sin 40^\circ}{\sin 20^\circ}=\frac{2\sin 60^\circ\cos 20^\circ-2\sin 40^\circ}{\sin 20^\circ}$

$=\frac{\sin 80^\circ+\sin 40^\circ-2\sin 40^\circ}{\sin 20^\circ}=\frac{\sin 80^\circ-\sin 40^\circ}{\sin 20^\circ}$

$=\frac{2\sin 60^\circ\sin 60^\circ}{\sin 20^\circ}=2\cos 60^\circ=2\times\frac{1}{2}=1$

Ans : −"1"

19. Squaring and adding,

$$\cos(\alpha-\beta)=\frac{a^2+b^2}{2}-1=\cos 2\theta$$

$$\cos^2\theta=\frac{a^2+b^2}{4}$$

$$\frac{\cos 3\theta}{\cos\theta}=a^2+b^2-3a$$

20. Let $x=\tan A, y=\tan B, z=\tan C$

Given $\tan A+\tan B+\tan C=\tan A\cdot\tan B\cdot\tan C$

$\tan A+\tan B=-\tan C+\tan A\cdot\tan B\cdot\tan C$

$\tan A+\tan B=-\tan C(1-\tan A\cdot\tan B)$

$$\frac{\tan A+\tan B}{1-\tan A\cdot\tan B}=-\tan C$$

$\tan(A+B)=-\tan C$

$A+B+C=\Pi$

$2A+2B+2C=2\Pi$

$2A+2B=2\Pi-2C$

$\tan(2A+2B)=\tan(2\Pi-2C)$

$$\frac{\tan 2A+\tan 2B}{1-\tan 2A\cdot\tan 2B}=-\tan x$$

$\tan 2A+\tan 2B+\tan 2C=\tan 2A\cdot\tan 2B\cdot 2\tan C$

$$\frac{2\tan A}{1-\tan^2 A}+\frac{2\tan B}{1-\tan^2 B}+\frac{2\tan C}{1-\tan^2 C}=\frac{2\tan A}{1-\tan^2 A}\cdot\frac{2\tan B}{1-\tan^2 B}\cdot\frac{2\tan C}{1-\tan^2 C}$$

$$\frac{2x}{1-x^2}+\frac{2y}{1-y^2}+\frac{2z}{1-z^2}=\frac{2x}{1-x^2}\cdot\frac{2y}{1-y^2}\cdot\frac{2z}{1-z^2}$$

21. $\alpha+B=90^\circ$ and $\beta+\gamma=\alpha$

Given $\beta+\gamma=\alpha$

$\mathrm{Tan}(\beta+\gamma)=\mathrm{Tan}\,\alpha=\mathrm{Tan}(90-\beta)$

$$\frac{\mathrm{Tan}\,\beta+\mathrm{Tan}\,\gamma}{1-\mathrm{Tan}\,\beta\mathrm{Tan}\,\gamma}=\cot\beta$$

$\mathrm{Tan}\,\beta+\mathrm{Tan}\,\gamma=\cot\beta(1-\mathrm{Tan}\,\beta.\mathrm{Tan}\,\gamma)$

$\mathrm{Tan}\,\beta+\mathrm{Tan}\,\gamma=\cot\beta-\cot\beta\cdot\mathrm{Tan}\,\beta\cdot\mathrm{Tan}\,\gamma$

$\mathrm{Tan}\,\beta+\mathrm{Tan}\,\gamma=\cot\beta-\tan\gamma$

$\mathrm{Tan}\,\beta+\mathrm{Tan}\,\gamma=\cot(90-\alpha)-\tan\gamma$

$\mathrm{Tan}\,\beta+\mathrm{Tan}\,\gamma=\tan\alpha-\tan\gamma$

$\tan\alpha=\mathrm{Tan}\,\beta+2\mathrm{Tan}\,\gamma$

22. $\frac{2\cos 5\theta\cdot\cos\theta+5\cdot 2\cos 3\theta\cdot\cos\theta+10\cdot 2\cos^2\theta}{\cos 5\theta+5\cos 3\theta+10\cos\theta}$

$= \frac{2\cos\theta(\cos 5\theta + 5\cos 3\theta + 10\cos\theta)}{(\cos 5\theta + 5\cos 3\theta + 10\cos\theta)} = 2\cos\theta$

23. $\tan\theta = \frac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha} = \frac{\tan\alpha - 1}{\tan\alpha + 1} = \tan\left(\alpha - \frac{\Pi}{4}\right)$

24. $A + B + C = 270°$

$\cos^2 A + \cos^2 B + \cos^2 C = 2\cos(A + B)\cos(A - B) + \cos^2 C$

$\left[\because \cos C + \cos D = 2\cos\left(\frac{C + D}{2}\right)\cos\left(\frac{C - D}{2}\right)\right]$

$= 2\cos(270 - C)\cos(A - B) + \cos^2 C = -2\sin C \cdot \cos(A - B) + 1 - 2\sin 2C$

$= 1 - 2\sin C[\cos(A - B) + \sin C] = 1 - 2\sin C[\cos(A - B) + \cos(A + B)]$

$= 1 - 2\sin C[2\sin A \cdot \sin B] = 1 - 4\sin A\sin B\sin C$

25. $\sin A - \sin 2B + \sin^2 C$

$= 2\sin(270 - B) \cdot \cos(A - C) - 2\sin B \cdot \cos B$

$= -2\cos B[\cos(A - C) + \sin B]$

$= 2\cos B[\cos(A - C) + \sin 270 - (A + C)]$

$= -2\cos B[\cos(A - C) - \cos(A + C)] = -4\sin A\cos B\sin C$

26. a) r b) s c) p d) q

27. $\cos(\beta - \gamma) + \cos(\gamma - \alpha) + \cos(\alpha - \beta) = \frac{-3}{2}$

$2\cos(\beta - \gamma) + \cos(\gamma - \alpha) + \cos(\alpha - \beta) + 3 = 0$

$2[\cos(\beta - \gamma) + \cos(\gamma - \alpha) + \cos(\alpha - \beta)]$

$+\sin^2\alpha + \cos^2\alpha + \sin^2\beta + \cos^2\beta + \sin^2\gamma + \cos^2\gamma = 0$

$\Rightarrow [\sin\alpha + \sin\beta + \sin\gamma]^2 + [\cos\alpha + \cos\beta + \cos\gamma]^2 = 0$

$\Rightarrow \sin\alpha + \sin\beta + \sin\gamma = 0$ and $\cos\alpha + \cos\beta + \cos\gamma = 0$

$\therefore$ Both I & II are true

28. $\sin 85° - \sin 35° - \cos 65° =$

$= 2\cos\left(\frac{120}{2}\right)\sin\left(\frac{50}{2}\right) - \cos 65° = 2 \cdot \frac{1}{2} \cdot \sin 25° - \sin 25° = 0$

29. Both statement I&II are true

$2\cos\frac{\pi}{13} + \cos\frac{9\pi}{13} + \cos\frac{3\pi}{13} + \cos\frac{5\pi}{13}$

$= \cos\left(\frac{\pi}{13} + \frac{9\pi}{13}\right) + \cos\left(\frac{\pi}{13} - \frac{9\pi}{13}\right) + \cos\frac{3\pi}{13} + \cos\frac{5\pi}{13}$

$= \cos\left(\frac{10\pi}{13}\right) + \cos\left(\frac{8\pi}{13}\right) + \cos\frac{3\pi}{13} + \cos\frac{5\pi}{13}$

$= \cos\left(\pi - \frac{\pi}{13}\right) + \cos\left(\pi - \frac{5\pi}{13}\right) + \cos\frac{3\pi}{13} + \cos\frac{5\pi}{13}$

$= -\cos\frac{\pi}{13} + \cos\frac{5\pi}{13} + \cos\frac{3\pi}{13} + \cos\frac{5\pi}{13} = 0$

30. we have 3 A = 2 A + A

$$\tan 3A = \tan(2A + A) = \frac{\tan 2A + \tan A}{1 - \tan 2A \cdot \tan A}$$
$$\Rightarrow \tan 3A \cdot \tan 2A \cdot \tan A = \tan 3A - \tan 2A - \tan A = 1$$

31. we have 5x = 3x + 2x

$\tan 5x = \frac{\tan 3x + \tan 2x}{1 - \tan 3x \cdot \tan 2x}$

$\tan 5x \cdot \tan 3x \cdot \tan 2x = \tan 5x - \tan 3x - \tan 2x = 1$

32. Given A + C = B

$\tan(A + C) = \tan B$

$\frac{\tan A + \tan C}{1 - \tan A \cdot \tan C} = \tan B$

$\tan A + \tan C = \tan B - \tan A \cdot \tan B \cdot \tan C$

33. $\frac{1+\sin A-\cos A}{1+\sin A+\cos A} = \frac{2\sin^2\frac{A}{2}+2\sin\frac{A}{2}\cos\frac{A}{2}}{2\cos^2\frac{A}{2}+2\sin\frac{A}{2}\cos\frac{A}{2}}$

$$= \frac{2\sin\frac{A}{2}\left[\sin\frac{A}{2} + \cos\frac{A}{2}\right]}{2\cos\frac{A}{2}\left[\cos\frac{A}{2} + \sin\frac{A}{2}\right]} = \tan\frac{A}{2}$$

34. Conceptual

35. $\tan A \cdot \tan B \cdot \tan C == \frac{3+\sqrt{3}}{\sqrt{3}-1}$

36. $\tan A\tan B + \tan B\tan C + \tan C\tan A = 5 + 4\sqrt{3}$

37. If $A = 45°, B = 60°$, then

$$C = 2 + \sqrt{3} = \frac{\sqrt{3}+1}{\sqrt{3}-1} = \left(\frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}}\right)$$

$\tan(45° + 30°) \Rightarrow C = 75°$

38. $\tan^2\left(\frac{\pi}{16}\right) + \tan^2\left(\frac{3\pi}{16}\right) + \tan^2\left(\frac{5\pi}{16}\right) + \tan^2\left(\frac{7\pi}{16}\right)$

$= \tan^2\left(\frac{\pi}{16}\right) + \tan^2\left(\frac{3\pi}{16}\right) + \tan^2\left(\frac{\pi}{2} - \frac{3\pi}{16}\right) + \tan^2\left(\frac{\pi}{2} - \frac{\pi}{16}\right)$

$= \left[\tan^2\left(\frac{\pi}{16}\right) + \cot t^2\left(\frac{\pi}{16}\right)\right]^2 + \left[\tan\left(\frac{3\pi}{16}\right) + \cot\left(\frac{3\pi}{16}\right)\right]^2 - 4$

$= \frac{1}{\sin^2\left(\frac{\pi}{16}\right)\cos^2\left(\frac{\pi}{16}\right)} + \frac{1}{\sin^2\left(\frac{3\pi}{16}\right)\cos^2\left(\frac{3\pi}{16}\right)} - 4$

$$= \frac{4}{\sin^2\left(\frac{\pi}{8}\right)} + \frac{4}{\sin^2\left(\frac{\pi}{8}\right)} - 4$$

$$= 4\left[\frac{1}{\sin^2\left(\frac{\pi}{8}\right)} + \frac{1}{\cos^2\left(\frac{\pi}{8}\right)}\right] - 4$$

$$= \frac{4}{\sin^2\left(\frac{\pi}{8}\right)\cos^2\left(\frac{\pi}{8}\right)} - 4$$

$$= \frac{8}{\left[2\sin\left(\frac{\pi}{8}\right)\cos\left(\frac{\pi}{8}\right)\right]^2} - 4$$

$$= \frac{8}{\sin^2\left(\frac{\pi}{4}\right)} - 4$$

$$\frac{8}{\frac{1}{2}} - 4 \Rightarrow 16 - 14 = 12$$

11. Mensuration

WORKSHEET – 1
Level – I

1) C 2) A 3) D 4) B 5) D 6) C 7) A

8) B 9) D 10) B

1. 48.75 cm^2



Given length of a rectangle = Breadth of a rectangle +1 cm

L = b + 1

Perimeter of a rectangle = $2(1 + b) = 28$

L + b = 14 (2)

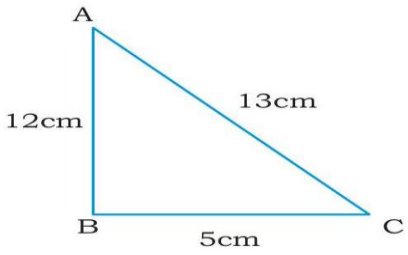
Substitute equation (1) in equation (2)

b + 1 + b = 14

b = 6.5 cm 1 = 7.5 cm

Area of a rectangle = $1 \times b = 6.5 \times 7.5 \Rightarrow 48.75\ cm^2$.

2. 0.0030 m^2



Given sides of right traingle are 5 cm, 12 cm, 13 cm

Area of a triangle = $\frac{1}{2} \times b \times h = \frac{1}{2} \times 5 \times 12 = 30\ cm^2 \Rightarrow 0.0030\ m^2$.

3. 24 cm^2

Given sides of a triangle are 4 cm, 13 cm and 15 cm

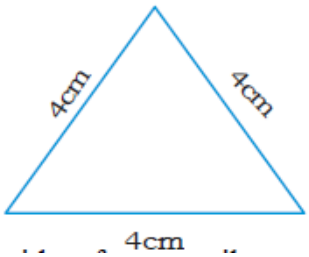
Semi perimeter (s) = $\frac{a+b+c}{2} = \frac{4+15+13}{2} = 16$ cm

$\Delta = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{16(16-4)(16-13)(16-15)}$

$\Delta = \sqrt{16 \times 12 \times 3 \times 1} = \sqrt{576}$

$\Delta = 24\ cm^2$.

4. Ans: (2) $4\sqrt{3}\ cm^2$



Given side of an equilateral triangle = 4 cm

Area of an equilateral triangle = $\frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4}(4)^2 \Rightarrow 4\sqrt{3}$ cm^2.

5. Diameter of circle = d = 14 cm ⇒ Area of circle = $\frac{\pi d^2}{4} = \frac{22}{7} \times \frac{14\times14}{4} = 154$ cm^2

6. Radius of semicircle = r = 21 cm

   Perimeter of semicircle = $\frac{36r}{7} = \frac{36}{7} \times 21 = 108$ cm

7. $d_1 = 14$ cm, $d_2 = 12$ cm ⇒ Area of Rhombus = $\frac{1}{2}\ d_1\ d_2 = \frac{1}{2} \times 14 \times 12 = 84$ cm^2

8. $h = \frac{\sqrt{3}}{2}\ a = 3\sqrt{3}$

   Area of equivaleral triangle = $\frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4} \times 6 \times 6 = 9\sqrt{3}$ cm^2

9. Diagonal of square = d = $14\sqrt{2}$ cm

   Area of square = $\frac{d^2}{2} = \frac{(14\sqrt{2})^2}{2} = \frac{196\times2}{2} = 196$ cm^2

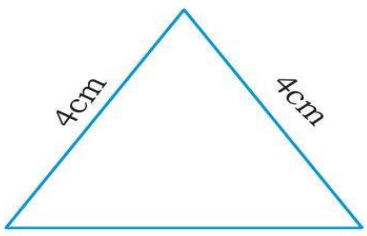
10. b = 14 cm, h = 21 cm

    Area of triangle = $\frac{1}{2}bh = \frac{1}{2} \times 14 \times 21 = 147$ cm^2

Level – II

1) B	2) A	3) C	4) D	5)C	6)48cm2
7)D	8)D	9)B	10)C	11)C	12)B
13)C	14)5cm	15)B	16)B	17)A	18)D
19)C	20)B	21)C	22)C	23)B	24)C
25)A,B	26)A,B,C,D	27)A	28)A	29)8cm	
30)a-s;b-r;c-q;d-p		31)A,B,C,D	32) A,B	33) D	34) a-q, b-s, c-r, d-p

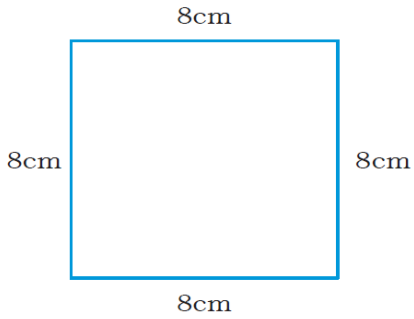
1. 12 cm



Given side of 4 cm equilateral triangle = 4 cm

Perimeter of an equilateral triangle = 3a = 3(4) ⇒ 12 cm.

2. Ans: (a) 64 cm^2



Given side of a square = 8 cm

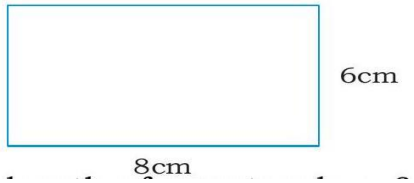
Area of a square = $s^2 = (8)^2 \Rightarrow 64\ cm^2$.

3. 210 cm^2

Given diagonals of a rhombus are 20 cm and 21 cm

Area of a rhombus = $\frac{1}{2}\ d_1\ d_2 = \frac{1}{2} \times 20 \times 21 = 210\ cm^2$.

4. 48 cm^2

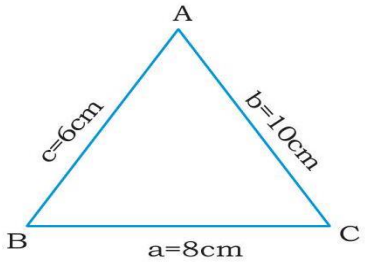


Given length of a rectangle = 8 cm

Breadth of a rectangle = 6 cm

Area of a rectangle = $1 \times b = 8 \times 6 = 48\ cm^2$.

5. 12 cm



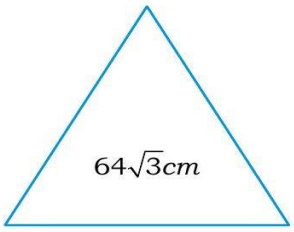
Given sides of a triangle are 6 cm, 8 cm and 10 cm

Semi perimeter (s) = $\frac{a+b+c}{2} = \frac{6+8+10}{2} \Rightarrow 12$ cm.

6. Given a = 10 cm, b = 12 cm

Area of isosceles triangle area of triangle = $\frac{1}{2} \times b \times h = \frac{1}{2} \times 12 \times 8 = 48$sqcms.

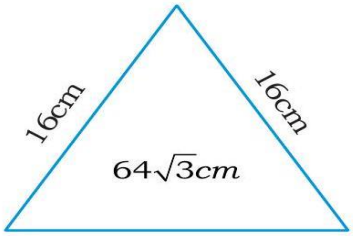
7. 16 cm



Given, area of an equilateral triangle = $64\sqrt{3}$ cm^2

$\frac{\sqrt{3}}{4}a^2 = 64\sqrt{3} \Rightarrow a^2 = 64 \times 4 \Rightarrow a = 16$ cm.

8. 48 cm



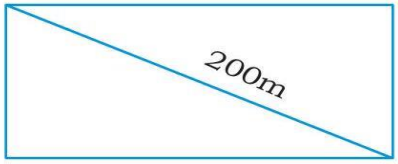
Given the perimeter of an equilateral triangle = 3a = 48

therefore a = 16 cm

Area of an equilateral triangle = $64\sqrt{3}$ cm^2

$\frac{\sqrt{3}}{4}a^2 = 64\sqrt{3} \Rightarrow a^2 = 64 \times 4 \Rightarrow a = 16$ cm

9. 19200 cm^2



Given $\frac{l}{b} = \frac{4}{3} \Rightarrow b = \frac{3l}{4} \Rightarrow$ Diagonal of a rectangle = 200 m

$\sqrt{l^2 + b^2} = 200$

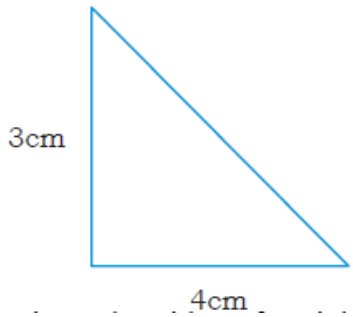
$\sqrt{l + \left(\frac{3l}{4}\right)^2} = 200 \Rightarrow \sqrt{l^2 + \frac{9l^2}{16}} = 200$

$\sqrt{\frac{25l^2}{16}} = 200 \Rightarrow \frac{5l}{4} = 200$

$l = 160$ cm $b = \frac{3(160)}{4} = 120$ cm

Area of a rectangle = $l \times b = 160 \times 120 \Rightarrow 19200$ cm^2.

10. 6 cm^2

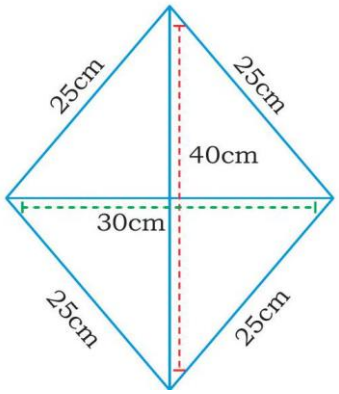


Given the sides of a right triangle are 3 cm and 4 cm

Area of a right triangle = kcm^2

$\frac{1}{2} \times b \times h = k \Rightarrow \frac{1}{2} \times 4 \times 3 = k \Rightarrow k = 6\ cm^2.$

11. 600 cm^2



Given perimeter of a rhombus = 100 cm

4 s = 100 cm

s = 25 cm

We know that, some of squares of the sides of rhombus is equal to sum of squares

of diagonals $(AB)^2 + (BC)^2 + (CD)^2 + (AD)^2 = (AC)^2 + (BD)^2$

Diagonal (AC) of a rhombus = 40 cm

$4(25)^2 = (40)^2 + (BD)^2$

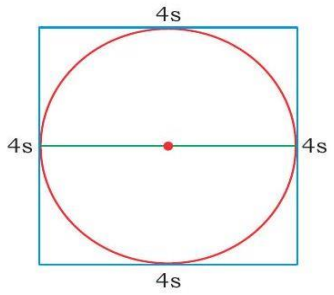
$4(625) = 1600 + (BD)^2$

$2500 = 1600 + (BD)^2 \Rightarrow (BD)^2 = 900 \Rightarrow BD = 30\ cm$

Area of a rhombus $= \frac{1}{2} \times AC \times BD = \frac{1}{2} \times 40 \times 30 \Rightarrow 600\ cm^2.$

12. 16.6 $cm^2 \Rightarrow Ar(\triangle BEC) = \frac{1}{2} Ar(\square ABCD)$

13. 16 s^2

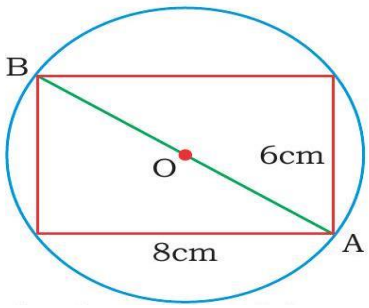


Given, area of a circle $4s^2\pi \Rightarrow \pi r^2 = 4s^2\pi \Rightarrow r = 2s \Rightarrow d = 4s$

Diameter of a circle = side of a square = 4 s

Area of a square = $a^2 \Rightarrow (4\ s)^2 \Rightarrow 16\ s^2$.

14. 5 cm



Given the length and breadth of a rectangle are 6 cm and 8 cm

$AB = \sqrt{l^2 + b^2} = \sqrt{(8)^2 + (6)^2} = \sqrt{100} \Rightarrow$ AB = 10 cm

AB is diameter and OA is radius. $\Rightarrow$ OA = $\frac{AB}{2} = \frac{10}{2} = 5$ cm.

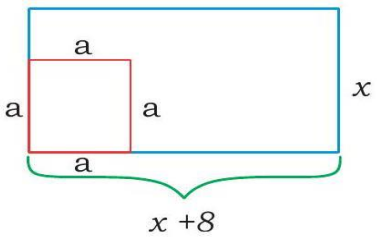
15. Area of square - Area of circle

$\Rightarrow a^2 - \pi r^2 = 49 - \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} = 49 - \frac{77}{2} = 49 - 38.5 = 10.5\ \text{cm}^2$

16. As we know, 1min = $6^0 \Rightarrow$ 10 min = 60°

Area swept by minute hand = $\frac{60}{360} \times \frac{22}{7} \times 14 \times 14 = 102.66\ \text{cm}^2$

17. Let $l = x + 8; b = x \Rightarrow$ A.T.P x = 2a



Remaining area = area of rectangle - area of square = $x(x + 8) - a^2$

$\Rightarrow 2a(2a + 8) - a^2 = 4a^2 + 16a - a^2 = 3a^2 + 16a$

18. 300%

Given, length and breadth are increased by 100%

The increased length = 100% of $l = \frac{100 \times l}{100} = l$

Total length $= l + l = 2l$

The increased breadth = 100% of b $= \frac{100 \times b}{100} = b$

Total breadth = b + b = 2 b

Area of actual dimensions $= l \times b$

Area of increased dimensions $= 2l \times 2b = 4lb$

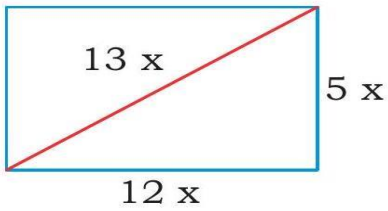
Increased area $= 4lb - lb = 3lb$

Increased area percentage $= \frac{3lb}{lb} \times 100 \Rightarrow 300\%$.

19. The length of diagonal = 169

$13x = 169$

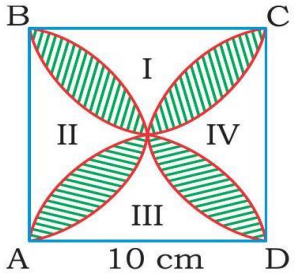
$x = 13$



Area of the field $= [12(13)] \times [5(13)] = 156 \times 65 = 10140\ m^2$

20. Area of unshaded (I & II)

Area of square -2 (area of semi circle) $\Rightarrow a^2 - 2\left(\frac{1}{2}\pi r^2\right)$



$10^2 - 3.14\left(\frac{10}{2} \times \frac{10}{2}\right) \Rightarrow 100 - 314 \times \frac{1}{2} \times \frac{1}{2} \Rightarrow 100 - 78.5 = 21.5\ cm^2$

As, similiary

Area of unshaded (II & IV) = Area of square -2(area of semi circle) = $21.5\ cm^2$

$\therefore$ Area of shaded = area of square - area of (I, II, III, IV)

$= (10)^2 - (21.5 + 21.5) = 100 - 43 = 57\ cm^2$

21. $\pi : 1 \Rightarrow$ Given, Radius of circle increased by 1 cm $\Rightarrow R = r + 1$

$\frac{\text{circumference of a circle}}{\text{Diameter of a circle}} = \frac{2\pi R}{2R} \Rightarrow \pi : 1.$

22. Decreases by 4%

Given, Length of rectangle increased by 20%

Increased length = 20% of $l = \frac{20}{100} \times l \Rightarrow \frac{l}{5} \Rightarrow$ Total length $= l + \frac{l}{5} = \frac{6l}{5}$

breadth of rectangle decreased by 20%

Decreased length = 20% of b $= \frac{20}{100} \times b \Rightarrow \frac{b}{5} \Rightarrow$ Total breadth $= b - \frac{b}{5} = \frac{4b}{5}$

Area of actual dimensions $= l \times b$

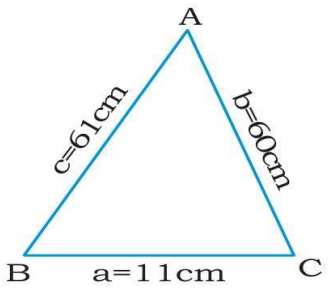
Area of changed dimensions $= \frac{6l}{5} \times \frac{4b}{5} \Rightarrow \frac{24}{25} lb$

The decreased area $= lb - \frac{24}{25} lb \Rightarrow \frac{lb}{25}$

The decreased area percentage $= \frac{\frac{lb}{25}}{lb} \times 100$

23. Area of square - 2(area of semi circle APD &BPC) = 10.5 cm^2

24. Area of shaded region = area of square −4(area of quadrant circle) = 42 cm^2



Given, sides of a triangle are 11 cm, 60 cm and 61 cm

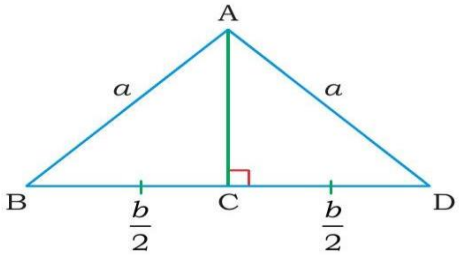
Semi perimeter $= \frac{11+60+61}{2} = 66$ cm $\Rightarrow \Delta = \sqrt{s(s-a)(s-b)(s-c)}$

$\Delta = \sqrt{66(66-11)(66-60)(66-61)} \Rightarrow \Delta = \sqrt{66 \times 55 \times 6 \times 5}$

$\Delta = \sqrt{(11)^2 \times (6)^2 \times (5)^2} \Rightarrow \Delta = 330$ cm^2

Perimeter of a triangle $= a + b + c = 11 + 60 + 61 \Rightarrow 132$ cm.

26.



1. Sides of isosceles triangle a, a, b

AC is perpendicular bisector and also height

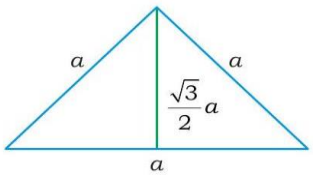
$BC = CD = \frac{b}{2} \Rightarrow AC^2 = a^2 - \frac{b^2}{4} \Rightarrow AC = \frac{\sqrt{4a^2 - b^2}}{2}$

Area of a traingle $= \frac{1}{2} \times b \times h = \frac{1}{2} \times b \times \frac{\sqrt{4a^2-b^2}}{2} \Rightarrow \frac{b}{4}\sqrt{4a^2 - b^2}$.

2. Given, diameter of a circle = du ⇒ Area of a circle = πr^2

$= \pi\left(\frac{du}{2}\right)^2 \left[\because r = \frac{d}{2}\right] = \frac{\pi d^2}{4}u^2$

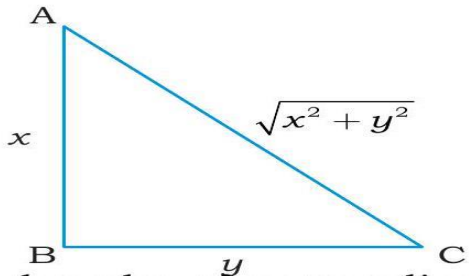
3.



Given, side of equilateral triangle = a; Area of equilateral triangle = $\frac{\sqrt{3}}{4}a^2$

$\frac{1}{2} \times a \times h = \frac{\sqrt{3}}{4}a^2 \Rightarrow h = \frac{\sqrt{3}}{2}a$.

4.



Given, lengths of perpendicular sides of right angled triangle are x and y

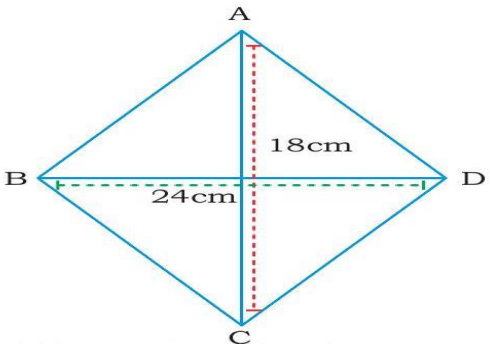
$(AC)^2 = y^2 + x^2$

$AC = \sqrt{x^2 + y^2}$

Perpendicular of a triangle = $AB + BC + AC = x + y + \sqrt{x^2 + y^2}$.

27. Area of rhombus $= \frac{1}{2} \times d_1 \times d_2 \Rightarrow \frac{x^2}{\sqrt{2}} = \frac{1}{2} \times x \times d_2 \Rightarrow \frac{2x}{\sqrt{2}} = d_2 \Rightarrow d_2 = \sqrt{2}x$

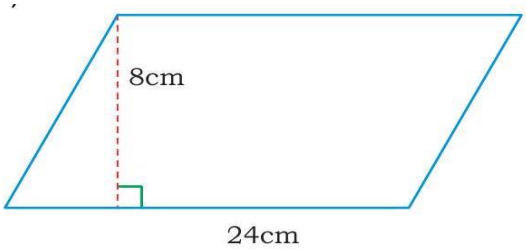
28. Ans: (a) 216 cm^2



Given, Diagonals of a rhombus are 24 cm and 18 cm

Area of a rhombus $= \frac{1}{2}\ d_1\ d_2 = \frac{1}{2} \times 24 \times 18 \Rightarrow 216\ cm^2$.

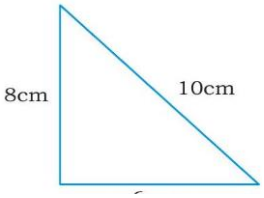
29. Ans: (b) h = 8 cm



Give, Base of parallelogram = 24 cm ⇒ Area of parallelogram = 192 cm^2

$b \times h = 192 \Rightarrow 24 \times h = 192 \Rightarrow h = \frac{192}{24} \Rightarrow h = 8$ cm.

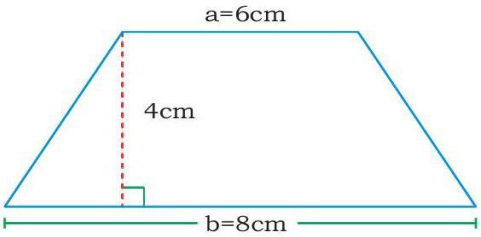
30. a)



Given, Triangle of sides 6 cm, 8 cm and 10 cm

Area of a triangle $= \frac{1}{2} \times b \times h = \frac{1}{2} \times 8 \times 6 \Rightarrow 24\ cm^2$.

b) Ans: (3)

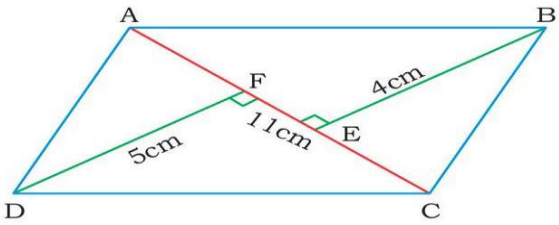


Given, Parallel sides of a trapezium are 6 cm and 8 cm a = 6 cm, b = 8 cm

Height of a trapezium = 4 cm

Area of a trapezium $= \frac{1}{2}\ h(a + b) = \frac{1}{2} \times 4 \times (6 + 8) \Rightarrow 28\ cm^2$.

c)



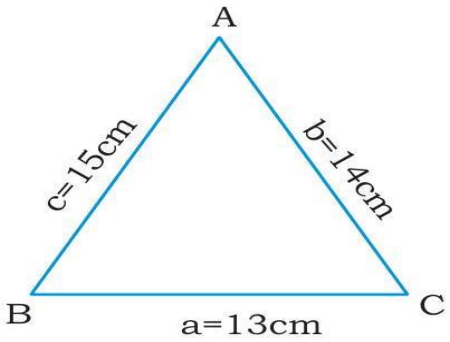
Given, Diagonal of a quadrilateral = 10 cm

AC = 10 cm, BE = 4 cm, DF = 5 cm

Area of a quadrilateral = Area of △ ABC + area of △ ADC

$= \frac{1}{2} \times AC \times BE + \frac{1}{2} \times AC \times DF = \frac{1}{2} \times 10 \times 4 + \frac{1}{2} \times 10 \times 5 = 20 + 25 \Rightarrow 45\ cm^2$.

d) Ans: (1)



Given, Sides of a triangle are 13 cm, 14 cm and 15 cm

Semi perimeter $= \frac{13+14+15}{2} = 21$ cm $\Rightarrow \Delta = \sqrt{s(s-a)(s-b)(s-c)}$

$\Delta = \sqrt{21(21-13)(21-14)(21-15)} \Rightarrow \Delta = \sqrt{21 \times 8 \times 7 \times 6}$

$\Delta = \sqrt{(3)^2 \times (7)^2 \times (2)^4} \Rightarrow \Delta = 84$ cm^2.

31. conceptual 32. conceptual

32. side of triangle

$\frac{\sqrt{3}}{4}a^2 = s \Rightarrow a = \frac{\sqrt{4} \times s}{\sqrt{\sqrt{3}}} = \frac{2\sqrt{s}}{3^{1/4}} = 3a = 3\left(\frac{2\sqrt{s}}{3^{1/4}}\right) = 2\sqrt{s} \cdot 3^{3/4}$

parameter=circumference.

$2\sqrt{s} \cdot 3^{3/4} = 2\pi r \Rightarrow \pi r^2 = \frac{3\sqrt{3}\, s}{\pi}$.

34. Conceptual

WORKSHEET – 2
Level – I

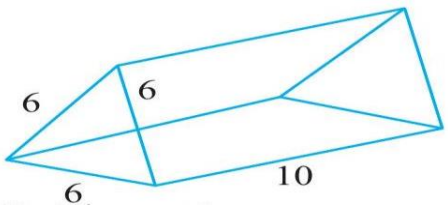
1) C 2) C 3) C 4) D 5) A 6) D 7) A

8) B 9) C 10) B

1. Lengths are 5 cm, 10 cm, 15 cm $\Rightarrow$ l = 5 cm, b = 10 cm, h = 15 cm

TSA of cuboid $= 2(lb + hl + bh) = 2(50 + 150 + 75) = 2(275) \Rightarrow 550$ cm^2.

2. Edge = 6 cm = a $\Rightarrow$ Volume of cube $= a^3 = (6)^3 = 36 \times 6 \Rightarrow 216$ cm^3.



Height = 10 cm

Volume of prism $=? = \frac{\sqrt{3}}{4}a^2 \times h \Rightarrow \frac{\sqrt{3}}{4} \times 36 \times 10 \Rightarrow 90\sqrt{3}$ cm^3.

4. Diameter = 5 cm, Depth = 14 m, Radius $= \frac{5}{2} = 2.5$ m

Volume of cylinder $= \pi r^2 h = \frac{22}{7} \times \frac{5}{2} \times \frac{5}{2} \times 14 \Rightarrow 275$ m^3.

5. h = 21 cm, d = 18 cm, r = 9 cm

   Volume $= \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 9 \times 9 \times 21 = 22 \times 81 \Rightarrow 1782$ cm^3.

6. r = 2 m, h = 14 m ⇒ Volume of cylinder $= \pi r^2 h$

   Cost of digging per 1 m^3 = rs1.20/−

   $= \frac{22}{7} \times 2 \times 2 \times 14 = 88 \times 2 \Rightarrow 176$ m^3 $= 176 \times 1.20 = 211.2$

   Total cost of digging = 211.2Rs

7. r = 7 cm ⇒ Volume of hemisphere = ?

   $= \frac{2}{3}\pi r^3 = \frac{2}{3} \times \frac{22}{7} \times 7 \times 7 \times 7 = \frac{44 \times 49}{3} \Rightarrow 718.6$ cm^3.

8. Surface area $= 6a^2$ $\therefore 6a^2 = 864 \Rightarrow a^2 = 144 \Rightarrow a = 12$.

   Volume $= a^3 = (12)^3$ cm^3 $\therefore d = (12)^3$

   $\sqrt[3]{(12)^3} = 12$.

9. $2\pi r = 12$ cm $\Rightarrow h = \pi$

   Volume of cylinder $= \pi r^2 h = \pi \times \pi \left(\frac{6}{\pi}\right)^2 = \frac{\pi^2}{\pi^2} \times 36 \Rightarrow 36$ cm^3.

10. $r_1 = 4x$ $r_2 = 5x$

    $h_1 = 5$ $h_2 = 6$

    Volume of cylinder $= \pi r^2 h$

    ratio, $\frac{\pi r_1^2 h_1}{\pi r_2^2 h_2} = \frac{(4)^2}{(5)^2} \times \frac{5}{6} = \frac{16 \times 5}{25 \times 6} = \frac{8}{15} \Rightarrow 8:15$

Level – II

1)C	2)A	3)A	4)D	5)C	6) D	7) 60cm3
8)A	9)D	10)C	11)C	12)A	13)120m3	14) B
15)B	16)D	17)A	18)14cm	19) B	20) B	21) C
22)B	23)B	24)21175	25) A,B,C,D	26)A	27) B	28) B
29)a-r;b-t;c-q;d-s		30)A,C	31)A	32) B	33) C	34) B

35) a-q; b-s; c-p; d-r

1. $d_1 = 2x$ $d_2 = 3x$

   $h_1 = ?$ $h_2 = ?$

   Volume of 1 st cylinder = Volume of 2 nd cylinder

   $\pi r_1^2 h_1 = \pi r_2^2 h_2 \Rightarrow \frac{h_1}{h_2} = \frac{r_2^2}{r_1^2} = \frac{\left(\frac{3}{2}\right)^2}{(1)^2} = \frac{9}{4} \Rightarrow 9:4$

2. Diagonal $= \sqrt{3}a$ $\therefore \sqrt{3}a = \sqrt{12} \Rightarrow a = \frac{\sqrt{3} \times \sqrt{4}}{\sqrt{3}} \Rightarrow a = 2$ cm.

3. $1 = 25$ cm, b = 20 cm, h = 15 cm $\Rightarrow$ Volume $= l \times b \times h = 25 \times 20 \times 15 \Rightarrow 7500$ cm^3.

4. T.S.A of cube $= 6a^2 \Rightarrow 6a^2 = 150 \Rightarrow a^2 = 25 \Rightarrow a = 5$

   Volume $= a^3 = 25 \times 5 \Rightarrow 125$ cm^3.

5. $V = 1200$ cm$^2, l = 15$ cm, $b = 10$ cm, $h = ?$

   Volume of cuboid $= l \times b \times h \Rightarrow 1200 = 15 \times 10 \times h \Rightarrow \frac{120}{15} = h \Rightarrow h = 8$ cm.

6. a = 16 cm $\Rightarrow$ Volume of cube $= ? = a^3 \Rightarrow (16)^3 \Rightarrow 4096$ cm^3.

7. h = 10 cm $\Rightarrow$ Sides $= 3 \times 4 \times 5$

   Volume of traingle prism $= ? = h \times$ area of base $\Delta^{le} = 10 \times \frac{1}{2} \times 3 \times 4 = 60$ cm^2

8. h = 3 m, d = 8 m, r = 4 m v $= \frac{1}{3}\pi r^2$ h $\Rightarrow$ v $= \frac{352}{7}$ m^3

9. $h = 10m, b = 8m, l = 12m$

   Volume of salt pit $= l \times b \times h = 10 \times 8 \times 12 \Rightarrow 96 \times 10 \Rightarrow 960$ m^3.

10. $\frac{\text{Volume of cylinder x}}{\text{Volume of cylinder y}} = ? \frac{\pi r_1^2 h_1}{\pi r_2^2 h_2} = \frac{(7)^2 \times 2}{(4)^2 \times 7} = \frac{98}{112} = \frac{7}{8}$

11. a = 5 cm $\Rightarrow 5 \times$ Volume of cube = Volume of sphere $\Rightarrow 5 \times (5)^3 = \frac{4}{3}\pi r^3$

    Volume of sphere = 625.

12. R = 7 cm, h = 2 cm

    Volume of cylinder $= \pi r^2$ h $= \frac{22}{7} \times 7 \times 7 \times 2 = 22 \times 14 \Rightarrow 308$ cm^3.

13. 1 = 4 m, b = 5 m, h = 6 m $\Rightarrow$ Volume of cuboid $= l \times b \times h = 4 \times 5 \times 6 \Rightarrow 120$ m^3.

14. $\ell = 3$ units, b = 3 units, h = 2 units

    $2\begin{cases} b = 15 \\ l = h \end{cases}$

    Volume of cube 1 = Volume of cube 2

    $3 \times 3 \times 2 = 15 \times l \times h$

18. $\Rightarrow 18 = 15 \times l^2 = l = \sqrt{\frac{6}{5}}$

19. h = 3 cm, r = 14 cm

    Volume of right circular cone $= \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 14 \times 14 \times 3 = 44 \times 14 = 616$ cm^3

16. r = 2 inches, h = 7 inches

    Volume of cylinder $= \pi r^2 h = \frac{22}{7} \times 7 \times 2 \times 2 = 88$ inches.

17. Ans: (a) 396 cm^2, 704 cm^2

    Given, Radius of a cylinder = 7 cm

    Height of a cylinder = 9 cm

    C.S.A of a cylinder $= 2\pi$rh $= 2 \times \frac{22}{7} \times 7 \times 9 \Rightarrow 396$ cm^2.

T.S.A of a cylinder = $2\pi r(r + h) = 2 \times \frac{22}{7} \times 7 \times (7 + 9) = 44 \times 16 \Rightarrow 704$ cm^2.

18. 14 cm

Given, T.S.A of a cone = 770 cm^2 ⇒ Slant height = 4 (Radius)

T.S.A of a cone = $\pi r^2 + \pi rl$

$770 = \pi r^2 + \pi r(4r)$ [$\because l = 4r$]

$770 = 5\pi r^2$

$r^2 = 7 \times 7 \Rightarrow r = 7$ cm

Diameter of a cone = $2r = 2(7) = 14$ cm.

19. Area of one face of the cube = 64 cm^2 = $a^2 \Rightarrow a = \sqrt{64} \Rightarrow a = 8$

Volume of cube = $64 \times 8 \Rightarrow 512$ cm^3.

20. let the length be $x, h = 2x \Rightarrow$ b = $\frac{x}{2}$

⇒ Volume of cuboid = $l \times b \times h = \frac{x}{2} \times 2x \times x \Rightarrow x^3$.

21. 1 = 8 m, h = 6 m, b = 22.5 cm. ⇒ $\frac{800 \times 600 \times 22.5}{25 \times 11.25 \times 6} = 6400$

22. 1 = 30 cm, b = 25 cm, h = ?

Volume of water in cuboidel vessel = $1 \times b \times h$

$4.51 = 30 \times 25 \times$ h ⇒ 4.5×1000 cm^3 = $750 \times$ h ⇒ h = $\frac{4500}{750} \Rightarrow$ h = 6 cm

23. a = 7 cm(Cone) ⇒ Cone radius = 7 cm ⇒ Cone height = 7 cm

Volume of right circular cone = $\frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 7. \Rightarrow \frac{539}{6}$ cm^3

24. h = 11 cm, d = 7 cm, $r = \frac{7}{2}$ cm ⇒ 50 = n cylinder ⇒ Volume = $\pi r^2 h \times 50$

$= \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 11 \times 50 = \frac{22 \times 77}{4} \times 50 = 423.5 \times 50 \Rightarrow 21175$ cm^3.

25. h = 3 m, d = 8 m, r = 4 m.

a) $l^2 = h^2 + r^2 \Rightarrow l = \sqrt{9 + 16} \Rightarrow l = 5$ m.

b) C.S.A of cone = $\pi rl = \frac{22}{7} \times 4 \times 5 \Rightarrow \frac{440}{7}$ m^2.

c) T.S.A of cone = $\pi r(l + r) = \frac{22}{7} \times 4(9) = \frac{792}{7} \Rightarrow 113.1$ m^2.

d) Volume of cone

$= \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 4 \times 4 \times 3 \Rightarrow \frac{352}{7} m^3$.

26. h = 8 cm, r = 2 cm

Volume of cone = Volume of sphere

$\frac{1}{3}\pi r^2 h = \frac{4}{3}\pi r^3 \Rightarrow r_1^2 h = 4r^3 \Rightarrow (2)^2(8) = 4r^3$

$\frac{4 \times 8}{4} = 4r^3 \Rightarrow \sqrt[3]{8} = r \Rightarrow r = 2cm$.

27. r = 8 cm r = 6 cm, h = ?

Volume of hemisphere = Volume of cone

$\frac{2}{3}\pi r^3 = \frac{1}{3}\pi r^2 h \Rightarrow 2(8)^3 = (6)^2 \times h$

$\frac{2\times8\times8\times8}{6\times6} = h \Rightarrow h = 28.44$ cm.

28. radius of hemisphere (r_1) = (r_2) radius of cone

Volume of hemisphere = Volume of cone $\Rightarrow \frac{2}{3}\pi r^3 = \frac{1}{3}\pi r^2 h \Rightarrow 2r = h.$

29. a) T.S.A of hemisphere =? $= 3\pi r^2 \Rightarrow r = 7$ cm

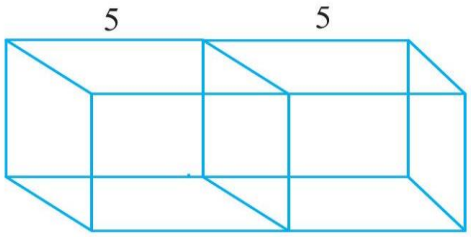
T.S.A $= 3 \times \frac{22}{7} \times 7 \times 7 = 21 \times 22 \Rightarrow 462$ cm^2.

b) $\frac{r}{h} = \frac{5}{7}$ { cylinder } $\Rightarrow$ Volume of cylinder $= \pi r^2 h = 550 \Rightarrow r = 1$ cm

c) Volume of cylinder $= 448\pi$cm^3 $\Rightarrow$ h = 7 cm $\Rightarrow$ r = ?

$$\pi r^2 h = 448\pi \Rightarrow r^2 h = 448 \Rightarrow r^2 \times 7 = 448 \Rightarrow r^2 = \frac{448}{7} \Rightarrow r^2 = 64 \Rightarrow r = 8cm$$

d) Volume of cube $= a^3 = 125 \Rightarrow a = 5$ cm



Length and heights are equal $\Rightarrow$ length $= (5 + 5)$cm $\Rightarrow 10$ cm.

30. $r_1 = 6$ cm, $r_2 = 8$ cm, $r_3 = 10$ cm. $\Rightarrow$ Volume of sphere $= \frac{4}{3}\pi r^3$

$$\frac{4}{3}\pi r^3 = \frac{4}{3}\pi r_1^2 + \frac{4}{3}\pi r_2^3 + \frac{4}{3}\pi r_3^3 \Rightarrow r_1^3 + r_2^3 + r_3^3 = r^3$$

$(6)^3 + (8)^3 + (10)^3 = r^3 \Rightarrow 216 + 512 + 1000 = r^3 \Rightarrow \sqrt[3]{1728} = r.$

$r = 12$

31. C.S.A of cone $= 1159\frac{5}{7}$; Area of base $= 254\frac{4}{7}$

$$\pi r\ell = \frac{8118}{7} \; ; \; \pi r^2 = \frac{1782}{7}$$

$\Rightarrow \ell = \frac{8118}{7} \times \frac{7}{22} \times \frac{1}{9} = 41$; $\Rightarrow r^2 = \frac{1782}{7} \times \frac{7}{22}$

$\Rightarrow \ell = 41$; $\Rightarrow r = 9$

$\therefore \ell^2 = r^2 + h^2 \Rightarrow h^2 = \ell^2 - r^2 = 41^2 - 81$

$h^2 = 1600$

$\Rightarrow h = 40$

Volume of cone $= \frac{1}{3}\pi r^2 \text{ h} = \frac{1}{3} \times \frac{1782}{7} \times 40 = 3394\frac{2}{7}$ cm^3

32. Here, $a^3 = 64$ cm$^3 \Rightarrow a = 4$ cm

So, $\ell = 8$ cm, b $= 4$ cm, h $= 4$ cm

T.S.A of cuboid $= 2(\ell b + bh + h\ell) = 160$ cm^2

33. Volume of cuboid $= \ell \times \text{b} \times \text{h} = 50 \text{ m} \times 25 \text{ m} \times 3 \text{ m} = 3750$ m^3 $= 3750$KL

34. 693 kg

Weight of the pillar = Volume of pillar × density

volume of pillar=volume of cylinder+volume of cone

$= \pi r^2 h + \frac{1}{3}\pi r_1^2 h_1 = \frac{22}{7} \times (10)^2 \times \left(280 + \left(\frac{1}{3} \times 42\right)\right) = 693000g = 693$ kg

35. a-q, b-s, c-p, d-r

12. Arithmetic

WORKSHEET – 1

Level – I

1) D 2) C 3) C 4) A 5) C 6) A 7) B
8) A 9) B

1. $P:Q = 85000:15000 = 85:15\ 17:3$
2. Ratio of their shares $= 22{,}500:35{,}000 = 9:14$
   deepaks share $= \text{Rs } 13800 \times \frac{14}{23} = \text{Rs}8400$
3. $(A+B)$'s 1 day work $= \frac{1}{4} \Rightarrow A$'s 1 day work $= \frac{1}{2}$
   B's 1 day work $= \frac{1}{4} - \frac{1}{12} = \frac{1}{6} \Rightarrow$ Hence B alone can complete the work in 6 days .
4. Aditya speed $= 20$ km/hr $= 20 \times \frac{5}{18} m \left| \text{sec} = \frac{50}{9} m \right| \text{sec}$
   Time taken to cover 400 m $= 400 \times \frac{9}{50} \text{sec} = 72\text{sec} = 1\frac{12}{60} mn = 1\frac{1}{5} mn$
5. speed $= \left(\frac{750}{150}\right) \text{m|sec} = 5 \text{ m|sec} = \left(5 \times \frac{18}{5}\right) \text{km/hr} = 18 \text{ km/hr}$
6. $72 \times \frac{5}{18} m|\text{sec} = 20m|\text{sec}$
7. A man complete one work in 10 days $\Rightarrow$ A man complete work in one day $= \frac{1}{10}$
8. $25 \text{ m} \mid \text{sec} = 25 \times \frac{18}{5} = 90 \text{ km/hr}$
9. A:B:C $= 7:8:11$. Hire charges paid by B $= \text{Rs}520 \times \frac{8}{26} = \text{Rs } 160$

Level – II

1) C 2) A 3) D 4) B 5) D 6) C 7) 384
8) D 9) C 10) Rs.4000 11) B 12)B 13) B
14)1hr 15)B 16) D 17)D 18) C 19)B 20) A
21)D 22)Rs.400 23)A,B,C 24) C 25) C 26)C 27) A
28)405,486 29) A 30)a-q;b-p;c-s,t;d-r 31) A 32) A 33) 1hr
34) A 35) A 36)a-q;b-p;c-s;d-r

1. Whole work will be done by x in $10 \times 4 = 40$ days
   whole work will be done by y in $40 \times \frac{100}{40} = 100$ days
   Whole work will done by Z in $13 \times 3 = 39$ days
   z will be complete the work first.
2. Ratio of rates of working of A and $B = 2:1$
   so ratio of times takes $= 1:2$
   A's 1 day's work $= \frac{1}{6}$ B's 1 day's work $= \frac{1}{12}$
   $(A+B)'$'s 1 day's work $= \frac{1}{6} + \frac{1}{12} = \frac{3}{12} = \frac{1}{4}$
   so A and B together can finish the work in 4 days
3. speed $= \frac{200}{24} m \left| \text{sec} = \frac{25}{3} m \right| \text{sec} = 30 \text{ km/hr}$
4. A person crosses a 600 m distance at speed $= \frac{600}{5 \times 60}$ m/sec

$= 2$ m/sec $= 2 \times \frac{18}{5} = 7.2$ km/hr

5. Speed $= 9$ km/hr $= 9 \times \frac{5}{18}$ m/sec $\Rightarrow$ Total distance $= 35 \times 4$ m $= 140$ m
   time taken $= 140 \times \frac{2}{5}$sec $= 56$sec
6. Speed $= 108$ km/hr $= 108 \times \frac{5}{18}$ m/sec $= 30$ m/sec
   distance covered in 15sec $= (30 \times 15)m = 450$ m
7. Let each side of the square be xkm and let the avarage speed of the place around the square field by ykm/hr, thus. $\frac{x}{200} + \frac{x}{400} + \frac{x}{600} + \frac{x}{800} = \frac{4x}{y}$
   $\Rightarrow \frac{25x}{2400} = \frac{4x}{y} = y = \frac{2400 \times 4}{25} = 384 \Rightarrow$ average speed $= 384$ km/hr
8. Speed from A to B $= 250 \times \frac{2}{11} mph = \frac{500}{11}$mph
   speed from B to A $= 200 \times \frac{2}{9} mph = \frac{500}{9}$mph
   average speed $= \frac{2 \times \frac{500}{11} \times \frac{500}{9}}{\frac{500}{11} + \frac{500}{9}}$mph $= \frac{5,00,000}{4500+5500}$mph $= 50$mph
9. Let the distance covered by the cyclist be x and the time taken be y, their required ratio $= \frac{\frac{1}{2}x}{2y} : \frac{x}{y} = \frac{1}{4} : 1 = 1 : 4$
10. A:B:C $= 500 : 300 : 200 = 5 : 3 : 2$
    let their capital be $5x, 3x, 2x$, respectively
    Then $5x = 10000 \Rightarrow x = 2000 \Rightarrow C'$ s capital $= 2x \Rightarrow = 2 \times 2000 = 4000$
11. Let 1 man's in 1 day's work = x and 1 boy's 1 day work = y
    then $5x + 2y = 4(x + y) \Rightarrow$ x = 2y $\Rightarrow \frac{x}{y} = \frac{2}{1}$ x: y $= 2 : 1$
12. Ratio of times taken by A and B $= 100 : 130 = 10 : 13$
    suppose B takes x days to do the work then $10 : 13 : : 23 : x \Rightarrow x = \frac{23 \times 13}{10} = \frac{299}{10}$
    A's 1 day work $= \frac{1}{23}$ B's 1 day work $= \frac{10}{299}$
    $(A + B)$'s 1 day work $= \frac{1}{23} + \frac{10}{299} = \frac{1}{13} \Rightarrow (A + B)$ can do the work in 13 days.
13. Ratio of shares A, B and C = ratio of their investmenst
    $A : B : C = 250000 : 300000 : 350000 = 5 : 6 : 7$
    share of $B =$ Rs $187200 \times \frac{6}{18} = 62400$
14. Let the duration of the flight be x hours
    then $\frac{600}{x} - \frac{600}{x + \frac{1}{2}} = 200 \Rightarrow$ x$(2x + 1) = 3 \Rightarrow 2x^2 + x - 3 = 0 \Rightarrow (2x + 3)$(x $- 1) = 0$
    x = 1hr ($\because$ neglecting the - ve value of x)
15. $\frac{\left(\frac{1}{2}\right)x}{21} + \frac{\left(\frac{1}{2}\right)x}{24} = 10 \Rightarrow \frac{x}{21} + \frac{x}{24} = 20 \Rightarrow 15x = 168 \times 20 \Rightarrow x = \frac{168 \times 20}{15} = 224$ km
16. Let the speed of two trains be 7x and 8xkm/hr
    then $8x = \frac{400}{4} = 100 \Rightarrow x = \frac{100}{8} = 12.5$
    Speed of first train $= 7 \times 12.5$ km/hr $= 87.5$ km/hr
17. Time taken 1hr40 min48sec $= 1$hr$40\frac{4}{5}$min $= 1\frac{51}{75}$hr $= \frac{126}{75}$hr
    let the actual speed be xkm/hr, then $\frac{5}{7}x \times \frac{126}{75} = 42 \Rightarrow$ x $= \frac{42 \times 7 \times 75}{5 \times 126} \Rightarrow$ x $= 35$ km/hr

18. (A + B) 's 1 day's work $= \frac{1}{15} + \frac{1}{10} = \frac{1}{6}$
work done by A and B in 2days $= \frac{1}{6} \times 2 = \frac{1}{3} \Rightarrow$ Remaining work $= \left(1 - \frac{1}{3}\right) = \frac{2}{3}$
Now $\frac{1}{15}$ work is done by A in 1 day $= \frac{2}{3}$, work will be done by A
$\left(\frac{15}{x} \times \frac{2}{3}\right) = 10$ days $\Rightarrow$ Hence the total time taken $= 10 + 2 = 12$ days
19. (20×16) woman can complete the work in 1 day, 1 woman 's 1 day's work $= \frac{1}{320}$
(16×15) man can complete the work in 1 day $\Rightarrow$ 1 man's 1 day's work $= \frac{1}{240}$
So required ratio $= \frac{1}{240} : \frac{1}{320} = \frac{1}{3} : \frac{1}{4} = 4:3$
20. 162 km/hr $= 162 \times \frac{5}{18} = 45$ m/sec
21. A's 1 day's work $= \frac{1}{15} \Rightarrow$ B's 1 day's work $= \frac{1}{20}$
$(A + B)$'s 1 day's work $= \frac{1}{15} + \frac{1}{20} = \frac{7}{60} \Rightarrow$ (A + B) 's 4 days work $= \frac{7}{60} \times 4 = \frac{7}{15}$
Therefore remaining work $= 1 - \frac{7}{15} = \frac{8}{15}$
22. C's 1 day work $= \frac{1}{3} - \left(\frac{1}{6} + \frac{1}{8}\right) = \frac{1}{3} - \frac{7}{24} = \frac{1}{24}$
A's work : B's work: C's work $= \frac{1}{6} : \frac{1}{8} : \frac{1}{24} = 4:3:1$
C's share for 3 days $= \text{Rs}\left(3 \times \frac{1}{24} \times 3200\right) = 400$
23. 25 m/sec $= 25 \times 18 = 90$ km/hr
24. use 'p' profit $= \frac{A_1 \times P}{A_1 + A_2} = \frac{2000}{5000} \times 14000 = 4000$
25. use 'q' profit $= \frac{A_2 \times q}{A_1 + A_2} = \frac{5000}{7000} \times 14000 = 10000$
26. use 'p' profit $= \frac{A_2 \times P \times t_1}{A_1 + A_2} = \frac{6000 \times 3000 \times 2}{7500} = 4800$
27. Let 1 man 1 day's work = x and 1 boy's 1 day work = y
Then $6x + 8y = \frac{1}{10}$ and $26x + 48y = \frac{1}{2} \Rightarrow$ solve these two we get $x = \frac{1}{100}$ $y = \frac{1}{200}$
(15 men +20 boys) 's 1 days $= \frac{15}{100} + \frac{20}{200} = \frac{1}{4}$
15 men and 20 boys can do the work in 4 days
28. $1500:2500:3000 \Rightarrow \therefore 3:5:6 \Rightarrow$ A's profit $= 369$
$\frac{1}{10}x + \frac{3}{14} \times \frac{9x}{10} = 369 \Rightarrow x = 1260$
Share of $B = 1260 \times \frac{9}{10} \times \frac{5}{14} = 405/- \Rightarrow$ Share of $C = 1260 \times \frac{9}{10} \times \frac{6}{14} = 486/-$
29. $126 \times \frac{5}{18} = 35$
30. a)profit A:Profit $B = 2500:3000$
$= 5:6$
b) $90 \times \frac{5}{18} = 25$ m/s
c) Profit(P) $\rightarrow \frac{2}{5} \times 10000 = 4000 \Rightarrow$ profit(Q) $\rightarrow \frac{3}{5} \times 10000 = 6000$
d) work done in one day by A = 1/ no of working days $= \frac{1}{5}$
31. Formula $\frac{s_1 \times s_2}{s_2 - s_1} \times \frac{t_1 + t_2}{60} = \frac{4 \times 6}{2} \times \frac{16}{60}$ $S_1 = 4$ $T_1 = 8$ $S_2 = 6$ $T_2 = 8$ $= \frac{24}{2} \times \frac{16}{60} = 8 \times \frac{2}{5} = \frac{16}{5} = 3\frac{1}{5}$ KM

32.

33. Let the speed of A and B are $6x$kmph and $9x$kmph
let the total distance to travel = ykm
$\frac{y}{6x} - \frac{y}{9x} = \frac{1}{3} = \frac{y}{x}\left(\frac{1}{18}\right) = \frac{1}{3}$ = y = 6x ⇒ Required time for A = $\frac{6x}{6x}$ = 1 hour

33. 90 km/hr = $90 \times \frac{5}{18} = 25$ 34. Distance = speed x time = 20x2 = 40 km

34. $125 \times \frac{18}{5} = 450$ km/hr

35. a) work done in one day by A = 1/ no of working days = $\frac{1}{5}$
b) profit of A :Profit of $B = x : y$
c) A and B together can complete in
$$\frac{1}{\frac{1}{4}+\frac{1}{5}} = \frac{1}{\frac{9}{20}} = \frac{20}{9} \Rightarrow c \cdot p = \left(\frac{100}{100 + p\%}\right) \times S.P$$
d) use average speed formula

WORKSHEET – 2

Level – I

1.	B	4.	B	7.	C	10.	C
2.	D	5.	A	8.	B		
3.	B	6.	C	9.	B		

Level – II

1) B	2) C	3) B	4) B	5) C	6) D	7) Rs.2646
8) B	9) C	10) Rs.600	11) D	12) B	13) A	14) B
15) D	6) B	17) C	18) B	19) C	20) B	
21) 9 1 %11		22) B	23) D	24) A	25) C	26) B
27) 25	28) C	29) a-s,b-r,c-t,d-q		30) B	31) 9575	32) C
33) B	34) A	35) B	36) a-r,b-p,c-q,d-t			

1. Raj got 35% discount, if there was no discount, Raj would pay Rs 224 this means giving 35% discount = Rs 224 off , 35% marked price = 224
$$\frac{35}{100} \times mp = 224 = \text{mp} = \frac{224 \times 100}{35} = Mp = 640$$

2. Let gurupreet pay Rs 'A' for the sofa we know, she got 20% discount had she gothen 25% discount she would have saved Rs 1000 that means 5% extra dis count means Rs 1000 ⇒ 5% of A = 1000
$$\frac{5}{100} \times A = 1000 \Rightarrow \text{A} = \frac{1000 \times 100}{5} \quad \text{A} = \text{Rs}20{,}000$$

3. 30% discount on 2000 = 30% of 2000 = Rs 600
25% discount on 2000 = 25% of 2000 = Rs 500
Remaining amount = 2000 − 500 = 1500
second discount of 5% = of 1500 = Rs 75
Total discount = 500 + 75 = Rs 575
so difference in dicount = 600 − 575 = 25

4. Let the initial price be Rs 100 , they incresed price by 40%
so, new price = 100 + 40% = 140% of 100 = Rs 140
new to have no profit no loss situations, chandrika must give Rs 40 off how much percent is Rs 40 of 140
chandrika must give $\frac{40}{140} \times 100 = 28.5\%$ DISCOUNT
5. A dozen pair means = 12 pair mp for 12 pairs = Rs 8000
10% discount, so fixed frice = 8000 − 10% of 8000 = 7200
price of one pair = $\frac{7200}{12} = Rs600$
how many pair of trousers can be bhought in Rs 2400
No of pairs of trousers = $\frac{2400}{600} = 4$ pairs
6. Equivalent discount = $\left(25 + 10 - \frac{25\times10}{100}\right)\%$ price by = $A = 35 - 2.5 = 32.5\%$
fixed price = $(100 - 32.5)\%$ of price by 405 = 67.5% of ⇒ $405 = \frac{67.5}{100} \times A$
$A = \frac{100 \times 405}{67.5} = 600$
7. 25% discount initially means price = $(100 - 25)\%$ of 360 = Rs 2700 2% more discount means 2% of 2700 = Rs 54
Simran pays = 2700 − 54 = Rs 2646
8. Equivalent discount = $\left(20 + 5 - \frac{20\times5}{100}\right)\%$ $25 - 1 = 24\%$
9. $\left(15 + 10 - \frac{15\times10}{100}\right)\% \Rightarrow (25 - 1.5)\% \Rightarrow 23.5\%$
10. MP of an article = Rs 800 ⇒ after discount = 90% of 800 = $\frac{90}{100} \times 800 = 720$
profit = 20% ⇒ CP of an article = $\frac{720\times100}{120} = 600$
11. Marked price = Rs 200
Discount % = 12.5%
Discount=Discount % x M.P = $12.5\% \times 200 = 25$
Selling price = marked price-discount = 200 − 25 = rs 175
profit % = 25%
cost price = $c \cdot p = \left(\frac{100}{100+p\%}\right) \times S \cdot P = C \cdot P = \left(\frac{100}{100+25}\right) \times 175 = 140$
12. CP gain S.p
100 10% 110
selling price = $800 - \frac{20}{100} \times 800 = 704 \Rightarrow \frac{100}{110} \times 704 = 640$
13. MP of the article = Rs 240
after the discount sp of the article = $240 \times \frac{90}{100} \times \frac{95}{100} = Rs205.20$
14. Let the cp of an article = Rs100 ⇒ mp of an aritcle = Rs110
sp of an article = Rs 99(10% discount)] ⇒ Difference % = $\frac{11}{110} \times 100 = 10$
15. Let cp of goods = Rs100 ⇒ MP of goods = Rs130(30% above)
SP of goods = 90% of 130 = Rs 117 ⇒ gain = 117 − 100 = 17
16. MP of cycle = Rs110 ⇒ after discount sp = $90\%110 = \frac{90}{100} \times 1100 = 990$
Gain = 10% ⇒ CP = $\frac{990\times100}{100+10} = \frac{990\times10}{110} \Rightarrow$ cp = 900

17. LP of an article = Rs 2000 ⇒ after discount s.p = $2000 \times \frac{80}{100} \times \frac{90}{100} = 1440$
18. 1st discount = 30%
    net two successive discount = $20 + 10 - \frac{20\times10}{100} = (30-2)\% = 28\% = \frac{2}{100} \times 550 = Rs11$
19. let the cost price = $4x$ and selling price = $5x$
    profit = $s \cdot p - c \cdot p = 5x - 4x = x$
    profit % = $\frac{\text{profit}}{c \cdot p} \times 100\% == \frac{x}{4x} \times 100\% = 25\%$
20. 75% of 40 = 25% of $x + 21$
    $\frac{75}{100} \times 40 = \frac{25}{100} \times x + 21$
    $3000 = 25x + 2100$
    $25x = 900$
    $\therefore X = 36$
21. $9\frac{1}{11}\%$
    $c \cdot p = \left(\frac{100}{100 + p\%}\right) \times S \cdot P = C \cdot P = \left(\frac{100}{100+10}\right) \times 8000 = 7272.72$
23. C. P = 10100 and loss % = 5
    s.p = c.p-loss = 10100 − 5% of 10100 = 9595
24. Profit = s.p-c.p = 15000 − 12500 = 2500
    Profit % = $\frac{\text{profit}}{c \cdot p} \times 100\% = \frac{2500}{12500} \times 100\% = \frac{1}{5} \times 100\% = 20\%$
25. $\frac{Sp}{400}$ $\frac{\text{Discount}}{100}$ $\frac{MP}{500} \Rightarrow D\% = \frac{100}{500} \times 100 = 20\%$
26. $\frac{SP}{100}$ $\frac{SP}{125}$ $\frac{MP}{100}$

100×18	$25 \times 5 \times 100$	25×100
1800	2250	2500

    $2500 \rightarrow \frac{900 \times 2500}{1800} = Rs1250$
27. Let the larger number = x x − 20 = 20% of x ⇒ 80%x = 20 ⇒∴ X = 25
28. Quantity of milk purchase be x ⇒ Quantity of water added by y ratio of water to milk = $y : x \Rightarrow CP = 64x \Rightarrow SP = 8(x + y)$ profit percent = 9.696 ⇒ water % added is $\frac{(9.6-8)\times100\%}{8}$ $1.6 \times 100 : 8 = 160 : 8 = 1 : 20$
    a) $\frac{20}{100} \times 750 = 150$
    b) Profit = SP − CP = 850 − 600 = 250
    c) CP = 56.25, gain 20% then SP ⇒ SP = $\frac{100+\text{ gain \%}}{100} \times CP$
    $= \frac{100+20}{100} \times 56.25 = \frac{120}{100} \times 56.25 = \frac{6}{5} \times 56.25 = 67.5$
    d) $SP = \frac{100-\text{ loss \%}}{100} \times CP \Rightarrow 51.7 = \frac{100-12}{100} \times CP$
    $51.7 \times 100 = 88 \times CP \Rightarrow CP = \frac{5170}{88} = 58.75$
30. Let the cos price be = 100 ⇒ then M.P = 140
    Discount = 25% of mp ⇒ $\frac{25}{100} \times 140 = 35$
    S.P = M.P -discount ⇒ 140 − 35 = 105 ⇒ gain % = (105 − 100)% = 5%

Hence the trades gains 5%

31. Cost price of machine = 7660 gain % = 10%

$\therefore SP = \frac{(100 + \text{ gains \%})}{100} \times CP = \frac{100+}{100} \times 7660 = 8426$

Let the MP be x ⇒ discount = 12% of x ⇒ $x \times \frac{12}{100} = \frac{3x}{25}$

$\therefore SP = MP -$ discount $= x - \frac{3x}{25} = \frac{22x}{25}$

But SP = 8426 $= \frac{22x}{25}8426 \Rightarrow x = \frac{8426\times25}{22}$ x = 9575 ⇒ MP of the machine = 9575

32. Let the cost price be = 100 ⇒ gain % = 20% ⇒$\therefore SP = 100 + 20 = 120$

Let the MP = x ⇒ discount = 25% of x = $x \times \frac{25}{100} = \frac{x}{4} \Rightarrow \therefore$ SP = MP-discount

$= x - \frac{x}{4} \Rightarrow sp = \frac{3x}{4} \Rightarrow \frac{3x}{4} = 120\ \frac{3x}{4} = 120 \Rightarrow x = \frac{120 \times 4}{3} = x = 160$

Hence the parked price is 60% above cost price

33. Selling price = Marked price-discount = 1250 − 105 = 1145

34. Discount % = $\frac{\text{discount}}{\text{markedprice}} \times 100\% = \frac{350}{1500} \times 100\% = 63\frac{1}{3}\%$

35. Average discount = $\left(x + y - \frac{xy}{100}\right)\% = \left(15 + 10 - \frac{150}{100}\right)\% = 23.5\%$

36. Formula

WORKSHEET – 3

Level – I

1) A	2) C	3) B	4) A	5) A	6) A	7) D
8) D	9) D	10) C	11) B	12) C	13) A	14) B
15) A						

1. Conceptual

2. $\frac{n+1}{2} = \frac{40+1}{2} = \frac{41}{2}$

3. Conceptual

4. 7 multiplus $= \frac{7+14+21+28\ldots\ldots+140}{20} = \frac{7(1+2+3+\cdots\ldots+20)}{20}$

$\frac{7}{20} \times \frac{20 \times 21}{2} = \frac{147}{2} = 73.5$

5. $\frac{8.55+10.95+14.35+5.15}{4} = \frac{39}{4} = 9.75$

6. $\frac{3+6+9+12+15}{5} = \frac{3}{5}(1 + 2 + 3 + 4 + 5) = \frac{3}{5} \times \frac{3}{5} = 9$

7. Average speed $= \frac{\frac{3}{4}+\frac{11}{12}+\frac{5}{6}}{3} = \frac{9+11+10}{12} = \frac{30}{12} \times \frac{1}{3} = 5/6$ km per minut

8. Average of 8,10,11,15,16 ⇒ $\frac{8+10+11+15+16}{5} = \frac{60}{5} = 12$

9. The average of first 50 natural No's $= \frac{n+1}{2} = \frac{50+1}{2} = \frac{51}{2} = 25.5$

11. $(A + B + C)'$ s are 6 years ago $= 90 - (6 \times 3) = 90 - 18 = 72$

C' s age 6 years ago $= \frac{3}{1+2+3} \times 72 = 36 \Rightarrow C's$ product age $= 36 + 6 = 42$ years

Level – II

1) B	2) C	3) C	4) D	5) A	6) B	7) D
8) C	9) B	10)A	11)C	12)A	13)B	14)416

15)C 16)D 17)76 18)D 19)B 20)A 21)Rs.768
22)A 23)C 24)8 25)C 26)B 27)D
28)a-t,b-p,c-q,d-r 29)C,D 30)A,B,C 31) B 32) 5 33)D
34)D 35)C 36)35 37)a-r,b-p,c-t,d-q

1. $\frac{(n+1)(2n+1)}{6}$ here n = 7 $\frac{1^2+2^2+3^2+4^2+5^2+6^2+7^2}{7} = \frac{8\times15}{6} = 4 \times 5 = 20$

2. Let be the third number be x the second number = $3x$, first number = $\frac{3x}{2}$

   $x + 3x + \frac{3x}{2} = 44 \times 3\ \frac{11}{2} = 44 \times 3 \Rightarrow x = 24$

   so largest no. = 3x = $3 \times 24 = 72$

3. $\frac{1+4+9+x+25+36+49}{7} = 20, \frac{124+x}{7} = 20$ x = 16

4. x, 8,16
   $8^2 = 16 \times x$
   $16 \times x = 64$
   $x = \frac{64}{16} = 4$
5. Given ratio = $2: 72x, 7x \Rightarrow$ their different = $40 \Rightarrow 5x = 40\ x = 8$
   no's = $2 \times 8 = 16, 7 \times 8 = 56$
6. $3: 4$ can not be equal to $5: 6$
7. Diagonal of a square = $\sqrt{2} \times$ side $\Rightarrow \sqrt{2} \times 30: 30 \Rightarrow \sqrt{2}: 1$
8. present ratio = $4: 5 \Rightarrow$ increase of 10% and 20%
   new ratio of salary will be $4 \times 1.1: 5 \times 1.2 = 11: 15$
9. $x \times 5 = 0.75 \times 8 \Rightarrow x = \frac{6}{5} \Rightarrow x = 1.20$
10. $25 \times 18 - [(12 \times 14) + (12 \times 7)] 450 - [168 - 204] = 450 - 372$ thirteenth results = 78
11. five consecutive odd NO's = $2x + 1, 2x + 3, 2x + 5, 2x + 5/1, 2x + 9$
    avarage pf these 5 no's = 61

$$\frac{2x + 1 + 2x + 3 + 2x + 5 + 2x + 7 + 2x + 9}{5} = 61$$

    $10x + 25 = 61 \times 5 \Rightarrow 10x + 25 = 305 \Rightarrow 10x = 305 - 25 \Rightarrow 10x = 280 \Rightarrow X = 28$
    smallest no = $2x + 1$ largest no = $2x + 9$
    $2(28) + 1$
    $= 56 + 1 = 57$

$$= 2(28) + 9$$

    their difference = $65 - 57 = 8$
    $= 56 + 9 = 65$
12. Let be the $no = x \Rightarrow A/C$ problem $\frac{x+x^2}{2} = 5x, x + x^2 = 10x$
    $x^2 = 10x - x \Rightarrow 9x = x^2 \Rightarrow x = 9$

13. Let 5 consecutive No's $= x, x+1, x+2, x+3, x+4$
    average $\frac{x+x+1+x+2+x+3+x+4}{5} = \frac{5x+10}{5} = \frac{5(x+2)}{5} = x+2$
    if two no's are increased $\frac{5x+10+x+5+x+6}{7} = \frac{7x+21}{7} = \frac{7x(x+3)}{7} = x+3$ so the average increased by 1
14. Let the number of boys and girls be $8x, 5x$ Total no.of students $= 13x5x = 160$
    $x = 32, 13x = 13 \times 32 = 416$
15. Total age of 5 members 3 years ago $= 17 \times 5 = 85$ years
    total age of 5 members now $= 85 + 3 \times 5 = 100$ years
    total age of 6 members now $= 17 \times 6 = 102$ years
    age of the baby $= 102 - 100 = 2$ years
16. weight of the teacher
    $= (35.4 \times 25 - 35 \times 24)\text{kg} = 45$ kg
17. avarage of 11 inings $= 36$
    Required number of runs $= (36 \times 11) - (3210) = 396 - 320 = 76$
18. Let the A, B, C represent their respective weights then we have
    $A + B + C = 45 \times 3 = 135 \rightarrow A + B = 40 \times 2 = 80 \Rightarrow B + C = 43 \times 2 = 86$
    $A + B = 80$
    $\underline{C} + \underline{B} = 86$
    $A - C = -6$
    $\Rightarrow C - A = 6$
    $C = 6 + A$ in equ(1)
    $A + B + 6 + A = 135 \Rightarrow 2A + B = 135 - 6 \Rightarrow 2A + B = 129 \Rightarrow 2A + B = 129$
    $\underline{A} + \underline{B} = 80$
    $A = 49 \Rightarrow A + B = 80 \Rightarrow 49 + B = 80 \Rightarrow B = 80 - 49 \Rightarrow B = 31$
19. AM of 75 numbers $= 35$
    sum of 75 no's $= 75 \times 35 = 2625$
    Increased sum $= 2625 + 375 = 3000$
    Increased average $= \frac{3000}{75} = 40$
20. Boys: Girls $= 4:3$
    Let $4x, 3x$
    $3x = 18 \Rightarrow x = 6$
    Boys $= 4 \times 6 = 24$
    Total no.of students $= 4x + 3x = 7x = 7 \times 6 = 42$
    Ratio of Boys and total students $= 24:42 = 4:7$
21. cost α weight
    $c^2 \alpha w^2 \Rightarrow c^2 = kw^2 \Rightarrow 4800 = k(20)^2 \Rightarrow 4800 = k(400)^2 \Rightarrow k = 12$
    Now $c = 12(8)^2 = 12 \times 64 = 768$
22. $\frac{5+8+6+2}{4} = \frac{21}{4} = 5.25$
23. Average of first 20 natural no's $= \frac{n+1}{2} = \frac{20+1}{2} = \frac{21}{2} = 10.5$
24. $\frac{2+4+6+x}{4} = 5 \Rightarrow x = 8$
25. $1:2 = x:8$

upon cross multiplication x = 4

26. $3:5 = 18:x$
    product of means = product of extremes.
    $3x = 5x18$
    $3x = 90$
    $x = 30$
27. $x:9 = 35:63$
    $\Rightarrow x = \frac{35}{63} \times 9 = 5$
28. a) $x:y::y:z$
    b) $7:12 = 7 \times 2:12 \times 2 = 14:24$
    c) 1.5 m to 10 cm = 150 cm to 10 cm = 15:1
    d)middle terms
29. $3:7 = 3 \times 3:7 \times 3 = 9:21$
30. $3600:300 = 12:1$
31. $\frac{x+x^2}{2} = 12x\ x + x^2 = 24x$
    $\Rightarrow x^2 = 24x - x\ x^2 = 23x$
    $x = 23$
32. $\frac{1+3+5+7+9}{5} = \frac{25}{5} = 5$
33. $\frac{4^2+5^2+6^2+8^2}{4} = \frac{141}{4} = 35.25$
34. $\frac{110}{100}(x_1 + x_2 + \cdots .. + x_{10}) \Rightarrow \frac{110}{100} \times 10\bar{x} = 11\bar{x}$
    Average increased by 10%
35. Required avarage $= \frac{(38.9\times10)-(42\times6)}{4} = \frac{137}{4} = 34.25$
36. Excluded number $= (27 \times 5) - (25 \times 4) = 135 - 100 = 35$
37. a) $\frac{12+x}{2} = 10 \Rightarrow 12 + x = 20 \Rightarrow x = 8$
    b) $\frac{17.5+3.5+3+7}{4} = \frac{31}{4} = 7.75$
    c) average $= \frac{6\cdot8+12\cdot2+121}{3} = \frac{140}{3} = 46\frac{2}{3}$
    d) $\frac{1+3+5+7+9}{5} = \frac{25}{5} = 5$

13. Progressions

WORKSHEET – 1
Level – I

1)C 2)D 3) A 4) A 5) A 6) B 7) D
8)C 9)B 10)B

1. $d = \frac{b-a}{n+1} = \frac{q-p}{5+1} = \frac{q-p}{6}$
2. $a = 8, d = a_2 - a_1 = 5 - 8 = -3 \Rightarrow a_7 = a + 6d = 8 + 6(-3) = 8 - 18 = -10$
3. $t_r = 7r - 3 \Rightarrow t_1 = 7(1) - 3 = 7 - 3 = 4 \Rightarrow t_2 = 7(2) - 3 = 14 - 3 = 11$
   $d = t_2 - t_1 = 11 - 4 = 7 \Rightarrow a = 4$
   $S_n = \frac{n}{2}(2a + (n-1)d) = \frac{n}{2}(2(4) + (n-1)7) = \frac{n}{2}(8 + 7n - 7) = \frac{n}{2}(1 + 7n) \Rightarrow S_n = \frac{n}{2}(7n + 1)$
4. $2x = \frac{-2}{7} + \frac{16}{7} \Rightarrow 2x = \frac{14}{7} \Rightarrow 2x = 2 \Rightarrow x = 1$
5. $54,51,48, \ldots\ldots \Rightarrow a = 54, d = 51 - 54 = -3$
   $S_n = 513 \Rightarrow \frac{n}{2}(2a + (n-1)d) = 513 \Rightarrow \frac{n}{2}(2(54) + (n-1)(-3)) = 513$
   $\frac{n}{2}(108 - 3n + 3) = 513 \Rightarrow \frac{n}{2}(111 - 3n) = 513 \Rightarrow n(111 - 3n) = 1026$
   $111n - 3n^2 = 1026 \Rightarrow 3n^2 - 111n + 1026 = 0 \Rightarrow n^2 - 37n + 342 = 0$
   $n^2 - 18n - 19n + 342 = 0 \Rightarrow (n-18)(n-19) = 0 \Rightarrow n = 18,19$
6. $2,5,8,11, \ldots . T_{60}$ $3,5,7,9,11, \ldots . T_{50}$
   2,5,8,11, ….179 3,5,7,9,11, … . ,101
   Identical terms are 5,11, … …
   $a = 5, d = 11 - 5 = 6$
   no.of.identical terms $a + (n-1)d \leq 101$
   $5 + (n-1)d \leq 101 \Rightarrow 5 + (n-1)6 \leq 101 \Rightarrow (n-1)6 \leq 96 \Rightarrow n - 1 \leq 16 \Rightarrow n \leq 17 \ \therefore n = 17$
7. $a - d + a + a + d = 33 \Rightarrow 3a = 33 \Rightarrow 11$
   $(a-d)(a)(a+d) = 792 \Rightarrow (11-d)11(11+d) = 792 \Rightarrow 11^2 - d^2 = \frac{792}{11}$
   $11^2 - d^2 = 72 \Rightarrow 121 - 72 = d^2 \Rightarrow 49 = d^2 \Rightarrow d = 7; a = 11 \Rightarrow a - d = 11 - 7 = 4$
   (least value)
8. $t_{17} = \frac{17(17+3)}{17+2} = \frac{17(20)}{19} = \frac{340}{19}$
9. $\frac{-9}{4}, -2, \frac{-7}{4}, \ldots\ldots \Rightarrow a = \frac{-9}{4}, d = -2 + \frac{9}{4} = \frac{-8+9}{4} = \frac{+1}{4}$

$t_n = 0 \Rightarrow a + (n-1)\left(\frac{+1}{4}\right) = 0 \Rightarrow \frac{-9}{4} + \frac{(n-1)(+1)}{4} = 0$

$-9 + n - 1 = 0 \Rightarrow -10 + n = 0 \Rightarrow n = 10$

10. $0.6, 1.2, 1.8, \ldots\ldots \Rightarrow a = 0.6,\ d = 1.2 - 0.6 = 0.6 \Rightarrow a_{13} = a + 12\ d = 0.6 + 12(0.6) = 0.6 + 7.2 = 7.8$

Level – II

1)B	2)C	3)B	4)A	5)17	6)C	7) D
8)A	9)C	10)A	11)D	12)C	13)A	14)A
15)D	16)B	17)D	18)6	19)0	20)C	
21) A,B,C,D	22)A,B,C	23)A	24)B	25)A	26)B	
27) $a \to q, b \to s, c \to t,\ d \to p$	28)B,C	29)C	30)B	31) A,B,C,D		
32) $a \to s, b \to s, c \to q,\ d \to r$		33)A	34)D	35)B		
36) $a \to r, b \to r, c \to p,\ d \to r$						

1. Given sequence is $2n^2 - 5$

   put n = 1,2,3 ... in the given sequence

   $2(1)^2 - 5, 2(2)^2 - 5, 2(3)^2 - 5 \Rightarrow -3, 3, 13 \ldots$

2. Given sequence is $6n - 3$

   18th term is $6 \times 18 - 3 = 105$

3. Clearly the given sequence following the pattern $+5, +7, +9 \ldots\ldots$

   i.e $7 + 5 = 12, 12 + 7 = 19.$

   Missing number $= 19 + 9 = 28$

4. The given series is arithmetic whose first term = 20,

   Common difference $= -\frac{2}{3}$

   As the common difference is negative, the terms will become negative after some stage. So the sum is maximum if only positive terms are added.

   Now, $t_n = 20 + (n-1)\left(-\frac{2}{3}\right) \geq 0$ if $60 - 2(n-1) \geq 0$ or $62 \geq 2n$ or $31 \geq n$

   $\therefore$ The first 31 terms are non-negative.

   $\therefore$ Maximum sum $= S_{31} = \frac{31}{2}\left\{2 \times 20 + (31-1)\left(-\frac{2}{3}\right)\right\} = \frac{31}{2}\{40 - 20\} = 310$

5. $d = (4x-1) - (5x+2) = (x+2) - (4x-1) \Rightarrow 4x - 1 - 5x - 2 = x + 2 - 4x + 1$

   $-x - 3 = -3x + 3 \Rightarrow\ -x + 3x = 3 + 3\ \Rightarrow 2x = 6\ \Rightarrow x = 3$

   $t_1 = 5x + 2 = 5 \times 3 + 2 = 17$

6. $a + d(n-1) = t_n, t_{10} = 9, t_9 = 10$

   put $n = 10, a + d(10-1) = 9,\ a + 9\ d = 9, a = 9 - 9\ d$

   $n = 9a + d(9-1) = 10 \Rightarrow a + 8\ d = 10 \Rightarrow a = 10 - 8\ d$

$9-9d=10-8d \Rightarrow 9-10=-8d+9d \Rightarrow\ d=-1$

7. a_1, a_2, a_3, a_4 are a_5 are in A.P

$a_1+a_5=a_2+a_4=a_3+a_3 \Rightarrow a_1+a_5=a_2+a_4=4 \Rightarrow \therefore a_3=2$

$a_1+a_2+a_3+a_4+a_5=4+2+4=10 \Rightarrow \sum_{i=1}^{5} a_i=10$

8. $t_m=\frac{1}{n}$ and $t_n=\frac{1}{m}$

$a+(m-1)d=\frac{1}{n}$ and $a+(n-1)d=\frac{1}{m}$

$\rightarrow 1 \ \rightarrow 2$

$\Rightarrow a+(m-1)d-a-(n-1)d=\frac{1}{n}-\frac{1}{m} \Rightarrow a+md-d-a-nd+d=\frac{m-n}{mn}$

$d(m-n)=\frac{m-n}{mn} \Rightarrow d=\frac{1}{mn}$ from 1

$a+(m-1)\left(\frac{1}{mn}\right)=\frac{1}{n} \Rightarrow a+\frac{1}{n}-\frac{1}{mn}=\frac{1}{n} \Rightarrow a=\frac{1}{mn}$

$t_{mn}{}^{th}$ term $=a+(mn-1)d=\frac{1}{mn}+(mn-1)\frac{1}{mn}=1$

9. a, b, c, are in A.P

let $a=A-d, b=A, c=A+D \Rightarrow a=b-D, c=b+D$

$abc=4 \Rightarrow (b-D)b(b+D)=4 \Rightarrow b(b^2-D^2)=4 \Rightarrow b^2-D^2<b^2 \Rightarrow b(b^2-D^2)<b^3$

$4<b^3 \Rightarrow b \geq (4)^{\frac{1}{3}} \Rightarrow b \geq (2)^{\frac{2}{3}}$ $\therefore$ Hence the minimum value of b is $(2)^{2/3}$

10. $t_n=(2n-1)^2=4n^2-4n+1 \Rightarrow S_n=\sum n^2-4\sum n+\sum 1$

$=4\frac{n(n+1)(2n+1)}{6}-4\frac{n(n+1)}{2}+n=\frac{n}{3}(4n^2+6n+2-6n-6+3)=n\frac{(4n^2-1)}{3}$

$=\frac{n(2n+1)(2n-1)}{3}$

11. Let $2a+1, 2a+3, 2a+5$ be the A.P

$m^4=(2a+1)+(2a+3)+\cdots$ upto m terms $=\frac{m}{2}\{2(2a+1)+(m-1)2\}=m(2a+$

$1+m-1)$

$m^3=(2a+1)+m-1 \Rightarrow m^3-m+1=2a+1$ (first integer)

12. $2\log a(1+2+3+\cdots.+n) \Rightarrow 2\log a\frac{n(n+1)}{2}=\log a\cdot(n)(n+1)$

13. $a-d, a, a+d$ are in A.P $\Rightarrow 3a=69 \Rightarrow a=23$

$3a=69 \Rightarrow a=23 \Rightarrow (23-d)23=483 \Rightarrow 529-23\ d=483 \Rightarrow 23\ d=46=d=2$

third part $a+d=23+2=25 \Rightarrow$ first part $\Rightarrow a-d=23-2=21$

second part $\Rightarrow a=23 \Rightarrow$ third part $=25$

14. $T_n=3n^2+4n+5 \Rightarrow S_n=\sum T_n=3\sum n^2+4\sum n+\sum 5$

$=3\cdot\frac{n(n+1)(2n+1)}{6}+4\cdot\frac{n(n+1)}{2}+5n=\frac{n}{2}[(n+1)(2n+1)+4(n+1)+10]$

$= \frac{n}{2}[2n^2 + n + 2n + 1 + 4n + 4 + 10] = \frac{n}{2}[2n^2 + 7n + 15]$

15. given $\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$ are in A.P

now multiplying by $(a+b)(b+c)(c+a)$ to each term

$(a+b)(c+a), (a+b)(b+c), (b+c)(c+a)$ are in A.P

$\Rightarrow a^2 + ab + bc + ca, b^2 + ab + bc + ca, c^2 + ab + bc + ca$ are in A.P

subtracting $ab + bc + ca$ from each term now a^2, b^2, c^2 are in A.P

16. $a - 3d, a - d, a + d, a + 3d \Rightarrow a - 3d + a - d + a + d + a + 3d = 20 \Rightarrow 4a = 20 \Rightarrow a = 5$

$\frac{(a-3d)(a+3d)}{(a-d)(a+d)} = \frac{2}{3} \Rightarrow \frac{a^2 - a^2}{a^2 - d^2} = \frac{2}{3} \Rightarrow \frac{25 - 9d^2}{25 - d^2} = \frac{2}{3} \Rightarrow 75 - 27d^2 = 50 - 2d^2$

$\Rightarrow -25\ d^2 = -25 \Rightarrow d = \pm 1$

when $a = 5, d = 1$, the four parts are 2,4,6,8

$a = 5, d = -1$, the four patr are 8,6,4,2

17. Sum of the unterior angles of a polygon on ' n ' sides (n-2) $180, a = 120, d = 5$

$\frac{n}{2}(2a + (n-1)d) = (n-2)180 \Rightarrow \frac{n}{2}(2(120) + (n-1)5) = (n-2)180$

$n^2 - 25n + 144 = 0 \Rightarrow (n-9)(n-16) = 0 \ \therefore n = 16, 9$

$n = 16 \Rightarrow \frac{n}{2}(2a + (n-1)d) = (n-2)180 \Rightarrow \frac{n}{2}(2(120) + (n-1)5) = (n-2)180$

$n^2 - 25n + 144 = 0 \Rightarrow (n-9)(n-16) = 0 \ \therefore n = 16, 9$

$T_{16} = a + (16-1)d = 120 + 15 \times 5 = 195$ this is not possible as interior angle cannot be greater than 180

$\therefore n = 9$ is the correct answer

18. $S_{2n} = 3\ S_n \left(\because S_n = \frac{n}{2}(2a + (n-1)d) \Rightarrow \frac{2n}{2}(2a + (2n-1)d) = \frac{3n}{2}(2a + (n-1)d) \right.$

$4a + 4nd - 2d = 6a + 3nd - 3d \Rightarrow 6a - 4a + 3nd - 4nd - 3d + 2d = 0$

$2a - nd - d = 0 \Rightarrow 2a = d(n+1) \Rightarrow S_{3n} : S_n$

$\frac{S_{3n}}{S_n} = \frac{\frac{3n}{2}(2a + (3n-1)d)}{\frac{n}{2}(2a + (n-1)d)} = \frac{3(nd + d + 3nd - d)}{nd + d + nd - d} = \frac{3(4nd)}{2nd} = 6$

19. $S_{n+3} - 3\ S_{n+1} - S_n = (S_{n+3} - S_{n+2}) - 2(\ S_{n+3} - S_{n+1}) + (S_{n+1} - S_n)$

$= T_{n+3} - 2T_{n+2} + T_{n+1} = (T_{n+3} - T_{n+2}) + (T_{n+2} - T_{n+1}) = d - d = 0$

20. (3) Sol : The numbers between 100 and 500 that are divisible by 7 are 105, 112, 119,126,133,140,147, ……..483,490,497.

Let such numbers be n. Then $497 = 105 + (n-1) \times 7$; or $n = 57$.

The number between 100 and 500 that are divisible by 21 are

105,126,147 … … 483.

Let such number be m. Then $483 = 105 + (m - 1) \times 21$; or $m = 19$.

Hence, the required number $= n - m = 57 - 19 = 38$.

21. all the four options are correct

22. $\sum(2n) = 2 + 4 + \cdots \ldots + 2n; 2(1 + 2 + \cdots . + n) = 2\left(\frac{n(n+1)}{2}\right) = n(n + 1)$

23. $t_n = 1 + 2 + 3 + \cdots \ldots + n$ a $= 1$, d $= 1; t_n = \frac{n(n+1)}{2}$

Statement (1) is true

Statement (2) is true

24. Here a $= 3$, d $= 3 \& l = 3$

$l = a + (n - 1)d \Rightarrow 99 = 3 + (n - 1)3 \Rightarrow n = 33$

$\therefore S_{33} = \frac{33(3 + 99)}{2} = \frac{33 \times 10_2}{2} = 1683$

25. $S_{100} = \frac{100}{2}[2 + (100 - 1) \times 1] = 5050$

26. $t_n = a + (n - 1)d \Rightarrow t_2 = a + (2 - 1)d = 2 \Rightarrow a + d = 2$

$t_7 = a + (7 - 1)d = 22 \Rightarrow a + 6d = 22$

from (1) & (2) $\Rightarrow$ d $= 4$, a $= -2$; Now $S_n = \frac{n}{2}[2a + (n - 1)d]$

$S_{35} = \frac{35}{2}[-4 + (35 - 1)4] = \frac{35}{2} \times 132 = 2{,}310$

27. a) $a_n = a + (n - 1)d = 1 + 2(n - 1) = 2n - 1$

b) $a_n = a + (n - 1)d = 2 + (n - 1)(2) = 2n$

c) $a = 1, d = 4$

$a_n = a + (n - 1)d = 1 + (n - 1)(4) = 4n - 3$

28. conceptual

d) $2k - k = 3 - 2k, k = 1$

29. $a = 3x - 5y, d = 3x - 9y - (3x - 5y) = 4y$

$t_{90} = a + 89$ d $= 3x - 5y + 89(-4y) = 3x - 361y$

$t_{91} = a + 90$ d $= 3x - 5y + 90(-4y) = 3x - 365y$

30. $t_8 = a + 7$ d $= 31 \Rightarrow a + 7 \times 4 = 31 \Rightarrow a + 28 = 31$

$\Rightarrow a = 3 \Rightarrow t_{15} = a + 10$ d $= 16 \Rightarrow (a + 14$ d$) - (a + 10$ d$) = 16 \Rightarrow 4$ d $= 16 \Rightarrow$ d $= 4$

The first four terms in A.P are $a, a + d, a + 2d, a + 3d$

$3, 3 + 4, 3 + 2 \times 4, 3 + 3 \times 4 = 3, 7, 11, 15$

31. d = 4 months $= \frac{4}{12}$ years $= \frac{1}{3}$ years ; a = 8 years

S_n = 168 years, n =? ; $\frac{n}{2}[2a + d(n-1)] = s_n$

$\frac{n}{2}\left[2 \times 8 + \frac{1}{3}(n-1)\right] = 168 \Rightarrow \frac{n}{2}\left[16 + \frac{(n-1)}{3}\right] = 168 \Rightarrow \frac{n}{2}\left[\frac{48+n-1}{3}\right] = 168$

$\Rightarrow n(47+n) = 168 \times 6 \Rightarrow n^2 + 63n - 16n - 1008 = 0 \Rightarrow n(n+63) - 16(n+63) = 0$

$(n+63)(n-16) = 0 \Rightarrow n = 16$

32. a) $24 + (n-1)(-3) < 0 \Rightarrow 24 - 3n + 3 < 0 \Rightarrow 27 < 3n \Rightarrow 9 < n \Rightarrow n = 10$

b) $210 = 21 + (2-1)21 \Rightarrow 10 = n.$

c) $(a + 19d) - (a + 13d) = 32 \Rightarrow 4d = 32 \Rightarrow d = 8.$

d) $113 + (n-1)(-5) < 0 \Rightarrow 118 - 5n < 0 \Rightarrow 118 < 5n \Rightarrow 23.6 < n \Rightarrow n = 24$

33. $S_n = 3n^2 + 5n \Rightarrow T_1 = 8, T_2 = 14, T_3 = 20 \Rightarrow$ Given terms are in A.P

34. $T_n = 5456 \Rightarrow 6_{n+2} = 5456 \Rightarrow 6_n = 5452 \Rightarrow n = 909$

35. $T_1^2 + T_2^2 + T_3^2 = 8^2 + 14^2 + 20^2 = 660$ 36 conceptual

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WORKSHEET – 2
Level – I

1)A	2)D	3)B	4) C	5) D	6) C
7)C	8)B	9)D	10) C		

1. conceptual

2. conceptual

3. $x^2 = \left(\frac{-2}{7}\right)\left(\frac{-7}{2}\right) \Rightarrow x^2 = 1 \Rightarrow x = 1$

4. $r = \frac{a_2}{a_1} = \frac{-1}{3}$

5. $a = 1, r = \frac{-1}{2} \Rightarrow a_7 = a \cdot r^6 = (1)\left(\frac{-1}{2}\right)^6 = \frac{1}{64}$

6. $a = 1, r = \frac{0.04}{1} \Rightarrow r < 1,\ s_\infty = \frac{a}{1-r} = \frac{1}{1-0.04} = \frac{100}{96} = \frac{25}{24}.$

7. $a_4 = p\ a_7 = q\ a_{10} = r$

a. $r^3 = p$ $ar^6 = q$ a. $r^9 = r$

→ (1) → (2)

$p \cdot r = (a.r^3)(a.r^9) = a^2 \cdot r^{12} = (a \cdot r^6)^2$

$pr = q^2$ [∴ from(2)]

8. In H.P $\Rightarrow t_2 = \frac{5}{4}; t_3 = \frac{1}{2} \Rightarrow \frac{1}{a+d} = \frac{5}{4}; \frac{1}{a+2d} = \frac{1}{2}$ $a + d = \frac{4}{5}$ are in A.P; $a + 2d = 2$ are in A.P → (2)

$(2) - (1) \Rightarrow a + 2d - a + d = 2 - \frac{4}{5} \Rightarrow d = \frac{10-4}{5} = \frac{6}{5}$. from (1) $\Rightarrow a + d = \frac{4}{5}.$

$a + d = \frac{4}{5} \Rightarrow a = \frac{4}{5}\frac{-6}{5}. = \frac{4-6}{5} = \frac{-2}{5} \Rightarrow a = \frac{-2}{5}$ are in A.P

$\frac{1}{a}=\frac{-5}{2}$ are in A.P (or) $a,\frac{5}{4},\frac{1}{2}$ are in .H.P

$\frac{a-\frac{5}{4}}{\frac{5}{4}-\frac{1}{2}}=\frac{a}{\frac{1}{2}}\Rightarrow\frac{4a-5}{5-2}=2a\Rightarrow 4a-5=6a(\text{ or })a=-5/2$

9. a,b,c are H.P $\Rightarrow \frac{1}{a},\frac{1}{b},\frac{1}{c}$ are in A.P $\Rightarrow \frac{1}{b}-\frac{1}{a}=\frac{1}{c}-\frac{1}{b}\Rightarrow\frac{a-b}{ab}=\frac{b-c}{bc}(\text{ or })\frac{a-b}{b-c}=\frac{ab}{bc}=\frac{a}{c}$.
10. Conceptual

Level - II

1)A	2)D	3) C	4)B	5)C	6) C
7)4050	8)A	9)D	10)C	11)A	12)C
13)A	14)A	15)A	16)9	17)B	18)A
19)C,D	20)A,C	21)A	22)D	23)B	24)A
25) $a\to r$, $b\to r$, $c\to q$, $d\to s$			26)A	27) C,D	28)A
29)D	30)D	31) $a\to r$, $b\to p$, $c\to s$, $d\to q$			32)B,C
33)D	34)C	35)A	36) $a\to q, b\to r, c\to p$		

1. a,b,c are in A.P $\Rightarrow 2b=a+c$

   $3^{2b}=3^{(a+c)}\Rightarrow(3^b)^2=(3^a)(3^c)\ \therefore 3^a,3^b,3^c$ are in G.P
2. x, 2x + 2,3x + 3, … … are in G.P $\Rightarrow$ a = x, $r=\frac{2x+2}{x}=\frac{3x+3}{2x+2}$

   $4x^2+4+8x=3x^2+3x\Rightarrow 4x^2-3x^2+8x-3x+4=0\Rightarrow x^2+5x+4=0$

   $(x+4)(x+1)=0\Rightarrow x=-4,-1$

   $x=-4, r=\frac{2(-4)+2}{-4}\Rightarrow r=\frac{-8+2}{-4}=\frac{-6}{-4}=\frac{3}{2}\Rightarrow a_4=a\cdot r^3=(-4)\left(\frac{3}{2}\right)^3=-4\left(\frac{27}{8}\right)$

   $=\frac{-27}{2}$

   $0_{f_{x=-1}}\Rightarrow r=\frac{2(-1)+2}{-1}=\frac{0}{-1}$ (does not exist)
3. Given H.P. in $\frac{4}{3},\frac{3}{2},\frac{12}{7}$ A.P $=\frac{3}{4},\frac{2}{3},\frac{7}{12},\ldots$

   First term $(a)=\frac{3}{4}; d=\frac{2}{3}-\frac{3}{4}=\frac{8-9}{12}=\frac{-1}{12}\Rightarrow$ In A.P, $t_4=a+3d=\frac{3}{4}+3\left(\frac{-1}{12}\right)=\frac{1}{2}$ the 4 th term in H.P. is 2
4. In A.P, $t_3=7, t_7=5$ i.e, a + 2 d = 7 → (1)

   $\underline{a+6\,d=5}$ → (2)

   $-4d=2\Rightarrow d=\frac{-1}{2}$ Substituting ' d ' in (1), $a+2\left(\frac{-1}{2}\right)=7\Rightarrow a=8$

   In A.P, $t_{15}=a+14\,d=8+14\left(\frac{-1}{2}\right)=8-7=1\ \therefore 15^{\text{th}}$ term in H.P. is 1
5. a = 2, r = 0.5
6. sum of infinite terms of g · p $=S_\infty=\frac{a}{1-r}=\frac{1}{1-\frac{1}{2}}=2$

7. $a_4 = ar^3 = 1350$

$\Rightarrow 50r^3 = 1350$

r = 3, $a_5 = a_4 r = 1350 \times 3 = 4050$

8. $3, \sqrt{3}, 1, \frac{1}{\sqrt{3}} \cdots\cdots$ a = 3, $r = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}} \Rightarrow a_n = \frac{1}{243} \Rightarrow a \cdot r^{n-1} = \frac{1}{243}$

(3). $\left(\frac{1}{\sqrt{3}}\right)^{n-1} = \frac{1}{243} \Rightarrow \left(\frac{1}{\sqrt{3}}\right)^{n-1} = \frac{1}{3^6} \Rightarrow \left(\frac{1}{3}\right)^{\frac{n-1}{2}} = \left(\frac{1}{3}\right)^6 \Rightarrow \frac{n-1}{2} = 6 \Rightarrow n = 13.$

9. $\frac{4}{3}, \frac{3}{2}, \frac{13}{7} \ldots\ldots\ldots$ are in H.P $\Rightarrow \frac{3}{4}, \frac{2}{3}, \frac{7}{13} \ldots\ldots\ldots$ are in A.P

$a = \frac{3}{4}, d = a_2 - a_1 = \frac{2}{3} - \frac{3}{4} \Rightarrow d = \frac{8-9}{12} = \frac{-1}{12}$

7 th term in A.P $= a + 6d = \frac{3}{4} + 6\left(-\frac{1}{12}\right) = \frac{3}{4} - \frac{1}{2} = \frac{3-2}{4} = \frac{1}{4}$ $\therefore$ 7 th term in H.P = 4

10. $5 + 55 + 555 + \cdots..$ upto nterms $= 5(1 + 11 + 111 + \cdots..$ .uptonterms)

$= \frac{5}{9}(9 + 99 + 999 + \cdots.) = .\frac{5}{9}((10-1) + (10^2 - 1) + (10^3 - 3) + \cdots..+n)$

$= \frac{5}{9}[(10-1) + (10^2 - 1) + (10^3 - 1) + \cdots. +$ nterms]

$= \frac{5}{9}[(10 + 10^2 + 10^3 + \cdots \ldots. + n) - (1 + 1 + \cdots \ldots + n)] = \frac{5}{9}\left[\frac{10(10^n - 1)}{10 - 1} - n\right]$

$= \frac{5}{9}\left[\frac{10^{n+1} - 10 - 9n}{9}\right] = \frac{5}{81}[10^{n+1} - 9n - 10]$

11. $0.21\overline{75} = 0.21757575 \ldots\ldots = 0.21 + 0.0075 + 0.000075$

$= 0.21 + \frac{75}{10000}\left[1 + \frac{1}{10^2} + \frac{1}{10^4} + \cdots \ldots.\right] = 0.21 + \frac{75}{10^4}\left[\frac{1}{1 - \frac{1}{10^2}}\right] = \frac{21}{10^2} + \frac{75}{10^4}\left(\frac{10^2}{99}\right)$

$= \frac{21}{100} + \frac{3}{4} \times \frac{1}{99} = \frac{359}{1650}$ or (short cut : $0.21\overline{75} = \frac{2175-21}{9900} = \frac{2154}{9900} = \frac{359}{1650}$)

12. given a, b, c are the pth , q^{th} , r^{th} terms of A.P AND G,P Then a = b = c $a^{b-c} \cdot b^{c-a} \cdot c^{a-b} = 1$

13. a, b, c are in A.P $\Rightarrow 2b = a + c \rightarrow (1)$

b, c, a are in G.P $\Rightarrow c^2 = ab \rightarrow (2)$ from (1)(2)

$2\left(\frac{c^2}{a}\right) = a + c \left[\therefore \frac{1}{b} = \frac{a}{c^2}\right]$

$\frac{2}{a} = \frac{a}{c^2} + \frac{c}{c^2}$ or $\frac{2}{a} = \frac{1}{c} + \frac{1}{b}$

$\frac{1}{c}, \frac{1}{a}, \frac{1}{b}$ are in A.P

14. Since a,b.c are in A.P,G.P and H.P as well

$b = \frac{a + c}{2} \rightarrow (1)$

$b^2 = ac \rightarrow (2)$

$b = \dfrac{2ac}{a+c} \rightarrow (3)$

from $(1),(2) \Rightarrow \frac{(a+c)^2}{4} = ac \Rightarrow (a+c)^2 = 4ac$

$(a+c)^2 - 4ac = 0 \Rightarrow (a-c)^2 = 0 \Rightarrow a = c \rightarrow$

on purtting c = a in equ (1), we get $b\frac{2a}{2} = a \rightarrow (5)$ $\therefore$ from (4) and (5) a = b = c.

15. since $|a| < 1$

$x = 1 + a + a^2 + \cdots . + \infty = \dfrac{1}{1-a} \Rightarrow \dfrac{1}{1-a} = x \Rightarrow \dfrac{1}{x} = 1 - a \Rightarrow a = \dfrac{x-1}{x}$

$y = 1 + b + b^2 + \cdots + \infty = \frac{1}{1-b} \Rightarrow b = \frac{y-1}{x} \Rightarrow$ L.H.S $= 1 + ab + a^2b^2 + \cdots \ldots . = \frac{1}{1-ab}$

$= \dfrac{1}{1 - \left(\frac{x-1}{x}\right)\left(\frac{y-1}{y}\right)} = \dfrac{xy}{x+y-1}$

16. $1 + 3 + 3^2 + \cdots + n \Rightarrow a = 1, r = 3$

$s_n = \dfrac{3^n - 1}{3-1} = \dfrac{3^n - 1}{2} \Rightarrow s_n > 7000 \Rightarrow \dfrac{3^n - 1}{2} > 7000 \Rightarrow 3^n - 1 > 14000 \Rightarrow 3^n > 14001$

the above condition satisfied $\Rightarrow$ when n = 9

$\therefore 3^8 = 6561$

$3^9 = 19683$

$s_n = \frac{3^9-1}{2} = \frac{19863-1}{2} = 9841 > 7000$ $\therefore$ n = 9 is the required value

17. $a_1, a_2, \ldots \ldots . a_n$ are in H.P $\frac{1}{a_1}, \frac{1}{a_2}, \ldots \ldots \ldots \frac{1}{a_n}$ are in A.P

$\dfrac{1}{a_2} - \dfrac{1}{a_1} = \dfrac{1}{a_3} - \dfrac{1}{a_2} = \cdots \ldots . . = \dfrac{1}{a_n} - \dfrac{1}{a_{n-1}} = d \Rightarrow \dfrac{a_1 - a_2}{a_1 a_2} = \dfrac{a_2 - a_3}{a_3 a_2} = \cdots . . = \dfrac{a_{n-1} - a_n}{a_n a_{n-1}}$

$= d$

$a_1 a_2 = \dfrac{a_1 - a_2}{d}, a_2 a_3 = \dfrac{a_2 - a_3}{d}, \ldots . a_{n-1} \cdot a_n = \dfrac{a_{n-1} - a_n}{d}$

adding all

$a_1 a_2 + a_2 \cdot a_3 + \cdots \ldots + a_{n-1} \cdot a_n = \dfrac{a_1 - a_2}{d} + \dfrac{a_2 - a_3}{d} + \cdots + \dfrac{a_{n-1} - a_n}{d}$

$= \dfrac{a_1 - a_n}{d} = \dfrac{a_1 a_n}{d}\left[\dfrac{1}{a_n} - \dfrac{1}{a_1}\right] = \dfrac{a_1 a_n}{d}\left[\dfrac{1}{a_1} + (n-1)d - \dfrac{1}{a_1}\right] = (n-1)a_1 a_n$

hence $a_1 a_2 + a_2 a_3 + \cdots \ldots + a_{n-1} a_n = (n-1) a_1 a_n$

18. $A.M \geq G \cdot M \geq H.M$

19. $(2x+2)^2 = x(3x+3) \Rightarrow x^2 + 5x + 4 = 0 \Rightarrow (x+1)(x+4) = 0 \Rightarrow x = -1, -4$

$x = -1 \Rightarrow -1, 0, 0$ is not G.P

$x = -4 \Rightarrow -4, -6, -9$ is G.P with $r = \frac{-6}{-4} = \frac{3}{2}; t_4 = -4\left(\frac{3}{2}\right)^3 = \frac{-27}{2}$

20. $\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$ are in A.P.

$$\frac{2}{c+a} = \frac{1}{b+c} + \frac{1}{a+b}$$

$$2ab + 2b^2 + 2ca + 2bc = ac + 2bc + c^2 + a^2 + 2ab + ac$$

$$b^2 = \frac{c^2 + a^2}{2}$$

$\therefore a^2, b^2, c^2$ are in A.P

21. statement 1: $a = 1, r = \frac{1}{3}$

$$s_\infty = \frac{a}{1-r} = \frac{1}{1-\frac{1}{3}} = \frac{3}{2}$$

statement 2 is true.

22. $a_{10} = 3 \Rightarrow a_1 + 9\,d = 3 \Rightarrow d = \frac{1}{9}$ 23. $a_4 = a_1 + 3\,d = 2 + \frac{1}{3} = \frac{7}{3}$

23. $\frac{1}{h_1}, \frac{1}{h_2}, \ldots\ldots \frac{1}{h_{10}} \Rightarrow \frac{1}{n_{10}} = \frac{1}{n_1} + 9\,d \Rightarrow d = \frac{-1}{54}$

where d is the common difference $\Rightarrow x\frac{1}{h_7} = \frac{1}{h_1} + 6\,d \Rightarrow \frac{1}{h_7} = \frac{1}{2} - \frac{6}{54} = \frac{7}{18}$

25. $r = \frac{t_2}{t_1}$, $S_n = a\left(\frac{1-r^n}{1-r}\right)$

26. $a = 2, r = \frac{2\sqrt{2}}{2} = \sqrt{2}$

$64 = a_x, 64 = 2(\sqrt{2})^{x-1} \Rightarrow 32 = (2)^{\frac{x-1}{2}}$

on comparing the terms both the sides, we get $x = 11$

27. Given a, b, c are in A.P $\Rightarrow 2b = a + c$

Also, given $a + 1, b, c$ are in G.P and $a, b, c + 2$ are also in G.P

$\Rightarrow b^2 = (a+1)c$

(2) and $b^2 = a(c+2)$

On solving (1), (2) and (3) we get, $a = 8$, $b = 12, c = 16$.

28. $\frac{2}{c} = \frac{a}{bc} + \frac{1}{b} \Rightarrow 2\,b = a + c$

29. $a^x = b^y = c^z = k \Rightarrow x = \log_a k, y = \log_b k, z = \log_c k$ use $\frac{y}{x} = \frac{z}{y}$

Let $\frac{a}{r}$, a, ar be the terms in G.P product $= a^3 = 27 \Rightarrow a = 3$

$\Rightarrow$ Sum $= a\left(\frac{1}{r} + 1 + r\right) = 13 \Rightarrow 3r^2 - 10r + 3 = 0 \Rightarrow (3r-1)(r-3) = 0 \Rightarrow r = \frac{1}{3}, 3$

Increasing G.P so $r = 3$ $G.P$ is 1,3,9,27,81 ... ...

Sum of first five terms = 121

31. $a = \frac{1}{3}\ ar^3 = 9; \frac{1}{3} \times r^3 = 9 \to r^3 - 27 \to r^3 = 3^3 \to r = 3$

$t_3 = ar^2 = x_2 = \frac{1}{3} \times (3)^2 = 3$

b) $ar^2 = 4 \Rightarrow a \cdot ar \cdot ar^2 \cdot ar^3 \cdot ar^4 = a^5r^{10} = (ar^2)^5 = (4)^5 = 1024$

c) $a = \frac{1}{3}\ r = \frac{1}{3} \Rightarrow \frac{a}{1-r} = \frac{\frac{1}{3}}{1-\frac{8}{3}} = \frac{\frac{1}{3}}{\frac{2}{3}} = \frac{1}{2}$

d) $a = 1, r = 3, \frac{a(r^n-1)}{r-1} = 364$

$1\frac{(3^n-1)}{3-1} - 364; 3^n - 1 = 728 \to 3^n = 729 \to 3^n = 3^6 \Rightarrow n = 6; \to n = 6$

32. $a = 1, ar^2 + ar^4 = 90$

$a(r^2 + r^4) = 90; 1(r^4 + r^2) = 90; r^4 + r^2 - 90 = 0; (r^2 + 10)(r^2 - 9) = 0$

$r^2 = 9; r = \pm 3$

33. Let the GP be $a, a^2, a^2 \ldots\ldots\ldots$, clearly $r < 1$

$\alpha_0 = a + ar + ar^2 + \cdots \ldots\ldots \alpha \Rightarrow 100 = a^2 + a^2r^2 + a^ra^4 \ldots\ldots\ldots\ldots \alpha$

$20 = \frac{a}{1-r}, 100 = \frac{a^2}{1-r^2}\left[\ s_\alpha = \frac{a}{1-r}\right]$

$400 = \frac{a^2}{1-r}, 100 = \frac{a^2}{(1-r^2)}; \frac{\frac{a^2}{(1-r)(1-r)}}{\frac{a^2}{(1-r)(1+r)}} = \frac{400}{100}; \frac{1+r}{1-r} = 4 \to 4 - 4r = 1 + r;\ r = \frac{3}{5}$

34. $\frac{a}{1-r} = 20 \Rightarrow \frac{a}{1-\frac{3}{5}} = 20 \Rightarrow \frac{\frac{a}{-12}}{5} = 20 \Rightarrow a = 20 \times \frac{2}{5} = 8$

35. first geometric mean is t_2 i.e., $ar = 8 \times \frac{3}{5} = 4.8$

36. a) $n = 0$

b) $n = \frac{-1}{2}$

c) $n = -1$

WORKSHEET-3
Level - I

1)B 2)C 3)A 4)C 5)B 6)D 7)B 8)A 9)D

1. $A.M = \frac{\frac{1}{2}+\frac{1}{3}}{2} = \frac{3+2}{6\times 2} = \frac{5}{12}$

2. $d = \frac{b-a}{n+1} = \frac{23-3}{4+1} = \frac{20}{5} = 4$

3. $G.M = \sqrt{6 \times 24} = 12$

4. 81

5. $A.M = 13 \Rightarrow \frac{a+b}{2} = 13 \Rightarrow a + b = 26$

G.M $= 12 \Rightarrow \sqrt{ab} = 12 \Rightarrow ab = 144$

$(a-b)^2 = (a+b)^2 - 4ab \Rightarrow (a-b)^2 = (26)^2 - 4(144) = 676 - 576 \therefore$ a,b are 18,8

$(a-b)^2 = 100 \Rightarrow a - b = 10$

from 1 and $2a + b = 26$

$$\begin{array}{r} a - b = 10 \\ - \; - \; - \; - \; - \; - \\ 2a = 36 \end{array}$$

$a = \frac{36}{2} = 18$ from 1, $b = 26 - 18 = 8$

6. $n\left(\frac{a+b}{2}\right) = 14\left(\frac{5+8}{2}\right) = 91$ 7. $a, g_1, g_2, g_3, b \Rightarrow \frac{g_1}{a} = \frac{g_2}{g_1} = \frac{g_3}{g_2} = \frac{b}{g_3} \Rightarrow {g_2}^2 = g_1 g_3$

7. $6 + 66 + 666 + \cdots . + n$

$S_n = \frac{a}{81}\{10(10^n - 1) - 9n\}, \forall a \in N. = \frac{6}{81}\{10(10^n - 1) - 9n\}$

$= \frac{60(10^n - 1)}{81} - \frac{6 \times 9n}{81} \Rightarrow \frac{20}{27}(10^n - 1) - \frac{2n}{3}$

9. $A = 15, G = 9, H = G^2 = A \cdot H \Rightarrow 81 = 15 \times H \Rightarrow H = 5.4$

Level – II

1)A	2)A	3)C	4)A	5)A	6)A	7)B
8)B	9)C	10)25	11)A	12)D	13)96	14)D
15)A	16)8	17)B,C	18)D	19)C	20)A	21)A
22) $a \to q, b \to s, c \to p, d \to r$		23)5	24)A,D	25) $a \to s, b \to p, c \to q, d \to r$		
26)A	27)C,D	28)D	29) $a \to s, b \to p, c \to q, d \to r$			30)D
31)A	32)C					

1. Let S_∞ denote sum to infinity $S_\infty = 1 + \frac{2}{3} + \frac{3}{3^2} + \frac{4}{3^3}$.

$\frac{1}{3} S_\infty = \frac{1}{3} + \frac{2}{3^2} + \frac{3}{3^3} + \frac{4}{3^4}$

(1) - (2) , we get $\frac{2}{3} S_\infty = 1 + \frac{1}{3} + \frac{1}{3^2} \ldots\ldots + \frac{1}{3^r} \ldots\ldots \propto$

$\frac{2}{3} S_\infty = \frac{a}{1-r}$, where $a = 1r = 1/3$

$\frac{2}{3} S_\infty = \frac{1}{1 - \frac{1}{3}} = \frac{3}{2} \Rightarrow \frac{2}{3} S_\infty = \frac{3}{2} \Rightarrow S_\infty = \frac{3}{2} \times \frac{3}{2} = \frac{9}{4}$

2. $1 \times 1 + 5 \times -1 + 9 \times 1 + 13 \times -1$

1,5,9,13 are in AP $1, -1, 1, -1$ are in GP

$a = 1d = 5 - 1 = 4, r = \frac{-1}{1} = -1 \Rightarrow a = 1d = 4r = -1$

3. Since a, b, c are in G.P.

$\therefore b^2 = ac$; taking logarithm on both sides with base ' x ' $\Rightarrow \log_x b^2 = \log_x ac$

$\Rightarrow 2\log_x b = \log_x a + \log_x c \Rightarrow \frac{2}{\log_b x} = \frac{1}{\log_a x} + \frac{1}{\log_c x}$

$\Rightarrow \log_a x, \log_b x, \log_c x$ are in H.P

4. Let the numbers be a and b. Then, $\frac{\frac{2ab}{a+b}}{\sqrt{ab}} = \frac{4}{5} \Rightarrow \frac{2\sqrt{ab}}{a+b} = \frac{4}{5}$

$\Rightarrow \frac{a+b}{2\sqrt{ab}} = \frac{5}{4} \Rightarrow \frac{a+b+2\sqrt{ab}}{a+b-2\sqrt{ab}} = \frac{5+4}{5-4}$ [Applying componendo and dividendo]

$\Rightarrow \left(\frac{\sqrt{a}+\sqrt{b}}{\sqrt{a}-\sqrt{b}}\right)^2 = \frac{9}{1} \Rightarrow \frac{\sqrt{a}+\sqrt{b}}{\sqrt{a}-\sqrt{b}} = \frac{3}{1} \Rightarrow \frac{\sqrt{a}}{\sqrt{b}} = \frac{3+1}{3-1} \Rightarrow \frac{a}{b} = \frac{4}{1}$

5. $1.2^2 + 2.3^2 + 3.4^2 + \cdots . + n_\text{term} = n(n+1)^2 = n^3 + 2n^2 + n$

$S_n = \sum T_n = \sum n^3 + 2\sum n^2 + \sum n$

$= \frac{n(n+1)}{2}\left\{\frac{n(n+1)}{2} + \frac{n(n+1)}{3} + 1\right\} = \frac{n(n+1)(3n^2+11n+10)}{12}$

$= \frac{n(n+1)(n+2)(3n+5)}{12}$

6. $S_\infty = 1 + 4x + 7x^2 + 10x^3$

$x.S_\infty = 0 + x + 4x^2 + 7x^3$

(2) - (1)we get ; $(1-x)S_\propto = 1 + 3x + 3x^2 + 3x^3 \ldots \ldots \propto$

$(1-x)S_\propto = 1 + \frac{3x}{1-x} \Rightarrow (1-x)\frac{35}{16} = \frac{1+2x}{1-x} \Rightarrow 35(1-x)^2 = 16 + 32x$

$\Rightarrow 35x^2 - 102x + 19 = 0 \Rightarrow (7x-19)(5x-1) = 0 \Rightarrow x \neq \frac{19}{7}(r > 1) \Rightarrow x = \frac{1}{5}(r < 1)$

7. conceptual

8. $1 + (2+3) + (4+5+6) + \cdots . + n$ terms number of terms in

$S = 1 + 2 + 3 + \cdots . + n = \frac{n(n+1)}{2}$

first term is 1, and common difference also

$S = \frac{\frac{n(n+1)}{2}}{2}\left\{2 \cdot 1 + \left(\frac{n(n+1)}{2} - 1\right) \cdot 1\right\} = \frac{n(n+1)}{8}(4 + n^2 + n - 2)$

$= \frac{n(n+1)(n^2+n+2)}{8}$

9. let $A^x = G^y = H^z = K$ then $A = K^{\frac{1}{x}}, G = K^{\frac{1}{y}}, H = K^{\frac{1}{z}} \Rightarrow G^2 = A \cdot H$

$\left(K^{\frac{1}{y}}\right)^2 = K^{\frac{1}{x}} \cdot K^{\frac{1}{z}} \Rightarrow \frac{2}{y} = \frac{1}{x} + \frac{1}{z} \Rightarrow \frac{1}{x}, \frac{1}{y}, \frac{1}{z} \Rightarrow$ Hence x, y, z are in H.P.

10. $\sum_{K=1}^{n} K^3 = \left(\frac{n(n+1)}{2}\right)^2 \neq 25$ for any $n \in N$

11 $\text{A.M} \geq \text{G} \cdot \text{M} \Rightarrow \frac{\log_d a+\log_b d+\log_a c+\log_c b}{4} \geq^4 \sqrt{\frac{\log a}{\log d} \times \frac{\log d}{\log b} \times \frac{\log c}{\log a} \times \frac{\log b}{\log c}} \geq 1$

$\log_d a + \log_b d + \log_a c + \log_c b \geq 4$

12. Since a, b, c are in A.P $\therefore 2b = a + c$ $\because \frac{a}{bc}, \frac{1}{c}, \frac{2}{b}$ are not in A.P

since $\frac{bc}{a} + \frac{b}{2} = \frac{2bc+ab}{2a} \neq c \Rightarrow$ given number are not in H.P

since $\frac{a}{bc} \cdot \frac{2}{b} = \frac{2a}{b^2c} \neq \frac{1}{c^2} \Rightarrow$ given numbers are not in G.P

so. option is (4)

13. $T_n = \frac{1^3+2^3+3^3+\cdots+n^3}{1+3+5+\cdots+(2n-1)} = \frac{\frac{n^2(n+1)^2}{4}}{\frac{n}{2}(1+2n-1)} \Rightarrow T_n = \frac{\frac{n^2(n+1)^2}{4}}{n^2} = \frac{1}{4}(n+1)^2 \Rightarrow T_n = \frac{1}{4}(n^2 + 2n + 1)$

$S_n = \sum_{n=1}^{n} T_n = \frac{1}{4}\left(\frac{n(n+1)(2n+1)}{6} + n(n+1) + n\right)$

$S_n = \frac{1}{4}\left(\frac{9 \times 10 \times 19}{6} + 9 \times 10 + 9\right) = \frac{1}{4}(285 + 90 + 9) = \frac{384}{4} = 96$

14. $S_\infty = 1 + 3(-x) + 5(-x)^2 + 7(-x)^3 + \cdots \infty \rightarrow (1)$

$(-x)S_\infty = (-x) + 3(-x)^2 + 5(-x)^3 + \cdots \infty \rightarrow (2)$

substracting (2) from (1), we get

$(1 + x)S_\infty = 1 + (-2x + 2x^2 - 2x^3 + \cdots \infty)$

$(1 + x)S_\infty = 1 + \left(\frac{-2x}{1-(-x)}\right) \Rightarrow (1 + x)S_\infty = 1 + \frac{2x}{1+x} = \frac{1-x}{1+x} \Rightarrow S_\infty = \frac{1-x}{(1+x)^2}.$

15. GIven that a, b, c are in G.P $\Rightarrow b^2 = ac$

also given that $a - b, c - a, b - c$ are in H.P $\Rightarrow \frac{2}{c-a} = \frac{1}{a-b} + \frac{1}{b-c}$

$\Rightarrow \frac{2}{c-a} = \frac{b-c+a-b}{(a-b)(b-c)} \Rightarrow 2(a-b)(b-c) = -(a-c)^2 - (a-c)^2$

$= 2(ab - 2b^2 + ac)$

$[\because b^2 = ac] \Rightarrow -(a-c)^2 = 2\,b(a - 2\sqrt{ac} + c)$

$-\{(\sqrt{a} - \sqrt{c})(\sqrt{a} - \sqrt{c})\}^2 = 2\,b(\sqrt{a} - \sqrt{c})^2 \Rightarrow 2\,b = -\{a + 2\sqrt{ac} + c\} \Rightarrow 2\,b$

$= -a - 2\,b - c$

$(\because b^2 = ac) \Rightarrow a + 4b + c = 0$

16. $2^3 + 4^3 + 6^3 + \cdots \ldots\ldots = 3528 = 2^3(1^3 + 2^3 + 3^3 + \cdots \ldots\ldots) = 3528$

$= \frac{2^3(n)^2(n+1)^2}{4} = 3528$

$n(n+1) = \sqrt{1764}$

$n(n+1) = 42$, for $n = 6$

17. Harmonic mean of a, b and $c = \frac{3}{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}}$

18. The AP is 1,2,3,4 … … ….

The GP is $1, x, x^2 \cdot x^3 \ldots\ldots\ldots$; t_n in AP $= a + d(n-1) = 1 + 1(n-1) = n$

t_n in GP $= ar^{-1} = 1 \times x^{n-1} = x^{n-1}$; t_n in AGP is nx^{n-1}.

19. $t_n = 2n + 3 \Rightarrow t_1 = 2 \times 1 + 3 = 5; t_2 = 2 \times 2 + 3 = 7, t_3 = 2 \times 3 + 3 = 9$

AP is 5,7,9

20. $a = 8, r = \frac{1}{16} \times 8 = \frac{1}{2}$; GP is $8, 8 \times \frac{1}{2}, 8 \times \left(\frac{1}{2}\right)^2, 8 \times \left(\frac{1}{2}\right)^3 \Rightarrow 8,4,2,1$

21. formula

22. formula

23. $1 + 6 + 27 + 324 + \cdots$. form A.G.P

$1 + 2 + (3)^2 + 4(3)^3 + \cdots$. form A.G.P

A.P is 1,2,3,4, ….

Sum of forms 100 natural numbers $= \in n = \frac{n(n+1)}{2}$

$= \frac{100^{50}(100+1)}{} = 50(101) = 5050$

24. $\frac{a+b}{2} = 20, \sqrt{ab} = 16; (\sqrt{ab})^2 = (16)^2$

$a + b = 40\ ab = 256$

Put $b = 40 - a \rightarrow a(40 - a) = 256 \rightarrow a^2 - 40a + 256 = 0 \rightarrow (a - 32)(a - 8) = 0$

$a = 32\ a = 8$; If $a = 32\ \ b = 8; a = 8\ \ b = 32$

25. $4 \cdot \frac{1}{3} + 9 \cdot \frac{1}{3r} + 14\frac{1}{3^3} + \frac{1}{3^4}$; 4,8,12,16 … … are in AP ; $t_1 = 4$

$t_2 - t_1 = 8 - 4 = 5; \frac{1}{3}, \frac{1}{3^2}, \frac{1}{3^3}$ … … are in GP ; $t_1 = \frac{1}{3} = \frac{1}{9} \times \frac{1}{3} = \frac{1}{27}$

26. (1) $2b = a + c, x = \sqrt{ab}, y = \sqrt{bc}$

$x^2 + y^2 = ab + bc = b(a + c) = 2\ b^2 \Rightarrow b^2 + c^2 > 2bc; \Rightarrow c^2 + a^2 > 2ac$

$2(a^2 + b^2 + c^2) > 2(ab + bc + ca) \Rightarrow ab + bc + ca < 1$

27. $T_r = \frac{1+2+3+\cdots\ldots+r}{1^3+2^3+\cdots\ldots\ldots+r^3} \frac{2}{r(r+1)} = 2\left(\frac{1}{r} - \frac{1}{(r+1)}\right)$

$\Rightarrow S_n = \sum_{r=1}^{n} T_r = 2\sum_{r=1}^{n}\left(\frac{1}{r} - \frac{1}{r+1}\right) = 2\left(1 - \frac{1}{n+1}\right)$

which can not be greater than 2

28. $A - G = \frac{3}{2}, G - H = \frac{6}{5}\ \ A - \frac{3}{2} = G, G - \frac{6}{5} = H$

$A - \frac{3}{2} = G, A - \frac{3}{2} - \frac{6}{5} = H\ A - \frac{3}{2} = G$ and $A - \frac{27}{10} = H$

$\left(A - \frac{3}{2}\right)^2 = G^2 = AH = A\left(A - \frac{27}{10}\right);\ \left(A - \frac{3}{2}\right)^2 = A\left(A - \frac{27}{10}\right)$

$A = \frac{15}{2}$ and $G = A - \frac{3}{2} = \frac{15}{2} - \frac{3}{2} = 6;\ \frac{a+b}{2} = \frac{15}{2}, \sqrt{ab} = 6$

$a + b = 15\ ab = 36$

How a, b are the roots of the equation $t^2 - 15t + 36 = 0$

$(t - 12)(t - 3) = 0; t - 12$ or $t = 3$; The numbers are 12,3

29. conceptual

30. $S_3(1 + 8\ S_1) = \left[\frac{n(n+1)}{2}\right]^2 \left(\frac{1+8n(n+1)}{2}\right)$

$= \left(\frac{n(n+1)}{2}\right)^2 (2n+1)^2 = \left(\frac{n(n+1)(2n+1)}{2}\right)^2 = 9 \cdot \left(\frac{n(n+1)(2n+1)}{6}\right)^2 = 9\ S_2{}^2$

31. $t_n = \sum(2n-1)^2$

$= \sum 4n^2 - \sum 4n + \sum 1 = \frac{4(n)(n+1)(2n+1)}{6} - \frac{4n(n+1)}{2} + n$

$= \frac{n}{3}(4n^2 + 6n + 2 - 6n - 6 + 3) = \frac{n(4n^2-1)}{3}$

32. $1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + \cdots \ldots + \infty \rightarrow (1)$

$S_\infty = 1 + 4\left(\frac{1}{5}\right) + 7\left(\frac{1}{5}\right)^2 + 10\left(\frac{1}{5}\right)^3 + \cdots . + \infty$

$\frac{1}{5}\ S_\infty = \frac{1}{5} + 4\left(\frac{1}{5}\right)^2 + 7\left(\frac{1}{5}\right)^3 + 10\left(\frac{1}{5}\right)^4 + \cdots . + \infty \rightarrow (2)$

$$\left(1 - \frac{1}{5}\right) S_\infty = 1 + 3\left[\frac{1}{5} + \left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^3 + \cdots \infty\right] \Rightarrow \frac{4}{5} S_\infty = 1 + 3\left(\frac{\frac{1}{5}}{1 - \frac{1}{5}}\right) = 1 + \frac{3}{4}$$

$$S_\infty = \frac{7}{4} - \frac{4}{5} = \frac{35}{16} \Rightarrow S_\infty = \frac{35}{16}$$

14. Plane Geometry

WORKSHEET – 1

Level – I

1) A	2) B	3) C	4) C	5) D	6) D	7) D
8) D	9) B	10) B				

Level – II

1) C	2) D	3) C	4) C	5) D	6) A	7) A
8) D	9) D	10) A	11) A	12) D	13) A	14) A
15) B	16) B	17) C	18) A	19) B	20) B	21) B,C
22) A	23) B	24) A	25) B	26) A	26)a-p; b-r; c-s; d-q	
27)132.50	28) D	29)	30) A,D	31) A	32) 2	

1. (3) $\angle AOB + \angle BOD = 180°$ (Linear pair)

   $\Rightarrow 100^o + \angle BOD = 180° \Rightarrow \angle BOD = 80°$ and given $\angle BOC : \angle COD = 3 : 5$

   $\therefore$ Total ratio = 3 + 5 = 8 $\Rightarrow \angle BOC = \frac{3}{8} \times 80 = 30°$

   and $\angle AOC = \angle AOB + \angle BOC = 100° + 30° = 130°$

2. (4) Let one of the angle be x

   supplement of x=3 times the complement

   of x $180\text{-}x=3(90\text{-}x) \Rightarrow 2x=90 \Rightarrow x=45°$

3. (3) C = G = 70_o (corresponding angles)

   G + H = 180_o (Linear pair) $\Rightarrow$ H = 110^0

4. (3) Given $\angle AOC = \angle ACO$ and $\angle BOD = \angle OBD$ But $\angle AOC = \angle$ BOD (vertically opposite angles) Therefore $\angle ACO = \angle OBD$

   These are alternate angles $\Rightarrow$ AC||BD

5. (4) (90 - x) : (180 - x) = 1 : 2 $\Rightarrow$ 180 - 2x = 180 - x $\Rightarrow$ x =0

6. sol. (1) Given p||n

   $\angle 1 = \angle 3$ (Corresponding angles) and $\angle 1 = 75° \Rightarrow \angle 3 = 75°$

   $\angle 3 + \angle 2 = 180°$ (sum of cointerior angles = 180°)

   $\Rightarrow 75 + \angle 2 = 180° \Rightarrow \angle 2 = 180° - 75° = 105º \Rightarrow \angle 2 = \angle 1 + \frac{1}{3}$ X90^0

   $\Rightarrow \angle 2 = \angle 1 + \frac{1}{3}$ × right angle

   (1) Total Ratio = 3 + 4 + 2 = 9 $\Rightarrow \angle BOC = \frac{3}{} \times 180^o =$ $60°$

7. (1) Total Ratio = 3 + 4 + 2 = 9 $\Rightarrow \angle BOC = \frac{3}{9} \times 180^o = 60°$

   $\angle COD \frac{4}{9} \times 180^o = 80° \Rightarrow AOD \frac{2}{9} \times 180^o = 40$

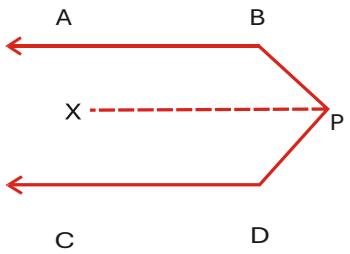
OP is the bisector of $\angle BOC \Rightarrow \angle POC = \frac{1}{2} \times 60^o = 30°$

OR is bisector of $\angle AOD \Rightarrow \angle DOR \frac{1}{2} \angle AOD \Rightarrow \angle DOR = \frac{1}{2} \times 40^o = 20°$

Now, $\angle POR = \angle POC + \angle COD + \angle DOR = 30° + 80° + 20^o = 130°$

$\therefore \angle POR = 130°$

8. Sol: (4) Draw PX||AB||CD



$\angle ABP + \angle BPX = 180°$ .....(1) (Sum of cointerior angles = 180^0)

$\angle CDP + \angle DPX = 180°$ ..... (2) (Sum of cointerior angles = 180°) Now, Adding (1) and (2)

$\angle ABP + \angle BPX + \angle DPX + \angle CDP = 360°$

$\Rightarrow \angle ABP + \angle BPD + \angle CDP = 360^0$ ($\angle BPX + \angle DPX = \angle BPD$)

9. sol: (4) $\angle ABC = \angle BCD$ (alternate angles are equal)

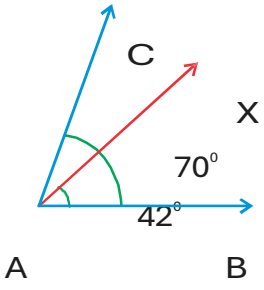
AB||CD $\Rightarrow \angle BCD = \angle CDG$ (alternate angles) $\therefore \angle ABC = \angle CDG$ But $\angle CDG = \angle FDE$(vertically opposite angles) $\therefore \angle ABC = \angle FDE$

10. (1) $\angle 2 = \angle 6$ (corresponding angles)

$1 + 2 = 180^o$, $1 + 6 = 180^o \Rightarrow 2x + y + 3x - y = 180 \Rightarrow 5x = 180 \Rightarrow x = 36$

$\angle 2 = 3x - y = 108 - y$

11. $\angle XAC = \angle BAC - \angle BAX = 70 - 42 = 28$



12. (4) Either equal or supplementry

13. (1) x + y = 90 x +50 = y - 20 (alternate interior angles) x - y = – 70

(x + y) + (x - y) = 90 + (- 70) 2x = 20

x = 10

y = 80

14. (1) $x + y = 90$ $(x + 47) + (y + 43) = 180$

15. (2) $x + y = 90$

_____(

1) $x - 50 = 10 + y$

$x - y = 60$_____(2)

from (1) and (2) $x = 75$ degrees and $y = 15$ degrees here cointerior angles are $10+y$ and $x+2y$ $10+y+x+2y=10+x+3y=10+75+3(15)=130$ degrees

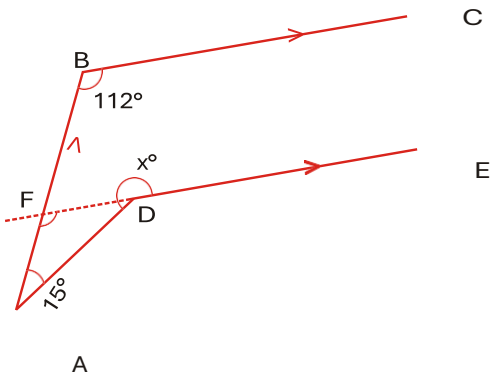
as the sum of cointerior angles is not 180 degrees .then lines are not parallel according to euclid's fifth postulate ,they are going to intersect as extended.

16. Sol. (2)

Given $\angle x = (5^2)° = 25°$, $\angle y = 10°$ and $\angle z = (4^3)°$ $= 64°$ $x + y + z = 25° + 10° + 64° = 99°$

$\therefore$ Reflex angle of $(x + y + z) = 360^o - 99^o = 261°$

17. sol:(3) Given, BC||DE



Draw FD parallel to BC through D

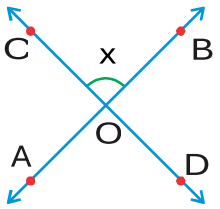
$\Rightarrow \angle AFD = \angle ABC = 112°$

($\because$ corresponding angles are equal)

$\therefore \angle FDA = 180° - (112° + 15°) = 53°$

$\Rightarrow x = 180° + 53° = 233° \Rightarrow \therefore \angle ADE = 360° - x = 127°$.

18. (1) ⍰COA ⍰ x $\Rightarrow$ $(180 - x) : (90 - x) = 31 : 13$



19. sol: (2) Join B and D, then $\angle BCD + \angle CBD + \angle CDB = 180°$ [angles in CBD]

$\Rightarrow b° + \angle CBD + \angle CDB = 180°$ ___(1)

But $\angle BAD + \angle ABC + \angle CBD + \angle BDC + \angle ADC = 180°$ [sum of angles in

$\Rightarrow a° + c° + \angle CBD + \angle BDC + \angle ADC = 180°$ __(2)

$\Rightarrow a° + c° + \angle ADC - b° = 0° \Rightarrow \angle ADC = b° - a° - c°$

20. sol: (2) Since, l, m,n are parallel and p||q, we have $4a° = 3b° = 5c°$ (corresponding angles are equal) and $4a + 120° = 180° \Rightarrow a = 15° \Rightarrow b = 20°, c = 12°$

$\therefore a + b - c = 15° + 20° - 12° = 23°.$

21. sol: (2,3) x : y = 3 : 7

Total ratio x + y = 10

$x + y = 180°$ ($\therefore$ Linear pair)

$\Rightarrow x = \therefore \frac{3}{10} \times 180^{\circ} = 54° \Rightarrow$ z = 54^0 (∵ Vertically opposite angles)

22. Sol: 1

$\underline{AOC} = 100^0$

As OM and ON are angle bisectors

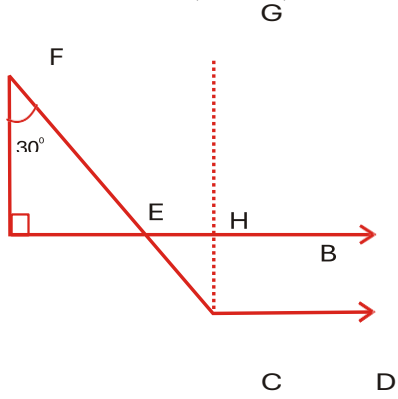
$\underline{AOM} \quad \underline{MOC} = \frac{100}{2} = 50^0$

$\therefore \underline{BON} = \underline{NOC} = 40^0$

23. (2) Through C, draw CG‖ AF.

Now, AF ‖ CG and tranversal FC cuts them at F and C respectively.

$\therefore \angle AFE = \angle GCE \Rightarrow \angle GCE = 30^0$, Now, $\angle FAE = \angle GHB \Rightarrow \angle GHB = 90^0$



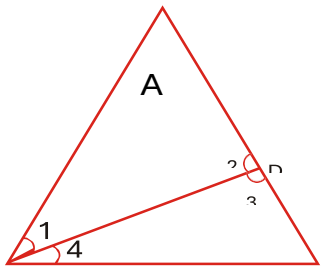
Now, AB ‖ CD and tranversal CG intersects them at H and C respectively.

$\therefore \angle DCH = \angle BHG \Rightarrow \angle DCH = 90^0 \therefore \angle ECD = \angle ECH + \angle DCH$

$\Rightarrow \angle ECD = \angle GCE + \angle DCH \Rightarrow \angle ECD = 30^0 + 90^0 = 120^0$

24. **to 25**: 24. [1] ; 25. [2] ;

From AC, cut AD = AB,



join AB = AD [construction] $\angle 2 = \angle 1$ ..... (1)

[Base angle of an isosceles triangle] ..... (2)

(exterior angle of triangle ABD) Δ $\angle 3 = \angle 1$

$\angle 2 >$ .....(3) [ext. $\angle$ of Δ BDC] $\Rightarrow \angle 3 >$ .....(4) [from (1) and (2)]

$\angle 4$ $\angle 2$

$\because$ $\angle 3 > \angle 4$ $\therefore$ BC > DC $\Rightarrow$ DC $< \Rightarrow$ AC − AD < BC

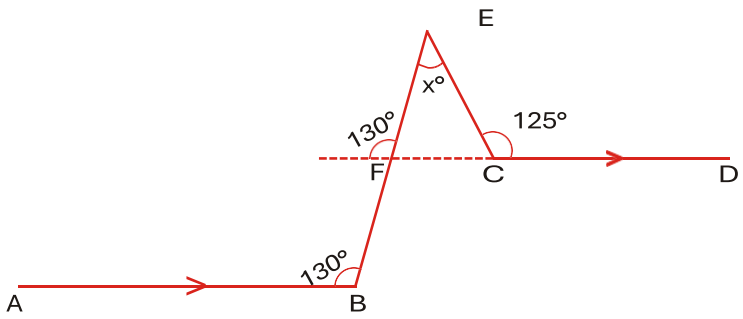
$\Rightarrow$ AC − AB < $[\because AD = AB]$

Similarly, AC – BC < AB, BC – AB < AC Hence proved.

26. Conceptual

27. $180^0 - 47.5^0 = 132.5^0$

28. (4) Given, AB||CD. Draw FC parallel to AB through C.



$\Rightarrow$ $\angle$CFE = 180° – 130° = 50°

30. (1,4) i)If two adjacent angles are equal, then each angle measures 90° is false because two equal adjacent angles may be measure other than 90°.

iv) Angles forming a linear pair can both be acute angles is false because the sum of angles forming a linear pair is 180°.

31. (1) In an equilateral triangle, every angle is 60^0 .
$\therefore$ $\angle$EFB = 60° $\Rightarrow$ $\angle$AFE = 120° $\Rightarrow$ $\angle$CGF = 120°

32. (2) Join E and G then $\angle$FEG = 30° and $\angle$FGE = $\Rightarrow EFG = 120^0 30^0$

WORKSHEET – 2

Level – 1

1) D	2) B	3) B	4) C	5) C	6) C	7) B
8) C	9) A	10) A	11) C	12) D	13) B	14) B
15) A						

Level – 2

1) A	2) C	3) D	4) B	5) B	6) D	7) B
8) A	9) B	10) C	11) B	12) C	13) D	14) C

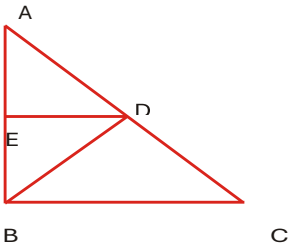
15) D	16) C	17) B	18) A	19) A	20) A,B,D	21) A,B,C
22) A	23) A	24) B	25) B	26) a-q;b-q;c-r;d-q		27) 1380
28) A	29) D	30) A	31) C	32) A	33) A	34) D

1. Sol: 1 $13 > x > 3$
2. 1:3:6 can not be the ratio of angles of right angled triangle let the angles x,3x,6x. x+3x+6x=180(angle sum property of a triangle) 10x=180 that means x=18 degrees

   $1x=18^0$,$3x=54^0$,$6x=108^0$
3. D is the midpoint of BC $\therefore$ AD is the median

   $\therefore$ AG:GD=2:1 ,AD:GD=3:1
4. sol: [2] $PQ = \frac{1}{2} MN = 2 \times PQ = 2 \times 1.6 = 3.2$ cm

   Also, $MN \frac{1}{2} BC = 2MN = 2 \times 3.2cm = 6.4$ cm
5. sol: [2] ; Drawn DE ||BC D is the mid point of AC



   By converse of mid point theorem E is the mid point of AC.

   Also $\angle AED = \angle EBC$ (corresponding angles)

   $\therefore \angle AED = 90$

   In $\Delta AED, \Delta DE$

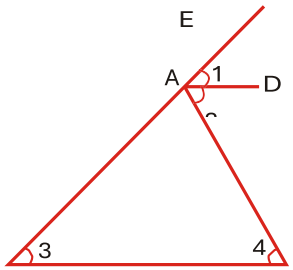
   AE = EB ($\because$ E is the midpoint of AB) $\angle AED = \angle DEB = 90$

   DE = ED (common)

   $\therefore \Delta AED \cong (ASA) \therefore AD = BD (CSTCE)$

   But AD = DC $\therefore$ B $D\frac{1}{2}AC$
6. sol:(4) Let ABC be a triangle such that the bisector AD OF $\angle CAE$ is parallel to the

   base BC as shown in We have,



$\angle 1 = \angle 3$ [Corresponding angles] And $\angle 2 = \angle 4$ (Alteranate angles)

$\angle 2 = \angle 4 \Rightarrow \angle 3 = \angle 4 \quad [\because \angle 1 = \angle 2] \Rightarrow AB = AC$

Hence, ΔABC, is isiosceles.

7. sol:(2) in $\Delta ABC, \angle C = 180 - 90 - 30 = 60^0 \therefore \angle DCE = \frac{60}{2} = 30^0$

Again in Δ DEC, $\angle CED = 180 - 90 - 30 = 60^0$

8. $x = \frac{x}{2} + y \Rightarrow \frac{x}{2} = y \Rightarrow x = 2y \Rightarrow y = 60°$

9. $|\underline{ACE} = 130^0 \Rightarrow |\underline{ACD} = 180 - 130 = 50^0$

$|\underline{DAC} = |\underline{DCA} = 50^0$

$|\underline{ADB} = |\underline{DAC} = 50^0 + 50^0 = 100^0$ *(EXTERIOR ANGLE PROPERTY)*

$\therefore |\underline{ABD} = 40^0$

10. As AB=BC

$|\underline{BCA} = |\underline{BAC} = Y$

AsAc=CD

$|\underline{BCA} = |\underline{CAD} = X$

$|\underline{ACB} = X + X = 2X$

$Y = 2|\underline{X}$

11. (2) Δ BDM $\cong \Delta$ CDN (By RHS axim) $\Rightarrow \angle B = \angle C$

12. (3)

13. 4

14. (3) $\angle$ ADC = $\angle$ BDE (vertically opposite angles) and BD = DC
$\angle$ DBE = $\angle$ DCA (By angle properties) $\Rightarrow$ Δ ADC $\cong$ Δ DBE (ASA) $\therefore$ AD = DE

15. [4] ; In Δ ABC :- $\angle B + \angle C = 70^0 + 50^0 = 120^0$

$\therefore \angle A = 180^0 - 120^0 = 60^0 \qquad \therefore \angle X \frac{60^0}{2} = 30^0$

$\therefore$ In Δ ABD :- $\angle B + \angle x = 70^0 + 30^0 = 100^0$

$\therefore \angle y = 180^0 - 100^0 = 80^0$

In Δ ABD :- $\angle Y > \angle B \qquad \therefore$ AB > AD - ---------- (1)

In Δ ABD :- $\angle Y > \angle B$ $\therefore$ AD > CD ----------(2)

From (1) and (2), AB > AD > CD

16. As DB=DC, Δ DBC is an isoceles triangle. $\Rightarrow \lfloor DBC = \lfloor DCB$ ---------(1)

As AB=AC , Δ ABC is an isoceles triangle. $\Rightarrow \lfloor ABC = \lfloor ACB$ ---------(2)

From (1) and (2) $\Rightarrow \lfloor ABC = \lfloor ACB$ $\therefore \lfloor ADB \neq \lfloor BDC$

17. (2) P = AB, AC = AR; $\angle$ PAC = $\angle$ BAR

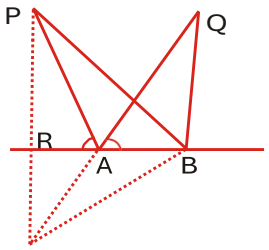
Δ PAC $\cong$ Δ BAR (SAS)

18. (1) Let $\angle$ BAD = x and $\angle$ ACD = y

Given BD = AD = CD $\Rightarrow \angle$ BAD = $\angle$ ABD = x and $\angle$ CAD = $\angle$ ACD = y

$\therefore$ $\angle$ BAC = x + y $= \frac{1}{2}$ 2(x + y) $= \frac{1}{2}$ ($\angle$ A + $\angle$ B + $\angle$ C)

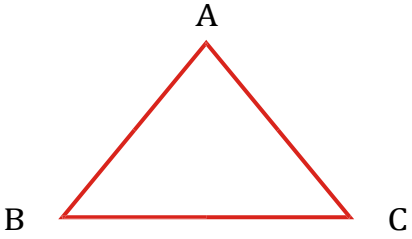
19. [1] ; Let PP′ $\perp$ XY , and QA produced cuts PP′ at P′



ΔPAR $\cong$ ΔP′AR (by RHS criterion) $\therefore$ PA + AQ = P′A + AQ = P′Q

ΔPBR $\cong$ ΔP′BR (by SAS criterion) $\therefore$ PB + BQ = P′B + BQ > P′Q

20. Sol: (1,2,4) Let ABC be an isosceles triangle such that AB=AC. Then, AB = AC $\angle C = \angle B = x$(say) We have

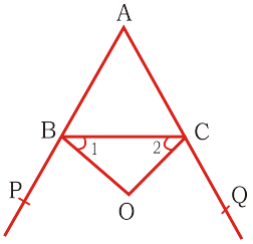


$\angle A = 2(\angle B + \angle X)$ [Given] $\Rightarrow \angle A = 2(\angle B + \angle B)$ [$\because \angle B = \angle X$]

$\Rightarrow \angle A = 4x$ Now, $\angle A + \angle B + \angle X = 1800 \Rightarrow 4x + x + x = 1800 \Rightarrow 6x = 1800 \Rightarrow x = 30^0$

Hence, $\angle A = 4x = 120^0, \angle B = 30^0$ And $\angle C = 30^0$

21. sol: [1,2,3] ; We have: $\angle B + \angle CBP = 1800$ (linear pair)



$\Rightarrow \frac{1}{2} \angle B + \frac{1}{2} \angle CBP = 90^0 \Rightarrow \frac{1}{2} \angle B + \angle 1 = 90^0$

$\Rightarrow \angle 1 = 90^0 - \frac{1}{2}\angle B$ _____(1)

Agin, $\angle C + + \angle BCQ 180^0 \Rightarrow \frac{1}{2}\angle C + \frac{1}{2}\angle BCQ \equiv 90^0 \Rightarrow \frac{1}{2}\angle C + \angle 2 = 90^0$

$\Rightarrow \angle 2 = 90^0 - \frac{1}{2}\angle C$ _____(2)

Now, in ΔOBC, we have: $\angle 1 + \angle 2 + \angle BOC = 180^0$

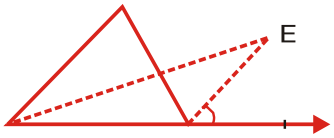
$\Rightarrow \left(90^0 - \frac{1}{2}\angle B\right) + \left(90^0 - \frac{1}{2}\angle C\right) + \angle BOC = 180^0$ (using (1) 1nd (2)

$\Rightarrow \angle BOC = \frac{1}{2}(\angle B + \angle C) = \frac{1}{2}(\angle A + \angle B + \angle C) - \frac{1}{2}\angle A$

Hence, $\angle BOC = 900 - \frac{1}{2}\angle A$

22. Given A ΔABC whose side BC is produced to D. The bisectors BE and CE of $\angle B$ and $\angle ACD$ respectively, meet at E.

To prove $\angle E = \frac{1}{2}A$ Proof Side BC of ΔABC is produced to D



$\therefore \angle ACD = \angle B + \angle$

$\Rightarrow \frac{1}{2}\angle ACD = \frac{1}{2}\angle B + \frac{1}{2}\angle A \Rightarrow \angle ECD = \frac{1}{2}\angle B + \frac{1}{2}\angle A$.....i

Again, side BC of ΔEBC is produced to D

$\therefore \angle ECD = \angle CBE + \angle BEC \Rightarrow \angle ECD = \frac{1}{2}\angle B + \angle E$.....ii

From (i) and (ii), we get:

$\frac{1}{2}\angle B + \angle E \frac{1}{2}\angle B + \frac{1}{2}\angle A$[each equal to $\angle ECD$] $\therefore \angle E \frac{1}{2}\angle A$

23. G divides BE in the ratio 2:1

$\frac{BG}{GE} = \frac{2}{1} \Rightarrow \frac{BG}{BE} = \frac{2}{3} \Rightarrow \frac{8}{BE} = BE = 12$

25. $\frac{CG}{FG} = \frac{2}{1} \Rightarrow \frac{CG}{5} = \frac{2}{1} \Rightarrow CG = 10$

26. $AD = 13.5 \Rightarrow \frac{AD}{GD} = \frac{3}{1} \Rightarrow \frac{13.5}{GD} = 3 \Rightarrow \therefore GD = 4.5$

27. a-q;b-q;c-r;d-q.

In ΔABC, we have $\angle B = 2\angle C$ or, $\angle B = 2y$, where $\angle C = y$

AD is the bisector of $\angle BAC$ So, let $\angle BAD = \angle CAD = x$

Let BP be the bisector of $\angle ABC$ join PD.

In ΔBPC, we have $\angle CBP = \angle BCP = y \Rightarrow BP = PC$

And $\Delta BPA \cong \Delta CPD$ (SAS Congruency).Then

$\angle D = 2y, DP = AP$(CPCT)

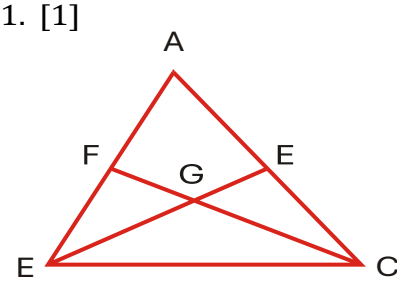
$\Rightarrow \angle PDA = y. From\ \Delta ADC, 4y + x = 180^0$

$From\ \Delta ABC, 3x + 2y = 180^0 \Rightarrow y = 36^0, x = 36^0$

28. Ans: 138^0 $\angle AEF = \angle EBD$ (corresponding angles)
$\angle AFE = \angle EDB$ $\therefore \angle BED = \angle EAF$

$\therefore \angle DBE + \angle DEB = \angle AEF + \angle AFE = 138°$

29,30,31. [1]



Let G be the point of their intersection.

Since BE = CF, we have $BG=\frac{2}{3}BE=\frac{2}{3}CF=CG$ ⍰ BG ⍰ CG

Also, GE = BE - BG = CF-CG = GF

Now in Δs BGF and CGE, we have

BG = CG GF = GE and $\angle BGF=\angle CGE$ [Vertically opposite angles]

$\therefore \Delta BGF=\Delta CGE$ [By SAS Congruence Axiom]

$\Rightarrow BF=CG \Rightarrow AB=AC$ [$\therefore BF=\frac{1}{2}AB, CE=\frac{1}{2}AC$]

WORKSHEET – 3

Level – I

1) A 2) D 3) A 4) A 5) C 6) A 7) D

Level – II

1) B	2) C	3) A	4) B	5) D	6) C	7) C
8) D	9) C	10) A	11) C	12) D	13) B	14) B
15) C	16) B	17) A	18) D	19) B	20)A,B,C,D	21) A
22) A	23) D	24) A	25) 5	26) C	27) a-s; b-r; c-q; d-p	
28) C	29) B	30)	31) a-s; b-r; c-q; d-p			

1. Given triangles are similar and triangle PQR is an equilateral

2. $\frac{AB}{PQ}=\frac{AC}{PR}\Rightarrow AC=\frac{AB}{PQ}.PR=\frac{3.6}{2.4}\times 5.4=8.1$

3. In Similar triangles,Ratio of the corresponding sides=ratio of angle bisectors=3:4

4. In a triangle ABC, D is a point on BC such that BD = DC and AD bisects $\angle BAC$

$\therefore \frac{AB}{AC}=\frac{BD}{BC}\Rightarrow \frac{AB}{AC}=1\Rightarrow AB=AC$

5. Conceptual

6. To prove that the triangles are similar using SAS similarity ,the angle

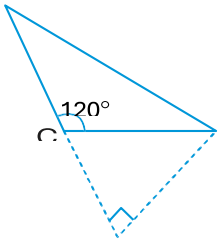
between the corresponding sides should be equal.

$\therefore \angle B = \angle E$

7. $\frac{\Delta ABC}{\Delta PQR} \quad \frac{AB^2}{PS^2} \quad \frac{AD}{PS} \quad \sqrt{\frac{49}{64}} \quad \frac{7}{8}$

8. $ar(\Delta DEF) = \frac{1}{4} ar(\Delta ABC) = \frac{1}{4} \times 24 = 6$

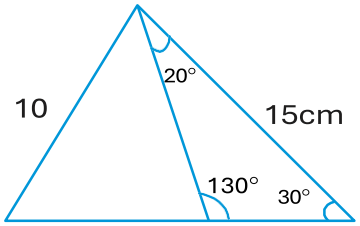
9.



10. $\frac{\triangle ABC}{\triangle PQR} = \frac{AD^2}{PS^2} = \frac{16}{81}$

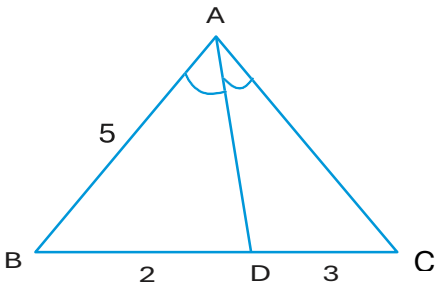
11. $\frac{AB}{BC} = \frac{10}{15} = \frac{2}{3} \Rightarrow \frac{AD}{CB} = \frac{3}{2}$

12. By AA similarity $\Delta ABC : \Delta DEF$



$\Delta ABC, \Delta DEF$ are not congruent since corresponding sides are not equal

13. $\frac{AB}{AC} = \frac{BD}{DC}$



$\frac{5}{AC} = \frac{3}{2} \Rightarrow AC = \frac{15}{2} = 7.5$

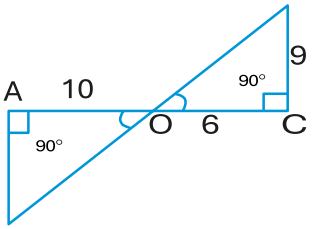
14. $\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD} \Rightarrow \frac{3}{DE} = \frac{2.5}{FD} \Rightarrow DE = 6$ DF = 5

$DE + EF + FD = 6 + 4 + 5 = 15$

15. $\Delta AOB \quad \Delta COD$

$\frac{AO}{OC} = \frac{AB}{CD}$

$\frac{10}{6} = \frac{AB}{9}$



$AD = \frac{10}{2} \times 9 = 15$

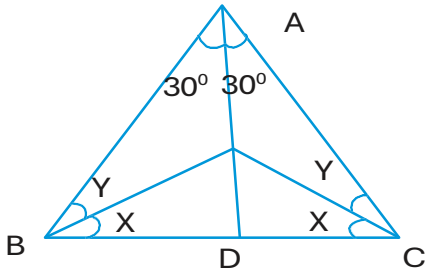
16. $\angle A+\angle B+\angle C=180$

$(x+y)+(x+y)+60=180$

$2(x+y)=120$

$x+y=60$

$\angle BPC=120$

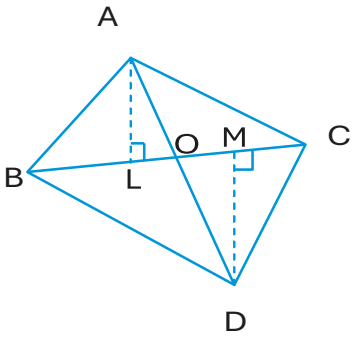


17. $\Delta AOL : \Delta DOM$

$\therefore = \frac{AL}{MD} = \frac{AO}{OD}$

$\frac{ar(\triangle ABC)}{ar(\triangle BCD)} = \frac{\frac{1}{2}\times BC\times AL}{\frac{1}{2}\times BC\times DM}$

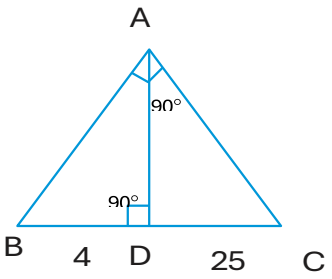
$\frac{AL}{MD} = \frac{AO}{OD}$



18. $AD^2 = BD\times CD$

$= 4\times 25$

AD=10



19. (1,2,3) Conceptual

20. (1,2,3,4) Conceptual

21. statement 1:As $DE \| BC$

In $\Delta ADE \text{ and } \Delta ABC$

$\angle ADE = \angle ABC$ (corresponding angles)

$\angle AED = \angle ACB$ (corresponding angles)

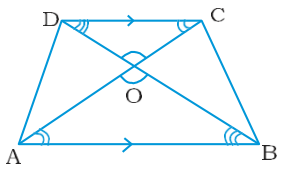
by AA similarity , $\Delta ADE \text{ and } \Delta ABC$ are similar

statement 2:according mid point theorem,the triangles are similar

22. $\frac{AB}{PQ} = \frac{BC}{QR} \Rightarrow \frac{5}{7} = \frac{15}{QR} \Rightarrow QR = 21$

23. $\frac{AB+BC+CA}{DE+EF+ED} = \frac{BC}{EF} \Rightarrow \frac{20}{DE+EF+FD} = \frac{1}{2} \Rightarrow DE+EF+FD=40$

24. $\frac{BC}{QR} = \frac{AD}{PS}$



25. $\frac{AB}{AC} = \frac{AD}{DC} \quad \frac{3/5}{2/5} = \frac{X}{Y} \Rightarrow X+Y = 3+2 = 5$

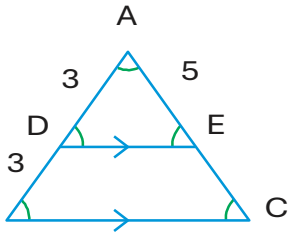
26. If two triangles are similar,then the ratio of the corresponding sides and corresponding altitudes are equal.

$\frac{AD}{PS} = \frac{3}{2} = \frac{BC}{QR}$

27. a-s; b-r; c-q; d-p

$AE=EC=5cm$

$AD=DB=3cm$



$DE=\frac{1}{2}BC=4cm$

B

8

a) $AB+AC+DE=6+10+4=20$

b) $AB+BC+CA=6+8+10=24$

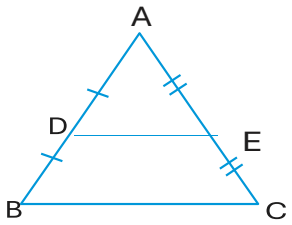
c) $(BC-DE)(BC+DE)=BC^2-DE^2 \Rightarrow 8^2-4^2=64-16=48$

d) $DE=\frac{1}{2}BC=4cm$

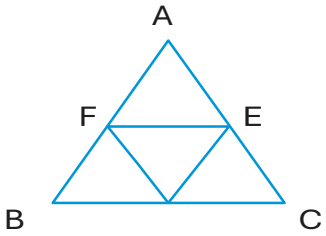
28. $\frac{\Delta ABC}{\Delta PQR}=\left(\frac{AB}{PQ}\right)^2=\frac{9}{49}$

29. $\left(\frac{AB}{PS}\right)^2=\frac{\Delta ABC}{\Delta PQR}=\frac{3}{5}\Rightarrow\frac{AD}{PS}=\sqrt{\frac{5}{5}}$

30.



31. a-s; b-r; c-q; d-p



$\Delta AEF \cong \Delta BDF \cong \Delta DCE \cong \Delta DEF$

$ar(\Delta ABC)$ $(EF)^2$ $(EF)^2$ 1 B) $\Delta DEF : \Delta ABC$

Ar(ΔABC) (BC) (2EF) 4 D) $\Delta DEF \cong \Delta CDE$

$\frac{\Delta DEF}{\Delta AEF}=\frac{1}{1}=1:1$

WORKSHEET-4

Level – I

1) B 2) A 3) B 4) C 5)A 6) D

Level – II

1) A 2) D 3) C 4) B 5) D 6) B 7) D

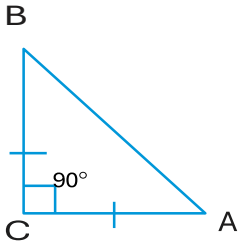
8) A 9) A 10) C 11) B 12) A 13) A 14) D

15) $\frac{\sqrt[5]{3}}{2}$ 16) A,B,C 17) A 18)B 19)C 20) a-4; b-3; c-2; d-1

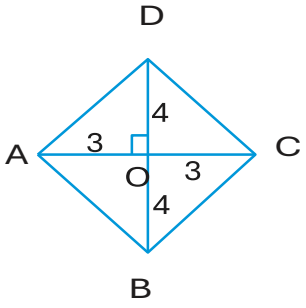
21) A 22) B

1. Conceptual

2. $AB^2=AC^2+BC^2=BC^2+BC^2=2BC^2$

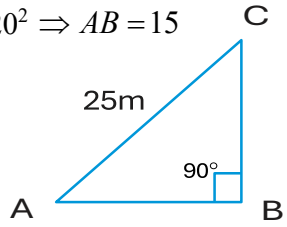


3. $11^2 > 5^2 + 8^2$
4. Conceptual
5.

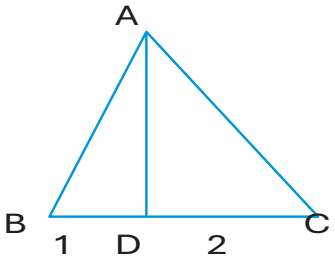


$AB^2 = OA^2 + OB^2 = 3^2 + 4^2 = 5^2 \Rightarrow$ AB = 5

6. In equilateral triangle, median=altitude $= \frac{\sqrt{3}}{2}$
7. $AC^2 + BD^2 = AB^2 + BC^2 + CD^2 + DA^2 = 4AB^2$
8. Conceptual
9. $h = \sqrt{3}, s = \frac{2}{\sqrt{3}} \times h \Rightarrow \frac{2}{\sqrt{3}} \times \sqrt{3} = 2 \Rightarrow \Delta = \frac{\sqrt{3}}{4}s^2 = \frac{\sqrt{3}}{4} \times 2^2 = \sqrt{3}$
10. $OA^2 + OC^2 = \left(OB^2 + OD^2\right) = 12^2 + 5^2 = 169$
11. $AB^2 = AC^2 - BC^2 = 25^2 - 20^2 \Rightarrow AB = 15$



12. $9AD^2 = 7AB^2 + BC^2 \Rightarrow \frac{AD^2}{AB^2} = \frac{7}{9}$



13. Conceptual
14. Shorter side=x,Hypotenuse=2x-3, Longer side=12

$\Rightarrow (2x-3)^2 = 12^2 + x^2$

$\Rightarrow 4x^2 + 9 - 12x = 144 + x^2$

$\Rightarrow 3x^2 - 12x - 135 = 0$

$\Rightarrow x^2 - 4x - 45 = 0$

$\Rightarrow (x-9)(x+5) = 0$

$\Rightarrow x = 9 \Rightarrow 2x - 3 = 2(9) - 3 = 15$

15. $\frac{5\sqrt{3}}{2}$ AD=$\frac{\sqrt{3}}{2}$XSIDE$\frac{\sqrt{3}}{2}$X5

16. Except (2,3,4) all the other are pythagorean triplets.

17. Conceptual

18. Conceptual

19. Conceptual

20. a) represents right triangle ,according to converse of pythagoras theorem

b) conceptual

c) conceptual

d) right angled isosceles triangle. $a^2 = b^2 + b^2$ .here two side lengths are equal to b, that means it is an isosceles triangle as well as right angled

21. $BD^2 = AD.DC$

$\Rightarrow AC^2 = 3^2 + 4^2 \Rightarrow AC = 5$

$AD = x; CD = 5 - x$

$In \quad \Delta ABD, AB^2 = AD^2 + BD^2 \Rightarrow BD^2 = AB^2 - AD^2$

$In \quad \Delta BCD, BC^2 = BD^2 + CD^2 \Rightarrow BD^2 = BC^2 - CD^2$

$\Rightarrow AB^2 - AD^2 = BC^2 - CD^2$

$\Rightarrow 9 - x^2 = 16 - (5 - x)^2$

$\Rightarrow 9 - x^2 = 16 - 25 - x^2 + 10x$

$\Rightarrow 10x = 9 - 16 + 25 = 18$

$\Rightarrow x = 1.8 \Rightarrow AD = 1.8; CD = 5 - 1.8 = 3.2$

$BD^2 = 1.8 \times 3.2 = \frac{18}{10} \times \frac{32}{10} = \frac{9 \times 2 \times 2 \times 16}{100} \Rightarrow BD = \frac{3 \times 2 \times 4}{10} = 2.4$

22. Shorter leg=x

hypotenuse=2x3

longer leg= x+7

$\Rightarrow (2x+3)^2 = (x+7)^2 + x^2$

$\Rightarrow 4x^2 + 9 + 12x = x^2 + 49 + 14x + x^2$

$\Rightarrow 2x^2 - 2x - 40 = 0$

$\Rightarrow x^2 - x - 20 = 0$

$\Rightarrow (x-5)(x+4) = 0$

$\Rightarrow x = 5 \Rightarrow x + 7 = 5 + 7 = 12$

WORKSHEET – 5

Level – I

1.	2.	3.	4.	5.	6.	7.	8.
9.	10.	11.	12.				

Level – II

1) C	2) B	3) B	4) C	5) A	6) B
7) A	8) A	9) A	10) A	11) D	12) D
13) C	14) C	15) A,B,C,D	16) B,C	17) B,C	18) A
19) C	20) C	21) A	22) a-r; b-q; c-p; d-s		
23) C	24) B	25) C	26) a-r; b- (r,s,t); c- p ; d-s		

27) A 28) C 29) B 30) B 31) A 32) B
33) B,C 34) A 35) 1800 36) A

1. EF $=\frac{1}{2}(AB+CD)=\frac{1}{2}(10+6)=8$

2. $\angle A=2x+30^0, \angle C=x+120^0$

We know that in a parallelogram ABCD $\angle A=\angle C$

$2x+30=x+120 \Rightarrow x=120-30 \Rightarrow x=90$

3. In Rhombus ,all the sides are equal.
diagonal divides it into two congruent triangles and two sides are equal as well.

4. a=6cm,b=8cm,h=4cm
$A=\frac{1}{2}(a+b)h=\frac{1}{2}(6+8)\times 4=28cm^2$

5. $\frac{(2n-4)\times 90^0}{n}=120^0 \Rightarrow 180n-360^0=120n \Rightarrow 60n=360 \Rightarrow n=6$

6. We know that in a parallel gram in which one angle is right angle is Rectangle

7. In Rhombus ,opposite angles are equal.

8. Quadrilaterals are classified into two types.1,convex ,2.concave quadrilateral.

9. EF $=\frac{1}{2}(AB+CD) \Rightarrow 10=\frac{1}{2}(10+CD) \Rightarrow CD=10$ cm

10. $n:\frac{n(n-3)}{2}=2:5 \Rightarrow 5n=5n=\frac{n(n-3)}{2}\times 2 \Rightarrow n-3=5 \Rightarrow n=8$

11. Sum of interior angles of a quadrilateral is 360 degrees.

12. A trapezium is a quadrilaterl in which one pair of opposite sides are parallel

13. As AB || CD and $\angle A+\angle D=180^0$ according to the angle sum property $\angle B+\angle C=180^0$

14. BD bisects $\angle B$ and $\angle D$

15. Sum of exterior angles of any polygon=360 degrees.

16. Conceptual

17. Conceptual

18. Conceptual

19. Conceptual

20. Conceptual

21. Conceptual

22. Conceptual

23. Conceptual

24. Use EF=$\frac{1}{2}$(AB+

25. Use EF = $_2$ (AB+CD)

26. Conceptual

27. N ΔABC,wehave $\angle CAB+\angle ABC=180^0$.and $\angle ABC=90^0 \Rightarrow 32^0+90^0+\angle ACB=180^0$
$\angle ACB=58^0$

28. We know that the diagonals of a rectangle are equal and bisect each other

$\therefore OA = OB \Rightarrow \angle OBA = \angle OBA = 32^0$

$\angle ABC = 90^0 \Rightarrow \angle OBA + \angle OBC = 90^0$

$32^0 + \angle OBC = 90^0$

$\therefore \angle OBC = 58^0$

29. In a rightangle triangle the square of hypotenuse is equal to sum of the squares of opposite side and adjacent side i.e. $AC^2 = AB^2 + BC^2$

30. EB=2AE (Given)

$\therefore \frac{AE}{EB} = \frac{1}{2}$ _____(1)

AD\\EF\\BC and tow transversals AB and DC interest them

$\frac{AE}{EB} = \frac{DF}{FC}$ _____(2)

Three parallel lines intersecting any two transversals cut off intercepts from them which are in the same ratio.

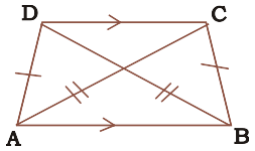
From (1) and (2), we get $\frac{DF}{FC} = \frac{1}{2} (or) \frac{1.5}{FC} = \frac{1}{2}$

FC = 2×1.5cm = 3cm

31. $LA + LB + LC + LD = 360^0 \Rightarrow$ 2x+1+4x-10+x+15+x+10=360

8x+16=360 $\Rightarrow$ 8x=344 $\therefore x = 43^0$

32. $OA^2 + OB^2 = OC^2 + OD^2$



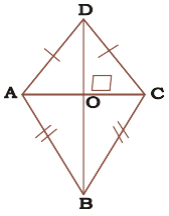
33.

In isosceles trapezium ABCD , $AB \parallel CD$,AB=BC

then the length of diagonals are equal

A kite ABCD with AB=AC and AD=CD

the diagonals of a kite are perpendicular to each other i.e, $BD \perp AC$



34. Conceptual

35.

$+ \angle A \ \angle B + \angle C + \angle D = 360^0$ (Angle sum property)

60+120+ $\angle C + \angle D = 360^0 \Rightarrow \angle C + \angle D = 180^0$.

36. Each interior angle of a regular polygon=$\left(\frac{2n-4}{n}\right)\times 90^0$

$K\left(\frac{2n-4}{n}\right)\times 90^\circ=\left(\frac{2n-4}{n}\right)\times 90^\circ \Rightarrow K=1$

WORKSHEET – 6

Level – I

1. A 2. B 3. C 4. D 5. A 6. C 7. B 8. B

Level – II

1) C 2) D 3) B 4) B 5) C 6) A 7) D

8) D 9) C 10) A 11) D 12) 13) A 14) C

15) A 16) A 17) D 18) A,B,C 19) A,B 20) A

21) a-q; b-p; c-s; d-r 22) C 23) B 24) A 25) a-s; b-r; c-q; d-p
26) a-s; b-r; c-p; d-q

1. $ar(\Delta ABC)=2\times ar(\Delta ABP)$

2. Radius and tangent are perpendicular to each other at the point of contact. $LAPO=90^0$

3. r < CP $\Rightarrow$ point lies exterior of the circle

4. $2\times$ *area of triangle* $=2\times ar(\Delta ABP)=$ *area of* Π*gm ABCD*

5. Quadrant

6. Triangle and rectangle on the same base and between same parallels ,then area

of triangle is equal to half of the area of the rectangle.

7. Length of arc= $\frac{\phi}{360^\circ}\times 2\pi r$

8. $\Delta ADP,\Delta BPC$ have equal bases and between same parallel lines.

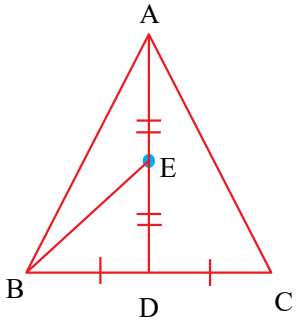
9. r = CP $\Rightarrow$ point lies on the circle

10. $2\pi r=44cm \Rightarrow r=44x\frac{7}{22}\times\frac{1}{2}=7 \Rightarrow \pi r^2=\frac{22}{7}\times 7x7=154$

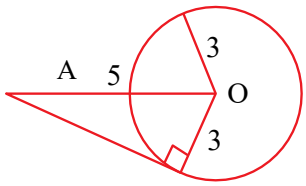
11. $ar(\Delta ABD)=\frac{1}{2}\times ar(\Delta ABC)$ 2

$ar(\Delta ABE)=\frac{1}{2}\times ar(\Delta ABD)$ 2

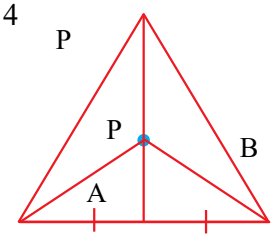
$\therefore ar(\Delta ABE)=C\times ar(\Delta ABC)$



12. OA=OB=3cm, OP=5cm



Then $PB^2 = OP^2 - OB^2 \Rightarrow PB^2 = 5^2 - 3^2 = 4^2 \Rightarrow PB = 4$



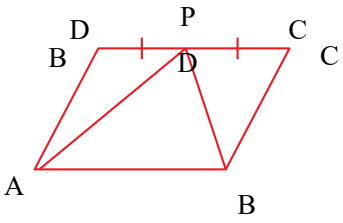
13. $ar(\Delta ABD) = ar(\Delta ACD)$

$ar(\Delta BPD) = ar(\Delta PCD)$

$ar(\Delta ABD) = ar(\Delta APB) + ar(\Delta BPD)$

$ar(\Delta ACD) = ar(\Delta APC) + ar(\Delta PCD)$

$\therefore ar(\Delta APB) = ar(\Delta APC)$



14. $ar(\Delta ABCD) = ar(\Delta ABPQ)$

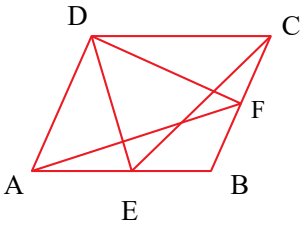
$\Rightarrow ar(\Delta ABC) = \frac{1}{2} ar(\square^{gm} ABCD) = \frac{1}{2} \times 40 = 20$

15. $ar(\Delta CDE) = \frac{1}{2}$ ar$(\square^{gm} ABCD)$

$ar(\Delta ADF) = \frac{1}{2}$ ar$(\square^{gm} ABCD)$

$\therefore ar(\Delta ADF) = ar(\Delta CDE)$

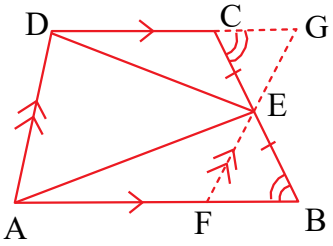
16. AD || GF



$\Delta CEG \cong \Delta BEF$ (ASA axion)

$ar(\Delta CEG) = ar(\Delta BEF)$

$ar(\Delta AED) = \frac{1}{2} ar(\square^{gm} AFGD)$



$= \frac{1}{2} [ar(\text{pentagon } AFECD) + ar(\Delta CEG)]$

$= \frac{1}{2} [ar(\text{pentagon } AFECD) + ar(\Delta BEF)]$

$= \frac{1}{2} [ar(\text{trapezium } AFECD)]$

17. $ar(\Delta ABC) = \frac{1}{2} BC.h$

$ar(\Delta ABD) = \frac{1}{2} BD.h = \frac{1}{2}.\frac{1}{3} BC.h$

$\Rightarrow \frac{1}{3}\left(\frac{1}{2}.BC.h\right) = \frac{1}{3}\left(ar(\Delta ABC)\right)$

15. ANALYTICAL GEOMETRY

WORKSHEET – 1

Level – I

1) C	2) A	3) D	4) B	5) A	6) D	7) C
8) C	9) A	10) B	11)D	12)C	13)B	14)D

Level – II

1) 1	2) 3	3) 4	4) 4	5) 4	6) 3	7) 3
8) 2	9) 1	10) 3	11) 3	12) 4	13) 3	14) 2
15) 4	16) 2	17) 2	18) 1	19) 2	20) 4	21) 3
22) 1	23) 1	24) 1	25) 1	26) 3	27) 4	
28) a-s; b-s; c-q; d-p,r		29) 1	30) 1	31) 1	32) 3	33) 4
34) 648	35) a-s; b-p; c-q,s; d-r					

1. $AB = \sqrt{(a\cos\theta - 0)^2 + (0 - a\sin\theta)^2} = \sqrt{a^2(\sin^2\theta + \cos^2\theta)} = |a|.$

2. Given $A = (1,1)$, $B = \left(\frac{2t^2}{1+t^2}, \frac{(1-t)^2}{1+t^2}\right)$

$$AB = \sqrt{\left(\frac{2t^2}{1+t^2} - 1\right)^2 + \left(\frac{(1-t)^2}{1+t^2} - 1\right)^2} = \sqrt{\left(\frac{2t^2 - 1 - t^2}{1+t^2}\right)^2 + \left(\frac{1 + t^2 - 2t - 1 - t^2}{(1+t^2)^2}\right)^2}$$

$$= \sqrt{\left(\frac{1+t^2}{1+t^2}\right)^2} = 1$$

3. Given $A(0,0)$, $B(1,0)$, $C = (0,1)$

$AB = 1$, $BC = \sqrt{2}$, $AC = 1$ $\therefore$ Perimeter $= 1 + 1 + \sqrt{2} = 2 + \sqrt{2}$

4. $AB = \sqrt{16 + 9} = 5$, $C = (-5,-2)$ then $AC = \sqrt{64 + 36} = 10$

$BC = \sqrt{144 + 81} = 15$. $\therefore AC + AB = BC \Rightarrow C = (-5,-2)$

5. $(x,y) = (2,3)$, $A(x,y) = 4,8$ $B(x_2, y_2) = (-2,-7)$

Ratio $= x_1 - x : x - x_2 = 4 - 2 : 2 + 2 = 2 : 4 = 1 : 2$ internally .

6. Required point $= \left[\frac{b(a+b)+a(a-b)}{a+b}, \frac{b(a-b)+a(a+b)}{a+b}\right]$

$= \left[\frac{a^2+b^2}{a+b}, \frac{a^2+2ab-b^2}{a+b}\right]$.

7. AP, AB, AQ are in H.P.

$\Rightarrow AB = \frac{2AP \cdot AQ}{AP+AQ} = \frac{2(2)6}{2+6} = 3.$

8. Given $A(0,-1), B(-2,0), C = (-1,-3)$

$AB = \sqrt{4+1} = \sqrt{5}$; $BC = \sqrt{1+9} = \sqrt{10}$; $AC = \sqrt{1+4} = \sqrt{5}$

$\therefore$ ABC is a right angled isosceles triangle.

9. Given $A(2,3), B(x,y), C = (3,4), m:n = 1:5$

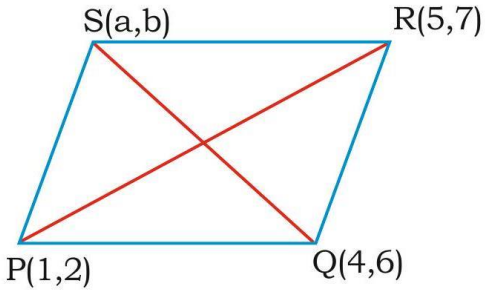
$A(2,3) = (x_1, y_1), B(x,y), C = (x_2, y_2)$

$1:5 = 2 - 3 : 3 - x \Rightarrow \frac{1}{5} = \frac{+1}{x-3}$ $x = 8$

$3 - 4 : 4 - y = 1:5 \Rightarrow \frac{1}{y-4} = \frac{1}{5} \Rightarrow 5 = y - 4 \Rightarrow y = 9$

$\therefore$ B(x, y) = (8,9).

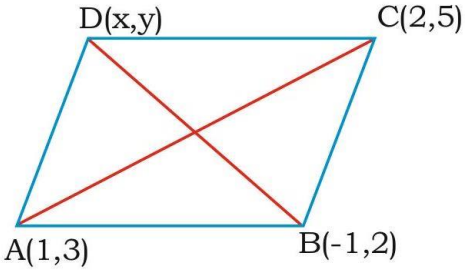
10. $(2-7+2, 1-4+5) = (-3,2)$
    Required point $= (-3,2)$
11. $A(1,1), B(-2,7), C(3,-3)$
    $AB = \sqrt{9+36} = \sqrt{45} = 3\sqrt{5}; BC = \sqrt{25+100} = \sqrt{125} = 5\sqrt{5}$
    $AC = \sqrt{4+16} = \sqrt{20} = 2\sqrt{5}$ $\therefore$ A, B, C are collinear
12. $A(a, 2), B(3,4)$
    $AB = 8 \Rightarrow \sqrt{(a-3)^2 + (2-4)^2} = 8 \Rightarrow a = 3 \pm \sqrt{60}$.
13. Midpoint PR = Midpoint of SQ



    $\Rightarrow S(a, b) = (2,3) \Rightarrow a = 2, b = 3$.
14. Midpoint AC = Midpoint BD

$$\left(\frac{x-1}{2}, \frac{y+2}{2}\right) = \left(\frac{1+2}{2}, \frac{3+5}{2}\right)$$

$\Rightarrow x - 1 = 3$
$\Rightarrow x = 4$ and $y + 2 = 8 \Rightarrow y = 6$



15. Here $5-23 : 23-y = 3:5 \Rightarrow \frac{-18}{23-y} = \frac{-3}{5} \Rightarrow -30 = -23 + y \Rightarrow y = -7$.
16. $SA = \sqrt{a(t^2-1) + 4a^2t^2} = \sqrt{a^2(t^2-1) + 4t^2} = a(t^2+1)$

$$SB = \sqrt{a\left(\frac{1}{t^2} - 1\right) + \left(\frac{2a}{t}\right)^2} = \sqrt{a^2(t^2+1)^2} = a\left(\frac{1}{t^2} + 1\right)$$

$$\frac{1}{SA} + \frac{1}{SB} = \frac{1}{a(t^2+1)} + \frac{1}{a\left(\frac{1}{t^2}+1\right)} = \frac{1+t^2}{a(1+t^2)} = \frac{1}{a}$$

17. If P divides QR externally in the ratio 1:2, then Q is midpoint of PR.
    Let R(x, y), then $(-2,-3) = \left(\frac{-5+x}{2}, \frac{4+y}{2}\right)$
    $\therefore x = 1 \& y = -10 \Rightarrow (x, y) = (1,-10)$.
18. Given AP = PQ = QB



    Let R be the midpoint of PQ such that $PR = PQ$

Now $AR = AP + PR, AR = QB + QR = BR$

$\therefore$ R is midpoint of AB $\Rightarrow R = \left(\frac{1}{2}, 3\right)$.

19. [2] Let x_1, x_2 are x-coordinates of A, B and y_1, y_2 are y-coordinates of A, B then
$x_1 + x_2 = -2a; x_1x_2 = -b^2; y_1 + y_2 = -2p; y_1y_2 = -q^2$
$AB^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2 = (x_1 + x_2)^2 - 4x_1x_2 + (y_1 + y_2)^2 - 4y_1y_2$
20. $a^2(\cos^2 48° + \cos^2 12^0) = d^2$
$\Rightarrow d^2 - a^2 = \frac{a^2}{2}\cos 36 \Rightarrow d^2 - a^2 = \frac{a^2(\sqrt{5}+1)}{8}$
21. $x = \frac{(1-t)x_1 + tx_2}{1-t+t}; y = \frac{(1-t)y_1 + ty_2}{1-t+t}$
22. Here $(5-2)^2 + (7-x)^2 = 25 \Rightarrow 9 + (7-x)^2 = 25 \Rightarrow (7-x)^2 = 16 \Rightarrow x = 3,11.$
23. $A(x_1, y_1) = (-5,1), B(x_2, y_2) = (P, 5), C(x_3, y_3) = (10,7)$
A, B, C are collinear $\Rightarrow$ Area of triangle is zero
$\Rightarrow \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| = 0$
$\Rightarrow -5(-2) + P(6) + 10(-4) = 0 \Rightarrow 6P = 30 \Rightarrow P = 5.$
24. A (5,2), B(7,9), C(9,16)
$AB = \sqrt{2^2 + 7^2} = \sqrt{4 + 49} = \sqrt{53};\ BC = \sqrt{2^2 + 7^2} = \sqrt{4 + 49} = \sqrt{53}$
$AC = \sqrt{4^2 + 14^2} = \sqrt{212} = 2\sqrt{53}.\ \therefore AC = AB + BC.$
25. $AB = \sqrt{36 + 16} = \sqrt{52};\ BC = \sqrt{36 + 16} = \sqrt{52};\ AC = \sqrt{12^2 + 8^2} = \sqrt{208} = 2\sqrt{2}$
$AC = AB + BC.$
26. If A, B, C are collinear $\Rightarrow$ area $(\triangle ABC) = 0$.
27. Slope of AB = Slope AC $\Rightarrow \frac{P+3}{2+1} = \frac{-1-P}{5-2} \Rightarrow P = 1.$
28. Hint: Use distance formula.
29. 4th vertex $= (2,-2) - (8,4) + (5,7) = (2 - 8 + 5, -2 - 4 + 7)$
$= (-1,1) = (x, y)\ \therefore 2x + 3 = 2(-1) + 3 = 1.$
30. Ratio in which $(4,-2)$ divides the join $A(2,-4)\&B(7,1)$ is $4 - 2 : 7 - 4 = 2 : 3$
Harmonic conjugate of $(4,-2)$ w.r.to AB is
$= \left(\frac{2(7) - 3(2)}{2 - 3}, \frac{2(1) - 3(-4)}{2 - 3}\right) = (-8,-14).$
31. $A = (-4,7), B = (1,2)$
32. Conceptual.
33. [4] 1) Is incorrect since A, B, C are collinear
2. Is false since $AB^2 \neq AC^2$
3. Is incorrect since $AB^2 \neq AC^2$ 4) Is correct since
$AB^2 = (1 - a - b)^2 + \left(\frac{b}{a} - b + a\right)^2$ and $AC^2 = (1 - a + b)^2 + \left(\frac{b}{a} - a - b\right)^2$
Note that $AB^2 = AC^2$.
34. $\frac{a+1+a^2+1}{2} = 11, \frac{a-1+a^2-1}{2} = 9 \Rightarrow a + a^2 - 20 = 0 \Rightarrow a = -5$ or 4 and $\frac{a^2+1+a^3+1}{2} =$
$41, \frac{a^2-1+a^3-1}{2} = 39 \Rightarrow a^2 + a^3 = 80$ which holds for a = 4
so, the given points are $A(5,3), B(17,15), C(65,63)$ and coordinates of F are
$\left(\frac{65+5}{2}, \frac{63+3}{2}\right) = (35,33)$
and $(BF)^2 = (35 - 17)^2 + (33 - 15)^2 = (18)^2 + (18)^2 = 648$

35. Length of AB = 2a

a) $BC = \sqrt{(2a)^2 + a^2} = \sqrt{5}a, CA = \sqrt{(2a)^2 + a^2} = \sqrt{5}a \Rightarrow BC = CA = AB$

$\triangle ABC$ is isosceles

b) $BC = \sqrt{3a^2 + a^2} = 2a, CA = \sqrt{3a^2 + a^2} = 2a, \triangle ABC$ is equilateral

c) $BC = \sqrt{(2a)^2 + (2a)^2} = 2\sqrt{2}a, CA = \sqrt{(2a)^2 + 0} = 2a = AB$

$So (BC)^2 = 8a^2 = (AB)^2 + (AC)^2, \Delta ABC$ is right angled and isosceles

d) $BC = \sqrt{a^2 + (3a)^2} = \sqrt{10}a, CA = \sqrt{a^2 + a^2} = \sqrt{2a}$

So $(AB)^2 + (AC)^2 < (BC)^2 \Rightarrow A$ is obtuse and thus $\triangle ABC$ is obtuse angled.

WORKSHEET – 2

Level – I

1) B	2) B	3) C	4) D	5) A	6) B
7) D	8) D	9) A	10) D		

Level – II

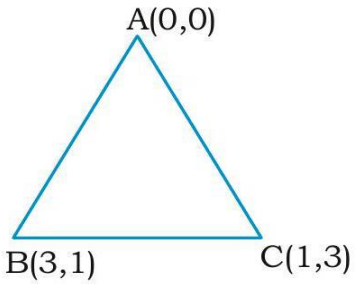
1) A	2) D	3) A	4) A	5) C	6) B	7) A
8) A	9) D	10) D	11) A	12) A	13) B	14) D
15) A	16) D	17) D	18) D	19) C	20) C	21) A
22) C	23) B	24) D	25) A,B,C	26) A	27) D	28) C
29) B	30) a-q; b-r; c-t; d-p		31) B	32) A,C,D	33) B	
34) A,B,C,D	35) a-s; b-p; c-q; d-r					

1. Centroid $= \left(\frac{1+2+4}{3}, \frac{-1-1-3}{3}\right) = \left(\frac{7}{3}, \frac{-5}{3}\right)$.

2. Given $(1,4) = \left(\frac{x+4-9}{3}, \frac{y-3+7}{3}\right) \Rightarrow (1,4) = \left(\frac{x-5}{3}, \frac{y+4}{3}\right)$

$\Rightarrow \begin{matrix} x-5=3 \\ x=8 \end{matrix}$ and $\begin{matrix} y+4=12 \\ y=8 \end{matrix} \Rightarrow$ Third vertex = (8,8)

3. $0 = \left(\frac{(y_1-y_2)(x_1x_2+y_1y_2)}{(x_2y_1-x_1y_2)}, \frac{(x_2-x_1)(x_1x_2+y_1y_2)}{(x_2y_1-x_1y_2)}\right)$



$= \left[\frac{-2\times 6}{-8}, \frac{-2\times 6}{-8}\right] = \left[\frac{3}{2}, \frac{3}{2}\right]$.

4. Given A (6,0), B(0,6), C(6,6)

$\therefore$ ABC is right angled triangle at C(6,6). $\therefore$ Orthocentre = (6,6).

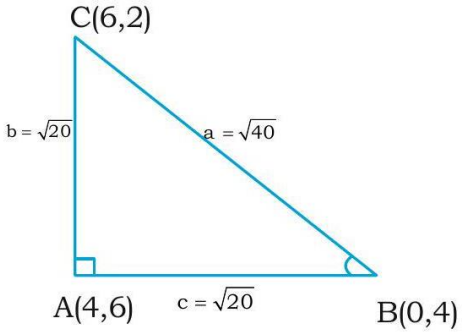
5. Given A(0,0) B(3,1) C(1,3)

$$\text{Circumcentre} = \left[\frac{y_2(x_1^2+y_1^2)-y_1(x_2^2+y_2^2)}{2(x_1y_2-x_2y_1)}, \frac{x_1(x_2^2+y_2^2)-x_2(x_1^2+y_1^2)}{2(x_1y_2-x_2y_1)}\right]$$

$$= \left[\frac{3(9+1)-1(9+1)}{2(9-1)}, \frac{3(1+10)-1(9+1)}{2(9-1)}\right] = \left[\frac{20}{2\times 8}, \frac{20}{2\times 8}\right] = \left[\frac{5}{4}, \frac{5}{4}\right].$$

6. Given $A(4,6), B(0,4), C(6,2)$

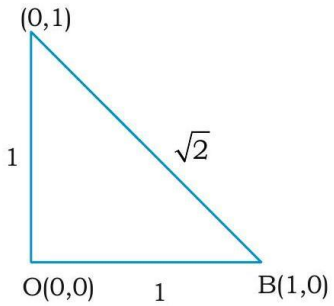
$b^2 + c^2 = a^2 \Rightarrow \angle A = 90 \therefore S =$ Midpoint of $BC = (3,3)$



$\therefore$ Circumradius $(R) = \frac{BC}{2} = \frac{\sqrt{40}}{2} = \sqrt{10}.$

7. Since $ON: NG = 3:1 \Rightarrow N = \left(\frac{3(3)+1(-3)}{3+1}, \frac{3(3)+(1)(3)}{3+1}\right) = \left(\frac{3}{2}, \frac{7}{2}\right).$

8. Incentre $= \left(\frac{ax_1+bx_2+cx_3}{a+b+c}, \frac{ay_1+b_2+cy_3}{a+b+c}\right)$



$$= \left(\frac{0+1+0}{2+\sqrt{2}}, \frac{1+0+0}{2+\sqrt{2}}\right) = \left(\frac{1}{2+\sqrt{2}}, \frac{1}{2+\sqrt{2}}\right) = \left(\frac{2-\sqrt{2}}{2}, \frac{2-\sqrt{2}}{2}\right)$$

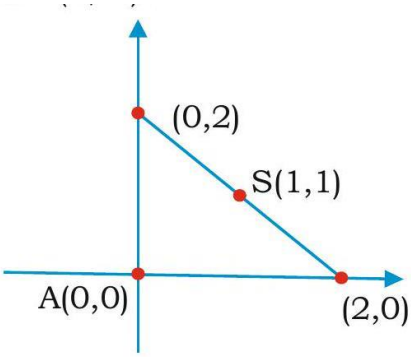
9. Centroid $G = \left(\frac{a+b+c}{3}, \frac{a+b+c}{3}\right) = (0,0)$

$$\Rightarrow a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = 3abc$$

10. $G(3,3), S(6,2)$. P divides $\overline{GS}$ in the ratio $2:3$

$$\therefore P = \left(\frac{2(6) - 3(3)}{2-3}, \frac{2(2) - 3(3)}{2-3}\right) = (-3,5).$$

11. Given $A(0,0), B(0,2)$ and $C(2,0)$.

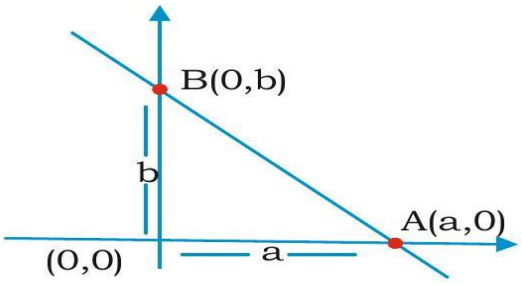


Distance between Circumcentre and Orthocentre $= \sqrt{1^2 + 1^2} = \sqrt{2}.$

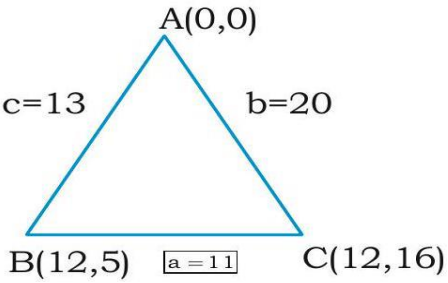
12. Given $A(-1,12), B(-1,0), C(4,0)$. Here $a = 5, b = 13, c = 12$

$$I = \left(\frac{-5 - 13 + 48}{30}, \frac{60 + 0 + 0}{30}\right) = (1,2).$$ x-coordinate $= 1.$

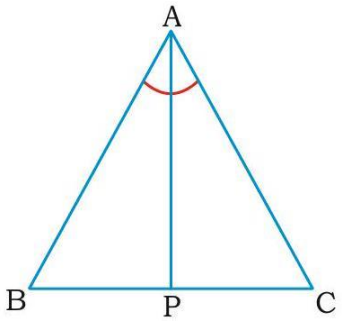
13. Incentre $= \left(\frac{ab}{a+b+\sqrt{a^2+b^2}}, \frac{ab}{a+b+\sqrt{a^2+b^2}}\right)$.



14. Incentre $= \left(\frac{0+120+156}{13+20+11}, \frac{0+100+208}{13+20+11}\right) = (9,7)$.



15. We know that $BP: PC = AB: AC = 4\sqrt{5}: 3\sqrt{5} = 4: 3$

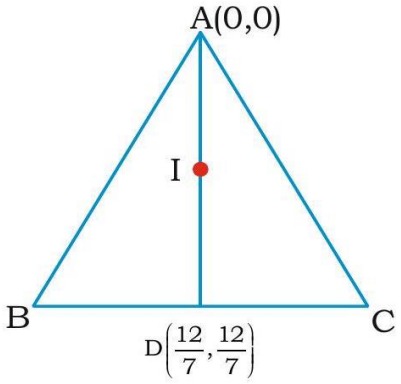


$$\therefore P = \left(\frac{4(-4)+3(-2)}{4+3}, \frac{4(6)+3(-5)}{4+3}\right) = \left(\frac{-22}{7}, \frac{9}{7}\right).$$

16. $P(0,1), G(2,3), S$ divides $\overline{PG}$ in ratio $3:1$ externally

$$S = \left(\frac{3(2)-1(0)}{3-1}, \frac{3(3)-1(1)}{3-1}\right) = (3,4)$$

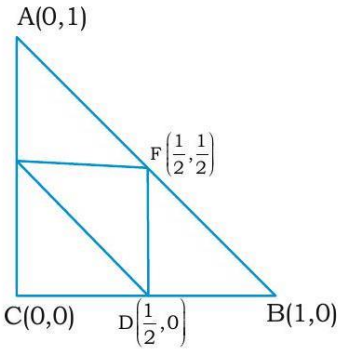
17.



I divides AD in the ratio $AB + AC: BC = 7: 5$

$$\therefore I = \left(\frac{7\left(\frac{12}{7}\right)+5(0)}{7+5}, \frac{7\left(\frac{12}{7}\right)+5(0)}{7+5}\right) = \left(\frac{12}{12}, \frac{12}{12}\right) = (1,1)$$

18. $\triangle ABC$ is a Right angled triangle at $C(0,0)$



$\therefore$ Circumcentre = midpoint of AB $= \left(\frac{1}{2},\frac{1}{2}\right)$.

19. If ' G ' is the centroid of $\triangle ABC$, then $AB^2 + BC^2 + AC^2 = 3(GA^2 + GB^2 + GC^2)$.

20. $D\left(0,\frac{1}{2}\right), E\left(\frac{1}{2},\frac{1}{2}\right), F\left(\frac{1}{2},0\right)$

Let D, E, F be the midpoints of $\overline{AB}, \overline{BC} \& \overline{CA}$

$A(1,0), B(0,0), (0,1) \Rightarrow a = BC = 1,\ b = CA = \sqrt{2}, C = AB = 1$

Incentre of $\triangle ABC$

$$= \left(\frac{1(1)+\sqrt{2}(0)+1(0)}{1+\sqrt{2}+1}, \frac{1(0)+\sqrt{2}(0)+(1)}{1+\sqrt{2}+1}\right)$$

$= \left(\frac{1}{\sqrt{2}(\sqrt{2}+1)}, \frac{1}{\sqrt{2}(\sqrt{2}+1)}\right) = \left(\frac{\sqrt{2}-1}{\sqrt{2}}, \frac{\sqrt{2}-1}{\sqrt{2}}\right) = \left(1-\frac{1}{\sqrt{2}}, 1-\frac{1}{\sqrt{2}}\right)$.

21. $OG : OS = 4 : 6 = 2 : 3$ (externally)

$$O = \left(\frac{2\left(\frac{5}{2}\right)-3\left(\frac{1}{3}\right)}{2-3}, \frac{2\left(\frac{5}{2}\right)-3\left(\frac{2}{3}\right)}{2-3}\right) = (-4,-3)$$

22. C is a point on the circle having AB as diameter

$\Rightarrow \angle ACB = 90° \Rightarrow$ Orthocentre = C. (3,2)

23. $A = (0,0), B(5,12), C = (16,12)$

$\Rightarrow a = 11,\ b = 20, c = 13$

Orthocentre of $\Delta I_1 I_2 I_3$ = Incentre of ΔABC

$= \left(\frac{11(0)+20(5)+13(16)}{11+20+13}, \frac{11(0)+20(12)+13(12)}{11+20+13}\right) = (7,9)$.

24. $A(5,-1), B(-2,3), C(-4,-7)$. Orthocentre $H = (0,0)$

Centroid $G = \left(\frac{5-2-4}{3}, \frac{-1+3-7}{3}\right) = \left(\frac{-1}{3}, \frac{-5}{3}\right)$

Nine point centre N divides HG in the ratio 3:1

$$N = \left(\frac{3\left(\frac{-1}{3}\right)+1(0)}{3+1}, \frac{3\left(\frac{-5}{3}\right)+(1)(0)}{3+1}\right) = \left(\frac{-1}{4}, \frac{-5}{4}\right)$$

25. Centroid $G = \left(\frac{a+b+c}{3}, \frac{a+b+c}{3}\right) = (0,0)$

$\Rightarrow a+b+c = 0 \Rightarrow a^3+b^3+c^3 = 3abc \Rightarrow a^2+b^2+c^2 = ab+bc+ca$.

26. [1] Conceptual

27. $\left(\frac{7+4+5}{3}, \frac{4-6+2}{3}\right) = (2,0)$.

28. $K = \frac{2+3+4}{3} \Rightarrow K = 3$.

29. Third vertex $C = 3G - A - B = 3(2,3) - (5,6) - (-1,4) = (2,-1)$.
30. Conceptual
31. Length of the diagonal $= \sqrt{(2+2)^2 + (3-1)^2} = \sqrt{4+4} = 2\sqrt{2}$

    Length of the side $= \frac{2\sqrt{2}}{\sqrt{2}} = 2$.
32. Conceptual.
33. $A(2,2), B(4,2), C = (2,4)$. Here $AB = 2, BC = 2\sqrt{2}, AC = 2$

    $BC^2 = AB^2 + AC^2 \Rightarrow \angle A = 90°$

    $\therefore$ Orthocentre $= (2,2)$, Circumcentre $= (3,3)$

    $\therefore OS = \sqrt{(5-1)^2 + (3-2)^2} = \sqrt{2} \Rightarrow a = 2$.
34. As H (Orthocentre), G (Centroid) and C (Circumcentre) are collinear, then

    $\begin{vmatrix} 4 & b & 1 \\ b & 2b-8 & 1 \\ -4 & 8 & 1 \end{vmatrix} = 0 \Rightarrow (8-b)(b-12) = 0$

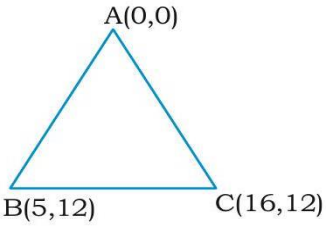
    Hence $b = 8$ or 12 which is wrong because collinearity does not explain Centroid, Orthocentre & Circumcentre

    Now $H, G\&C$ are collinear and $\frac{HG}{GC} = 2$ therefore,

    $\frac{-8+4}{3} = b$ (or) $b = \frac{-4}{3}$ and $\frac{16+b}{3} = 2\ b - 8$ or $b - 8$

    But no common value of b is possible.
35. a) Centroid is $\left(\frac{0+5+16}{3}, \frac{0+12+12}{3}\right) = (7,8)$

    b) Let the coordinates of circumcentre be $O(x,y)$ therefore $OA = OB = OC$



    $\Rightarrow x^2 + y^2 = (x-5)^2 + (y-12)^2 = (x-16)^2 + (y-12)^2$

    $x^2 + y^2 = (x-5)^2 + (y-12)^2, x^2 + y^2 = (x-16)^2 + (y-12)^2$

    $\Rightarrow 10x + 24y = 169$

    (or) $(x-5)^2 + (y-12)^2 = (x-16)^2 + (y-12)^2$

    $\Rightarrow 2x = 21 \Rightarrow x = \frac{21}{2}$ and $y = \frac{8}{3} \Rightarrow 0 = \left(\frac{21}{2}, \frac{8}{3}\right)$.

    c) $I = \left(\frac{0\times4+5\times20+16\times13}{13+20+11}, \frac{0\times11+12\times20+13\times12}{13+20+11}\right) = (7,9)$.

    d) $I_2 = \left(\frac{-5\times20+13\times16+11\times13}{-20+13+11}, \frac{-12\times20+0\times11+13\times12}{-20+13+11}\right) = (27,-21)$.

WORKSHEET – 3

Level – I

1)D	2)A	3)D	4)A	5)A	6)A	7)A
8)B						

Level – II

1)C	2)A	3)C	4)A	5)B	6)B	7)A
8)A	9)A	10)A	11)D	12)C	13)B	14)A

15)C 16)B 17)C 18)C 19)C 20)B 21)B

22)D 23)B 24)B,C 25)A 26)C 27)B 28)B

29)a-s;b-r;c-q;d-r 30)0 31)B,D 32) a-s; b-p; c-r; d-q 33)B,D

34)a-s;b-r;c-p;d-q 35)D

1. Given $A(1,2), B(2,3), C(x, 4)$ $40 = \frac{1}{2}\begin{vmatrix} -1 & -1 \\ (1-x) & (-2) \end{vmatrix}$

   $\Rightarrow |2 + 1 - x| = 80 \Rightarrow 3 - x = \pm 80 \Rightarrow x = -77,83$
2. $AB + BC = AC$ (or) $ar(\triangle ABC) = 0 \Rightarrow 3(-11 + k) + 4(-k + 8) + 5(3) = 0 \Rightarrow -k + 14 = 0$
3. Slope of AB = Slope of AC $\Rightarrow \frac{2k-2+2\,K}{-K+1-K} = \frac{6-2\,K-2+2\,K}{-4-K-K}$

   $\Rightarrow \frac{4\,K-2}{1-2\,K} = \frac{4}{-2(\,K+2)} \Rightarrow K = -1$
4. Given $A(1,1), B(3,4), C(5,-2), (4,-7)$

   Area $= \frac{1}{2}\begin{vmatrix} 1-5 & 3-4 \\ 1+2 & 4+7 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} -4 & -1 \\ +3 & 11 \end{vmatrix} = \frac{1}{2}|-44+3| = \frac{41}{2}$
5. Given $A(a,a), B(a+2,a), C(a+1,a+1)$

   Area $= \frac{1}{2}\begin{vmatrix} x_1 - x_2 & x_1 - x_3 \\ y_1 - y_2 & y_1 - y_3 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} a-a-2 & a-a-1 \\ a-a & a-a-1 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} -2 & -1 \\ 0 & -1 \end{vmatrix} = 1$
6. Given $A(-1,2), B(4,1), C(7,16)$.

   $D = (-1-4+7, 2-1+16) = (2,17)$

   area of ABCD $= \frac{1}{2}\begin{vmatrix} (-1-7) & (2-16) \\ (4-2) & (1-17) \end{vmatrix} = \frac{1}{2}\begin{vmatrix} -8 & -14 \\ 2 & -16 \end{vmatrix} = \frac{1}{2}|128+28| = 78$ sq.units
7. Given $A(2,-1), B(4,3), C(-1,2), D(-3,-2)$

   Area of ABCD $= \frac{1}{2}\begin{vmatrix} 2+1 & 4+3 \\ -1-2 & 3+2 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} 3 & 7 \\ -3 & 5 \end{vmatrix} = \frac{1}{2}(15+21) = \frac{36}{2} = 18$ sq.units
8. Given $A(-a,-b), B(0,0), C(a,b), D(a^2, ab)$

   Slope of $AB = \frac{b}{a}$; Slope of $AC = \frac{b}{a}$; Slope of $AD = \frac{ab+b}{a^2+a} = \frac{b}{a}$

   $\therefore A, B, C, D$ are collinear.
9. Given $A(2,1), B(8,5), C(x,7)$ are collinear

   Area $= \frac{1}{2}\begin{vmatrix} 2 & 1 & 1 \\ 8 & 5 & 1 \\ x & 7 & 1 \end{vmatrix} = 0 \Rightarrow 4x = 44 \Rightarrow x = 11$
10. Given $O(0,0), A(1,2), B(2,3)$. So, Area of $OAB = \frac{1}{2}|1 \cdot 3 - 2 \cdot 2| = \frac{1}{2}$ sq.units
11. Given $A(0,0), B(a^{x^2}, 0), C(0, a^{6x})$

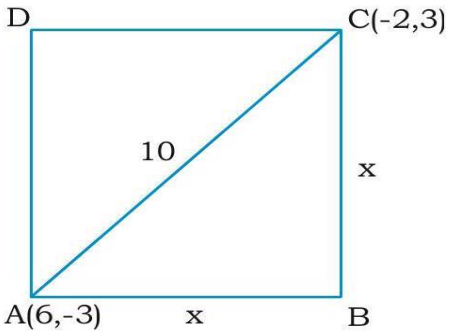
    Area of $\triangle ABC$ is $\frac{1}{2a^5} = \frac{1}{2}|a^{x^2} \times a^{6^x}| \Rightarrow a^{-5} = a^{x^2+6x}$

    $\Rightarrow x^2 + 6x + 5 = 0 \Rightarrow x = -1$ or $x = -5$
12. Area $= \frac{1}{2}\begin{vmatrix} 2a-a & 2a-a \\ a-a & a-2a \end{vmatrix} = 18 \Rightarrow \begin{vmatrix} a & a \\ 0 & -a \end{vmatrix} = 36 \Rightarrow a = \pm 6$

    Given triangle is right angled at (a,a)

    $\Rightarrow$ Circumcentre = Mid point of $(2a,a), (a,2a) \Rightarrow \left(\frac{3a}{2}, \frac{3a}{2}\right) = (\pm 9, \pm 9)$
13. Area $\frac{1}{2}\begin{vmatrix} x & 2x & 1 \\ -2 & 6 & 1 \\ 3 & 1 & 1 \end{vmatrix} = 5 \Rightarrow x = 2$ or $\frac{2}{3}$

14. Here $x\sqrt{2} = 10 \Rightarrow x = 5\sqrt{2}$. Area $\Rightarrow x^2 = 50$ sq.units



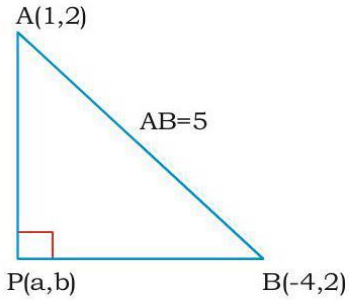
15. Area $= \frac{1}{2}\begin{vmatrix}(a+3-a+4) & (a-2-a-5)\\ (a+3-a) & (a-2-a)\end{vmatrix} = \frac{1}{2}\begin{vmatrix}7 & -7\\ 3 & -2\end{vmatrix} = \frac{1}{2}|-14+21| = \frac{7}{2}$ sq.units

16. Given $G(1,4), A(4,-3), B(-9,7)$

Area of $\Delta GAB = \frac{1}{2}\begin{vmatrix}-3 & 7\\ 10 & -3\end{vmatrix} = \frac{1}{2}|9-70| = \frac{61}{2}$

Required area $= 3 \times$ area of $\Delta GAB = 3 \times \frac{61}{2} = \frac{183}{2}$ sq.units

17. Here $25 = (a-1)^2 + (b-2)^2 + (a+4)^2 + (b-2)^2$



$2a^2 + 2b^2 + 6a - 8b = 0 \Rightarrow a^2 + b^2 + 3a - 4b = 0$

$\frac{1}{2}\begin{vmatrix}1 & 2 & 1\\ a & b & 1\\ -4 & 2 & 1\end{vmatrix} = \frac{25}{4} \Rightarrow 1(b-2) - 2(a+4) + 1(2a+4b) = \frac{25}{2}$

$\Rightarrow 5b - 10 = \frac{25}{2} \Rightarrow 2b - 4 = 5, -5 \Rightarrow b = \frac{9}{2}, \frac{-1}{2}$

If $b = \frac{9}{2}; (1) \Rightarrow a^2 + \frac{81}{4} + 3a - 4\left[\frac{9}{2}\right] = 0 \Rightarrow 4a^2 + 12a + 9 = 0 \Rightarrow (2a+3)^2 = 0 \Rightarrow a = \frac{-3}{2}$

If $b = \frac{-1}{2} \Rightarrow a^2 + \frac{1}{4} + 3a - 4\left[\frac{-1}{2}\right] = 0 \Rightarrow 4a^2 + 1 + 12a + 8 = 0 \Rightarrow (2a+3)^2 = 0 \Rightarrow a = \frac{-3}{2}$

$\therefore$ These are two points possible for P

18. Here $\Delta = 0 \Rightarrow \begin{vmatrix}a-0 & a-1\\ 0-b & 0-1\end{vmatrix} = 0 \Rightarrow \begin{vmatrix}a & a-1\\ -b & -1\end{vmatrix} = 0$

$\Rightarrow -a + ab - b = 0 \Rightarrow a + b = ab \Rightarrow \frac{1}{a} + \frac{1}{b} = 1$

19. Here Area of $\triangle ABC = \frac{1}{2}\begin{vmatrix}3 & -4\\ 7 & 2\end{vmatrix} = 12|6+28| = 17$ sq.units

Area of $\Delta GAB = \frac{17}{3}$ sq.units

20. Area $= 4(5) = 20$ sq.units

21. Given $A(6,3), B(-3,5), C(4,-2), D(x,3x)$

Area of $\triangle DBC = \frac{1}{2}\begin{vmatrix}x & 3x & 1\\ -3 & 5 & 1\\ 4 & -2 & 1\end{vmatrix} = \frac{1}{2}|28x - 14| = |14x - 7|$

Area of $|\Delta ABC| = \frac{1}{2}\begin{vmatrix} 6 & 3 & 1 \\ -3 & 5 & 1 \\ 4 & -2 & 1 \end{vmatrix} = \frac{49}{2} \Rightarrow \frac{|14x-7|}{\frac{49}{2}} = \frac{1}{2} \Rightarrow x = \frac{11}{8}, \frac{-3}{8}$

22. Area of $\triangle PBC$ = Area of $\triangle ABC$

$\Rightarrow \frac{1}{2}\begin{vmatrix} 3-4 & 5-2 \\ 3-\alpha & 5-\beta \end{vmatrix} = \frac{1}{2}\begin{vmatrix} 6-3 & 3-5 \\ 6-4 & 3-2 \end{vmatrix}$

$= \begin{vmatrix} -1 & 3 \\ 3-\alpha & 5-\beta \end{vmatrix}\begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix}$

$= |\beta - 5 + 3\alpha - 9||3 + 4| = |3\alpha + \beta - 14| : 7$

23. Area of $\triangle ABC$ = 2 area of $\triangle DCB$

$\Rightarrow \frac{1}{2}\begin{vmatrix} 9 & -2 \\ 2 & 5 \end{vmatrix} = 2 \times \frac{1}{2}\begin{vmatrix} -7 & 7 \\ -(X+3) & -(3x-5) \end{vmatrix}$

$\Rightarrow |45 + 4| = 2|21x - 35 + 7x + 21| \Rightarrow 2|28x - 14| = 49$

$\Rightarrow |8x - 4| = 7 \Rightarrow 8x = 4 \pm 7 \Rightarrow x = \frac{11}{8}$ or $\frac{-3}{8}$

24. Here ar($\triangle ABC$) = 2 · ar($\triangle BCD$)

$\Rightarrow \frac{1}{2}\begin{vmatrix} 6-3 & 46 \\ 35 & -23 \end{vmatrix} = 2\frac{1}{2}\begin{vmatrix} x & -3 & 4x \\ 3x+5 & -2 & 3x \end{vmatrix} \Rightarrow \frac{49}{2} = 2|14x - 7| \Rightarrow x = \frac{11}{8}(or)x = -\frac{3}{8}$

25. $\frac{1}{2}\begin{vmatrix} K+K & K+4+K \\ 2-2\,K+K & 2+2k-6+2\,K \end{vmatrix} = 0 \Rightarrow K = 2$

26. Given $A(3,-5), B(5,-1), C(2,1), D(-3,-2)$

or $ABCD = \frac{1}{2}\begin{vmatrix} 3-2 & 5+3 \\ -5-1 & -1+2 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} 1 & 8 \\ -6 & 1 \end{vmatrix} = \frac{1}{2}|48 + 1| = \frac{49}{2}$

27. Here $\Delta = \frac{1}{2}\begin{vmatrix} 1 & 7 & 7 & 12 & 0 & 1 \\ 1 & 21 & -3 & 2 & -3 & 1 \end{vmatrix} = \frac{1}{2}|21 - 7 - 21 - 147 + 14 + 36 - 36 - 0 + 0 + 3| = \frac{137}{2}$

28. Here $\Delta = \frac{1}{2}\begin{vmatrix} a & a & -a & a \\ b+c & b-c & c & b+c \end{vmatrix} = \frac{1}{2}|(ab - ac - ab - ac + ac + ab - ac - ab - ac - ac)|$

$= \frac{1}{2}|-4ac| = 2ac$

29. $a \to s; b \to r; c \to q; d \to r$

(a) $\Delta = \frac{1}{2}\begin{vmatrix} 3-4 & 3-5 \\ -8+11 & -8+14 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} -1 & -2 \\ 3 & 6 \end{vmatrix} = \frac{1}{2}|-6 + 6| = 0$

(b) $\Delta = \frac{1}{2}\begin{vmatrix} 1-4 & 1+4 \\ 2+3 & 2+1 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} -3 & 5 \\ 5 & 3 \end{vmatrix} = \frac{1}{2}|9 + 25| = 17$

(c) $\Delta = \frac{1}{2}\begin{vmatrix} 4-7 & 4+1 \\ 1-16 & 1+2 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} -3 & 5 \\ -15 & -1 \end{vmatrix} = \frac{1}{2}|+3 + 75| = \frac{78}{2} = 39$

(d) $\Delta = \frac{1}{2}\begin{vmatrix} -1-3 & -1-6 \\ -1-5 & -1-1 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} -4 & -7 \\ -6 & -2 \end{vmatrix} = \frac{1}{2}|8 - 42| = 17$

30. $\Delta = \frac{1}{2}\begin{vmatrix} -8+5 & -8+\frac{1}{2} \\ 8-6 & 8-3 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} -3 & -\frac{15}{2} \\ 2 & 5 \end{vmatrix} = \frac{1}{2}|-15 + 15| = 0$

31. Here $\frac{1}{2}\begin{vmatrix} P+1 & 2P+1 & 2P+2 & P+1 \\ 1 & 3 & 2P & 1 \end{vmatrix} = 0$

$\Rightarrow |3P + 3 - 2P - 1 + 4P^2 + 2P - 6P - 6 + 2P + 2 + 2P^2 = 2P| = 0$

$\Rightarrow 2P^2 - 3P - 2 = 0 \Rightarrow P = \frac{-1}{2}, P = 2$

32. (a) $r = \frac{\Delta}{s}$ and $R = \frac{5}{2} \Rightarrow r = \frac{6}{6} = 1, R = \frac{5}{2} \Rightarrow \frac{\Delta_1}{\Delta_2} = \frac{\lambda(1)^2}{\lambda\left(\frac{5}{2}\right)^2} = \frac{4}{25}$

(b) Area of $\triangle AGB = 5 \Rightarrow$ ar($\triangle ABC$) = 15

(c) area of $\triangle ABC = 20 \Rightarrow$ area of $\triangle PQR = \frac{1}{4}(\triangle ABC) = \frac{1}{4}(2Q) = 5$

(d) $A(K, 2-2K), B(-K+1, 2K), C(-4-K, 6-2K)$

$\Rightarrow$ Slope of AB = Slope of AC $\Rightarrow \frac{2K-2+2K}{-K+1-K} = \frac{6-2K-2+2K}{-4-K-K}$

$\Rightarrow \frac{4K-2}{-2K+1} = \frac{4}{-2K-4} \Rightarrow \frac{2K-1}{-2K+1} = \frac{1}{-2K-2} \Rightarrow -1 = -2K-2 \;\; K = \frac{1}{2}$

33. Here $\frac{1}{2}\begin{vmatrix} -5 & 3 & x & -5 \\ 0 & 0 & x-2 & 0 \end{vmatrix} = 20 \Rightarrow |0 + 3x - 6 - 0 + 0 + 5x - 10| = 40$

$\Rightarrow |8x - 16| = 40 \Rightarrow 8x - 16 = \pm 40 \Rightarrow x = 7$ or $x = -3 \;\; \therefore C(7,5)$ or $C = (-3, -5)$

34. $\Delta = \frac{1}{2}\begin{vmatrix} P+1 & 2P+1 & 2P+2 & P+1 \\ 1 & 3 & 2P & 1 \end{vmatrix} = \frac{1}{2}|2P^2 3P - 2|$

35. Let $\Delta \rightarrow$ be the area of $\triangle ABC, \Delta_1 \rightarrow$ be the area of $\triangle AGB$, $\Delta_2 \rightarrow$ be the area of ΔDEF

$\frac{\Delta_1}{\Delta_2} = \frac{\left(\frac{\Delta}{3}\right)}{\left(\frac{\Delta}{4}\right)} = \frac{4}{3} = \frac{K}{3} \Rightarrow K = 4$

WORKSHEET – 4

Level – I

1) B	2) B	3) C	4) B	5) D	6) B

Level - II

1) B	2) A	3) D	4) D	5) A	6) A	7) C
8) B	9) D	10) A	11) A	12) B	13) A	14) C
15) D	16) B	17) A	18) B	19) A	20) B	21) D
22) D	23) B	24) B	25) A	26) A	27) C	28) C
29) a-s; b-r; c-q; d-p		30) 0	31) a-s; b-p; c-q; d-r		32) C	33) 7
34) A	35) a-s; b-r; c-q; d-p					

1. Given point is $(-2,3)$ and distance $= 5$. Let $P(x, y)$ be the point
$(x+2)^2 + (y-3)^2 = 5^2 \Rightarrow x^2 + 4x + 4 + y^2 - 6y + 9 = 25 \Rightarrow x^2 + y^2 + 4x - 6y - 12 = 0$
2. Given point are $A(2,3)$ and $B(-2,5)$. Let P be the locus point (x, y)
$PA = PB \Rightarrow PA^2 = PB^2 \Rightarrow (x_1 - 2)^2 + (y_1 - 3)^2 = (x_1 + 2)^2 + (y_1 - 5)^2$
$8x_1 - 4y_1 + 16 = 0 \Rightarrow 2x_1 - y_1 + 4 = 0$. Equation of locus is $2x - y + 4 = 0$
3. Given point (3,4) and $(-4,3)$. Let the locus of point $P(x_1, y_1)$
It is collinear then area of $\triangle ABP = 0 \Rightarrow \frac{1}{2}\begin{vmatrix} x_1 - 3 & x_1 + 4 \\ y_1 - 4 & y_1 - 3 \end{vmatrix} = 0$
$\Rightarrow (x_1 - 3)(y_1 - 3) - (x_1 + 4)(y_1 - 4) = 0$
$\Rightarrow x_1 y_1 - 3x_1 - 3y_1 + 9 - x_1 y_1 + 4x_1 - 4y_1 + 16 = 0$
$\Rightarrow x_1 - 7y_1 + 25 = 0$. Equation of locus is $x - 7y + 25 = 0$.
4. $2x = 3y \Rightarrow 4x^2 = 9y^2 \Rightarrow 4x^2 - 9y^2 = 0$
5. $k = \frac{1}{2}\sqrt{h^2 + k^2}; \; h = \pm\sqrt{3}k \Rightarrow x^2 - 3y^2 = 0$
6. Conceptual
7. $x = \cos^2 t, y^2 = 4\sin^2 t \Rightarrow x + \frac{y^2}{4} = 1 \Rightarrow 4x + y^2 = 1$
8. Given $PA + PB = K$ and $K > AB \Rightarrow PA + PB > AB$

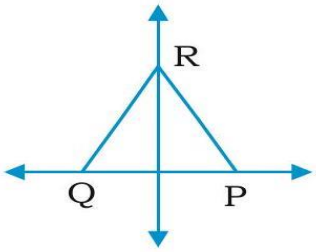
⇒ It satisfies the polygon of 3 sides ⇒ It represents an "Ellipse".

9. Given $A(4,0)$. Let the locus point be $P(x,y)$.
   $PA = 2|Y| \Rightarrow PA^2 = 4y^2 \Rightarrow (x-4)^4 + y^2 = 4y^2 \Rightarrow x^2 - 3y^2 - 8x + 16 = 0$
10. Let the locus point be $P(x_1 y)$
    $(\text{Distance of P from x-axis})^2 + (\text{Distance of P from y-axis})^2 = 25$
    $\Rightarrow |y|^2 + |x|^2 = 25 \Rightarrow x^2 + y^2 = 25$
11. Given point $A(-9,0) B(-1,0) \& PA:PB = 3:1$
    Let the locus of the point $P(x_1, y_1)$ then $\frac{PA}{PB} = \frac{3}{1} \Rightarrow PA^2 = 9PB^2$
    $\Rightarrow (x_1+9)^2 + (y_1-0)^2 = 9[(x_1+1)^2 + y^2] \Rightarrow x_1^2 + 18x_1 + 81 + y_1^2$
    $= 9x_1^2 + 18x_1 + 9 + 9y^2$
    $\Rightarrow 8x_1^2 + 8y_1^2 - 72 = 0 \Rightarrow x_1^2 + y_1^2 - 9 = 0 \Rightarrow x^2 + y^2 = 9$
12. Given $x = a + \lambda, y = b - \lambda$
    $x - a = \lambda^2 \rightarrow (1)$ and $y - b = -\lambda \Rightarrow \lambda = b - y \Rightarrow \lambda^2 = (b-y)^2 \rightarrow (2)$
    from (1)&(2) $x - a = (b-y)^2$
13. Given a, b, c are in $A.P \Rightarrow 2b = a + c$
    a, x, c are in $G \cdot P \Rightarrow x^2 = ab \Rightarrow a = \frac{x^2}{b}$; a, y, c are in $G \cdot P \Rightarrow y^2 = bc \Rightarrow c = \frac{y^2}{b}$
    $\Rightarrow 2b = \frac{x^2}{b} + \frac{y^2}{b} \Rightarrow x^2 + y^2 = 2b^2$
14. Given point (c, o) and $(-c, o)$
    Let P be (x,y) the locus point, then here $\sqrt{(x-c)^2 + y^2} + \sqrt{(x+c)^2 + y^2} = 4c^2$
    $\Rightarrow 2x^2 + 2y^2 = 2c^2 \Rightarrow x^2 + y^2 = c^2$
15. Given $A(2,-3) B(-2,1)$ are vertices of $\triangle ABC$
    Let c be the point $(x_1 y_1)$ then Centroid of $\triangle ABC = \left(\frac{2+(-2)+x_1}{3}, \frac{-3+1+y_1}{3}\right) = \left(\frac{x_1}{3}, \frac{y_1-2}{3}\right)$
    Here (x_1, y_1) satisfies $2x + 3y = 1 \Rightarrow 2\left(\frac{x_1}{3}\right) + 3\left(\frac{y_1-2}{3}\right) = 1 \Rightarrow 2x_1 + 3y_1 - 6 = 3$
    $\Rightarrow 2x_1 + 3y_1 = 9 \Rightarrow 2x + 3y - 9 = 0.$
16. Given $A(2,3)$ and $B(-2,3)$ and $|PA - PB| = 4$
    If $A(a,b)$ and $B(-a,b)$, then are locus of P such that-
    $|PA - PB| = K$ is $\frac{4x^2}{K^2} + \frac{4(y-b)^2}{k^2-4a^2} = 1$
    If $K = 4$ than $\frac{4x^2}{4^2} + \frac{4(y-3)^2}{4^2-4(2)^2} = 1$
    $\Rightarrow 4(y-3)^2 = 0 \Rightarrow y - 3 = 0$
17. Given perimeter of Δ'le is 14 cm and $a + b + c = 14$, then vertices are $(-3,0)$ and $(3,0)$ locus of another vertex $\frac{4x^2}{K^2} + \frac{4(y-b)^2}{K^2-4a^2} = 1$
    $\Rightarrow \frac{4x^2}{64} + \frac{4(y-b)^2}{64-4(3)^2} = 1 \Rightarrow \frac{x^2}{16} + \frac{y^2}{7} = 1.$
18. Given $A(0,4) B(0,4)$ and $|AP - PB| = 6$
    Let $A(a,b)$ and $B(a,-b)$ then are locus of P such that $|PA - PB| = K$
    is $\frac{4(x-a)^2}{K^2-4b^2} + \frac{4y^2}{K^2} = 1$
    Here $K = 0; \frac{4x^2}{36-4(4)^2} + \frac{4y^2}{6^2} = 1 \Rightarrow \frac{4x^2}{-28} + \frac{4y^2}{6^2} = 1 \Rightarrow 9x^2 - 7y^2 + 63 = 0$

19. Given $(\rho, v), (a\cos\theta, b\sin\theta), (b\cos\theta, a\sin\theta)$
Let the locus point be P(x, y).
Centroid $= \left(\frac{\rho+a\cos\theta+b\cos\theta}{3}, \frac{v+b\sin\theta+a\sin\theta}{3}\right) = \left(\frac{\rho(a+b)\cos\theta}{3}, \frac{v+(a+b)\sin\theta}{3}\right)$
$\Rightarrow \frac{\rho+(a+b)\cos\theta}{3} = x$ and $\frac{v+(a+b)\sin\theta}{3} = y \Rightarrow \cos\theta = \frac{3x-\rho}{a+b}, \sin\theta = \frac{3y-v}{a+b}$
Since $\cos^2\theta + \sin^2\theta = 1 \Rightarrow (3x-p)^2 + (3y-q)^2 = (a+b)^2$

20. Given $A = (a, 0) B(-a, 0)$ Let P(x, y) be the locus point
$\angle PAB - \angle PBA = \frac{\pi}{2}$, here $a^2 = x^2 - y^2 + 2xy\cot 2\alpha$
where $\alpha = \frac{\pi}{2}$ then $a^2 = x^2 - y^2 + 2xy - \cot$
Locus of P is $x^2 - y^2 - a^2 = 0$

21. Given $A(a, 0) B(-a, 0)$ and R or A variable point



Here $PR = \sqrt{(x-a)^2 + y^2}; RQ = \sqrt{(x+a)^2 + y^2}; PQ = 2a$
Here $\angle RPQ - \angle RQP = 2\alpha \Rightarrow \tan(P-Q) = \tan 2\alpha \Rightarrow \frac{\tan P - \tan Q}{1+\tan P \cdot \tan Q} = \tan 2\alpha$
from $\cos P = \frac{a-x}{\sqrt{(x-a)^2+y^2}}$, we have $\tan P = \frac{y}{a-x}$ and $\tan Q = \frac{y}{a+x}$
$$\Rightarrow \frac{\frac{y}{a-x} - \frac{y}{a+x}}{1 + \frac{y^2}{a^2-x^2}} = \tan 2\alpha \Rightarrow \frac{ay + xy - ay + xy}{a^2 - x^2 + y^2} = \tan 2\alpha \Rightarrow \frac{2xy}{a^2 - x^2 + y^2} = \frac{1}{(0 + 2\alpha)}$$
$\Rightarrow a^2 - x^2 + y^2 = 2xy\cot 2\alpha \Rightarrow a^2 = x^2 - y^2 + 2xy\cot 2\alpha$

22. Let $P(\alpha, \beta)$ be any point on the locus. by G.C
Let $A(\alpha, 0)$ and $B(o, b) \Rightarrow \frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{P^2} \rightarrow (1)$
Here P is the mid point A and B $\Rightarrow (\alpha, \beta) = \left(\frac{a}{2}, \frac{b}{2}\right)$
$\Rightarrow a = 2\alpha, b = 2\beta$; from (1) $\Rightarrow P^2(\alpha^2 + \beta^2) = \alpha^2\beta^2 \Rightarrow P^2(x^2 + y^2) = x^2y^2$.

23. Let $P(\alpha, \beta)$ be any point on the locus.
Here $\beta + m\alpha = \sqrt{a^2m^2 + b^2} \rightarrow (1)$ and $m\beta - \alpha = \sqrt{a^2 + b^2m^2} \rightarrow (2)$
$(1)^2 + (2)^2 \Rightarrow \alpha^2 + \beta^2 = a^2 + b^2 \Rightarrow x^2 + y^2 = a^2 + b^2$

24. Let the locus point be $P(x, y)$
Distance from P to x-axis is $2 \Rightarrow |x| = 2 \Rightarrow x^2 = 4 \Rightarrow x^2 - 4 = 0$

25. Given $A(a, 0) B(-\alpha, 0)$ Let $P(x, y)$ be a point on the Locus.
$\Rightarrow 2x^2 + 2y^2 + 2a^2 = 2c^2 \Rightarrow x^2 + y^2 + a^2 - c^2 = 0$

26. Given $A(2,3) B(-3,4)$. Let P(x, y) be a point on the Locus.
Here $\frac{1}{2}\begin{vmatrix} x-2 & x+3 \\ y-3 & y-4 \end{vmatrix} = \frac{17}{2}$.
Area for $\triangle PAB = 8.5 \Rightarrow \frac{1}{2}\begin{vmatrix} (2+3) & (3-y) \\ (2-x) & (3-4) \end{vmatrix} = \frac{17}{2} \Rightarrow \begin{vmatrix} 5 & 3-y \\ 2-x & -1 \end{vmatrix} = 17$
$\Rightarrow |15 - 5y + 2 - x| = 17 \Rightarrow (x + 5y - 17)^2 = 17^2 \Rightarrow x^2 + 10xy + 25y^2 - 34x - 170y = 0$

27. Let $P=(x,y)$ be a point on the Locus. Given that area of $\triangle POB = 2($ area of $\triangle POA)$
$\Rightarrow \frac{1}{2}\begin{vmatrix}2+4 & 3-4\\ 2-x & 3-y\end{vmatrix} = 12 \Rightarrow \begin{vmatrix}6 & 2\\ (2-x) & (3-y)\end{vmatrix} = 24 \Rightarrow |6(3-4)-(-2)(2-x)| = 24$
$\Rightarrow |22-6y-2y| = 24 \Rightarrow |-2(x+3y-11)| = 24 \Rightarrow |-(x+3y-11)| = 12$
$\Rightarrow (x+3y-11)^2 = 144 \Rightarrow x^2+6xy+9y^2-22x-66y-23=0$

28. Let $P=(x,y)$ be a point on the locus. Given that Area of $\triangle POB = 2$ (Area of $\triangle POA$)
$\Rightarrow \frac{1}{2}\begin{vmatrix}0-0 & 0-6\\ 0-x & 0-y\end{vmatrix} = 2\times\frac{1}{2}\begin{vmatrix}0-4 & 0-0\\ 0-x & 0-y\end{vmatrix} \Rightarrow \begin{vmatrix}0 & -6\\ -x & -y\end{vmatrix} = 2\begin{vmatrix}-4 & 0\\ -x & -y\end{vmatrix}$
$\Rightarrow |-6x| = 2|4y| \Rightarrow 36x^2 = 64y^2 \Rightarrow 9x^2 = 16y^2 \Rightarrow 9x^2-16y^2 = 0.$

29. a) Let the point on the Locus be $P(x,y)$.
Distance from P to x-axis is $5 \Rightarrow |y| = 5 \Rightarrow y^2 = 25 \Rightarrow y^2-25=0.$
b) Let the point on the Locus be $P(x,y)$
Distance from P to x-axis = Distance from P to y-axis
$\Rightarrow |y| = |x| \Rightarrow y^2 = x^2 \Rightarrow x^2-y^2=0$
c) Given $A(2,3)B(-2,5)$. Let the point on the Locus be $P(x,y)$
Here $PA = PB \Rightarrow PA^2 = PB^2$
$\Rightarrow (x-2)^2+(y-3)^2 = (x+2)^2+(y-5)^2 \Rightarrow -4x-6y+13 = 4x-10y+29$
$\Rightarrow 8x-4y+16=0 \Rightarrow 2x-y+4=0.$
d) Given $A(-3,4)B(3,4)$. Let the point on the Locus be (x,y)
Here $PA = PB \Rightarrow PA^2 = PB^2 \Rightarrow (x+3)^2+(y-4)^2 = (x-3)^2+(y-4)^2$
$\Rightarrow x^2+x+6x = x^2+9-6x \Rightarrow 12\text{x} = 0 \Rightarrow \text{x} = 0.$

30. $PA^2+PB^2 = 2PC^2$
$\Rightarrow (x-2)^2+(y-3)^2+(x-1)^2+(y-5)^2 = 2[(x+1)^2+(y-2)^2]$
$\Rightarrow x^2-4x+4+y^2-6y+9+x^2-2x+1+y^2-10y+25$
$= 2[x^2+2x+1+y^2-4y+4]$
$\Rightarrow 2x^2+2y^2-6x-16y+39 = 2x^2+2y^2+4x-8y+10$
$\Rightarrow -10x-8y+29=0 \Rightarrow 10x+8y-29=0$
$\Rightarrow a=10, b=8, c=-29 = 1+2a+b+c = 1+2(10)+8-29=0$

31. Conceptual

32. Let $p(x,y), \text{A}(a,0), B(-a,0)$ by the problem
$PA^2+PB^2 = 6a^2$
$\Rightarrow (x-a)^2+y^2+(x+a)^2+y^2 = 6a^2 = x^2+y^2 = 2a^2$

33. $\frac{a}{2}=h, \frac{b}{2}=k \Rightarrow a=2h, b=2k, AB=10$
$\Rightarrow \sqrt{a^2+b^2} = 10 \Rightarrow 4h^2+4k^2 = 100 \Rightarrow x^2+y^2 = 25 \Rightarrow k = 25$ sum of digits = 7.

34. Conceptual 35. Conceptual

WORKSHEET – 5
Level – I

1) C 2) A 3) D 4) A 5) B 6) D

Level – II

1) B	2) A	3) B	4) A	5) A	6) A	7) D
8) B	9) C	10) C	11) A	12) D	13) D	14) C
15) A	16) C	17) A	18) C	19) C	20) B	21) A
22) A	23) A	24) B, D	25) D	26) D	27) C	28) D
29) a-r; b-q; c-s; d-p		30) A	31) D	32) A, C	33) a-r; b-q; c-s; d-p	
34) A, B	35) a-s; b-s; c-q; d-p					

1. Here $P(h,k) = (x - X, y - Y) = (3 + 2, -2 - 1) = (5, -3)$.
2. Given Angle of Rotation $\theta = 45^\circ \Rightarrow \cos\theta = \cos 45^\circ = \frac{1}{\sqrt{2}}, \sin\theta = \sin 45^\circ = \frac{1}{\sqrt{2}}$.
   $P(X,Y) = (2\sqrt{2}, \sqrt{2}) \Rightarrow x = 2\sqrt{2}, y = \sqrt{2}$.
   We have $x = X\cos\theta - Y\sin\theta$ and $y = X\sin\theta + Y\cos\theta$
   $\Rightarrow x = \frac{2\sqrt{2}-\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{\sqrt{2}} = 1$ and $y = \frac{2\sqrt{2}+\sqrt{2}}{\sqrt{2}} = \frac{3\sqrt{2}}{\sqrt{2}} = 3 \therefore P(x,y) = (1,3)$.
3. Here $A(h,k) = (x - X, y - Y) = (3 - 6, 4 - 8) = (-3, -4)$.
4. Here $(X,Y) = (x - h, y - k) \Rightarrow (X,Y) = (-3,5)$.
5. Here $(X,Y) = (2,3); (h,k) = (-3,7)$
   We have $(x,y) = (X + h, Y + k)$.
6. Given $(x,y) = (2,0); (X,Y) = (0,4)$.
   New origin $= (x - X, y - Y)$.
7. Put $X = x + 1, Y = y - 1$
8. The point to which the origin has to be shifted to eliminate x and y terms in the equation $ax^2 + by^2 + 2gx + 2fy + c = 0$ is $\left(\frac{-g}{a}, \frac{-f}{b}\right)$.
   Hence $\left(\frac{-g}{a}, \frac{-f}{b}\right) = \left(\frac{-1}{1}, \frac{+2}{1}\right) = (-1,2)$
9. Given, new origin $= O' = (h,k) = (-5,3) \Rightarrow h = -5, k = 3$
   In the new system $P(x,y) = P(7,5) \Rightarrow X = 7, Y = 5$
   We have $x = X + h = 7 - 5 = 2$ and $y = Y + k = 5 + 3 = 8$.
   $\therefore P(x,y) = (2,8)$
10. To remove x, y terms in $x^2 + y^2 - 4x - 6y - 12 = 0$. New origin $O'(h', k')$.
    Comparing with Standard Form $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$
    Here $a = 1, b = 1, g = -2, f = -3, c = -12, h = 0$
    $\therefore O'(h', k') = \left(\frac{-g}{a}, \frac{-f}{b}\right) = (2,3)$.
    We have $x = X + h' \Rightarrow X = x - h' = 6 - 2 = 4$ and
    $y = Y + k' \Rightarrow Y = y - k' = 6 - 3 = 3 \therefore P(x,y) = (4,3)$.
11. The first degree terms are missing
    Hence, the new origin is $\left(\frac{-f}{h}, \frac{-g}{h}\right) = \left(\frac{-1}{\frac{1}{2}}, \frac{\frac{1}{2}}{\frac{1}{2}}\right) = (-2,1)$.
    The new equation is $xy - \frac{1}{2}(-2) + 1 - 6 = 0 \Rightarrow xy = 4 \therefore c = 4$
12. Given $xy = c^2$.
    Here $\theta = \begin{bmatrix} \frac{1}{2}\tan^{-1}\left(\frac{2h}{a-b}\right), \text{ if } a \neq b \\ (2n+1)\frac{\pi}{4}, n \in Z, \text{ if } a = b \end{bmatrix}$

$\therefore xy = c^2$ can be written as $0(x^2) + 1(xy) + 0(y^2) + 0(x) + 0(y) + c^2 = 0$
Comparing we have $h = 1, a = 0, b = 0, \Rightarrow a = b \ \therefore \theta = \frac{\pi}{4}$.

13. Given new origin $(h, k) = (1, -2)$. Given equation is $4x^2 + 9y^2 - 8x + 36y + 4 = 0$
We have $x = X + h = X + 1; y = Y + k = Y - 2$. Substituting in above equation
$4(x+1)^2 + 9(y-2)^2 - 8(x+1) + 36(y-2) + 4 = 0 \Rightarrow 4x^2 + 9y^2 - 36 = 0$

14. $\therefore$ The transformed equation is
$2(x-2)^2 + 4(x-2)(y+3) + 5(y+3)^2 - 4(x-2) - 22(y+3) + 7 = 0$
$\Rightarrow 2(x^2 - 4x + 4) + 4(xy + 3x - 2y - 6) + 5(y^2 + 6y + 9) - 4x + 8 - 22y - 66 + 7 = 0$
$\Rightarrow 2x^2 + 5y^2 - 8x + 8 + 4xy + 12x - 8y - 24 + 30y + 45 - 4x + 8 - 22y - 59 = 0$
$\Rightarrow 2x^2 + 5y^2 + 4xy - 22 = 0$

15. $x = X\cos\alpha - Y\sin\alpha, y = X\sin\alpha + Y\cos\alpha$

16. Given $\theta = 90°$, equation is $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$
Here $X = x\cos\theta + y\sin\theta \Rightarrow X = x(0) + y(1) \Rightarrow X = y$
and $Y = x\sin\theta + y\cos\theta \ \Rightarrow Y = -x(1) + y(0) \Rightarrow Y = -x$
Substituting X&Y values in equation (1), we get
$ay^2 - 2hxy + bx^2 + 2gy - 2fx + c = 0 \Rightarrow bx^2 - 2hxy + ay^2 + 2gy - 2fx + c = 0$

17. Given $\theta = 45°, \cos\theta = \frac{1}{\sqrt{2}}, \sin\theta = \frac{1}{\sqrt{2}} \cdot P(x, y) = (\sqrt{2}, \alpha)$
We have $x = X\cos\theta - Y\sin\theta; y = X\sin\theta + Y\cos\theta$
$\therefore x = \frac{X-Y}{\sqrt{2}} = \frac{\sqrt{2}-4}{\sqrt{2}} = 1 - 2\sqrt{2}$ and $y = \frac{X+Y}{\sqrt{2}} = \frac{\sqrt{2}+4}{\sqrt{2}} = 1 + 2\sqrt{2}$
$\therefore P = (1 - 2\sqrt{2}, 1 + 2\sqrt{2})$

18. Given $2x^2 + 4xy + \sqrt{y^2} - 4x - 22y + 7 = 0 \rightarrow (1)$
Here $a = 2, h = 2, b = 5, g = -2, f = -11, c = 7$
In order to eliminate the 1 st degree terms from the equation (1), we have to shift the origin to the point P.
$\therefore P = \left(\frac{hf-bg}{ab-h^2}, \frac{gh-af}{ab-h^2}\right) = \left(\frac{-12}{6}, \frac{18}{6}\right) = (-2, 3)$.

19. New coordinates are $(x + 2, y - 2)$.
The transformed equations is $2(x+2)^2 + (y-2)^2 - 8(x+2) + 4(y-2) + 1 = 1$
$\Rightarrow 2x^2 + 8x + 8 + y^2 - 4y + 4 - 8x - 16 + 4y - 8 + 1 = 0 \Rightarrow 2x^2 + y^2 - 11 = 0$.

20. Here $y^2 - 4y = 8x - 12 \Rightarrow y^2 - 4y + 4 = 8x - 8 \Rightarrow (y-2)^2 = 5(x-1)$
If the origin is shifted to (1,2), then the equation becomes $y^2 = 8x$
But the given equation is $y^2 = 4ax \Rightarrow a = 2$.

21. Let (x, y) be the original co-ordinates of P.
$x = \cos\alpha + 1$ and $y = \cos\beta + 1 \Rightarrow (x, y) = \left(2\cos^2\frac{\alpha}{2}, 2\cos^2\frac{\beta}{2}\right)$.
$(\because 1 + \cos 2\theta = 2\cos^2\theta)$

22. Given $x\cos\alpha + y\sin\alpha = p \ldots\ldots (1)$ changes to $X\cos\beta + Y\sin\beta = p$
We have $x = X\cos\theta - Y\sin\theta; \ y = X\sin\theta + Y\cos\theta$
Substituting these values in equation (1), we get
$\cos\alpha(X\cos\theta - Y\sin\theta) + \sin\alpha(X\sin\theta + Y\cos\theta) = P$
$\Rightarrow X(\cos\alpha \cdot \cos\theta + \sin\alpha \cdot \sin\theta) + Y(\sin\alpha\cos\theta - \cos\alpha \cdot \sin\theta) = P$
$\Rightarrow X\cos(\alpha - \theta) + Y\sin(\alpha - \theta) = p$
Substituting 2 and 3 equations which represent same equation we get $\cos\beta = \cos(\alpha - \theta)$.

23. We have $x = X\cos\theta - Y\sin\theta = \frac{X-\sqrt{3}Y}{2}$ and $y = X\sin\theta + Y\cos\theta = \frac{X\sqrt{3}+Y}{2}$
This point lies on $x + y = 1 \Rightarrow \left(\frac{1+\sqrt{3}}{2}\right)x + \left(\frac{1-\sqrt{3}}{2}\right)y = 1$
Comparing we have $p = \frac{2}{1+\sqrt{3}}, q = \frac{2}{1-\sqrt{3}} \Rightarrow \frac{1}{p^2} + \frac{1}{q^2} = 2$
24. Here $a = 9, b = 3, h = -\sqrt{3}$
Angle of Rotation $\theta = \frac{1}{2}\tan^{-1}\left(\frac{2h}{a-b}\right) = \frac{1}{2}\tan^{-1}\left(\frac{-2\sqrt{2}}{9-3}\right)$
$\theta = \frac{1}{2}\tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) = \frac{-\pi}{12}$. $\therefore \theta = \frac{n\pi}{2} - \frac{\pi}{12} = \frac{5\pi}{12}$ where $n = 1$.
25. $\theta = \frac{1}{2}\tan^{-1}\left(\frac{2h}{a-b}\right)$
26. Comparing the given equation with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$
We have, $a = 3, h = -2, b = -2, g = -3/2, f = -1$.
Required point is $\left(\frac{hg-bg}{ab-h^2}, \frac{gh-af}{ab-h^2}\right) = \left(\frac{2-3}{-6-4}, \frac{3+3}{-6-4}\right) = \left(\frac{1}{10}, \frac{-3}{5}\right)$
27. Here $a = 1, b = -1, h = 0, g = 1, f = 2$
Required point is $\left(\frac{hf-bg}{ab-h^2}, \frac{gh-af}{ab-h^2}\right) = \left(\frac{0+1}{-1-0}, \frac{0-2}{-1-0}\right) = (-1,2)$
28. Here $a = 1, b = 1, h = 0, g = -a, f = -2a$
Required point is $\left(\frac{hg-bg}{ab-h^2}, \frac{gh-af}{ab-h^2}\right) = \left(\frac{0+a}{+1+0}, \frac{0+2a}{+1+0}\right) = \left(\frac{a}{1}, \frac{2a}{1}\right) = (a, 2a)$
29. a -r; b-p; c-s; d-q
Let (X, Y) be the new coordinates of (x, y)
a) $(X, Y) = (x - h, y - k) = (7 - 1, 5 - 2) = (6,3)$
b) $(X, Y) = (-3 - 1, 1 - 2) = (-4, -1)$
c) $(X, Y) = (0 - 1, 5 - 2) = (-1,3)$
d) $(X, Y) = (-1 - 1, -2 - 2) = (-2, -4)$
30. Conceptual
31. Given $(h, k) = (1,2); (x, y) = (4,3), (X, Y) = ?$
Here $(X, Y) = (x - h, y - k) = (4 - 1, 3 - 2) = (3,1) = (p, q)$.
$p + q = 3 + 1 = 4$
32. Given $f(x, y) = x^2 - 4x - 4y + 16 = 0, \rightarrow x^2 - 4y = 0$
$f(X + h, Y + k) = X^2 + (2h - 4)X - 4Y = (h^2 - 4k - 4h + 16) = 0$
$\Rightarrow 2h - 4 = 0,\ h^2 - 4k - 4h + 16 = 0 \Rightarrow h = 2, k = 3.$
33. a-r, b-q, c-s, d-p
34. Given $x^2 - y^2 + 2x + 4y = 0$. Comparing with $ax^2 + by^2 + 2gx + 2fy + c = 0$, we have $a = 1, b = -1, g = 1, f = 2, c = 0$.
$O'(h, k) = \left(\frac{-g}{a}, \frac{-f}{b}\right) = \left(\frac{-1}{1}, \frac{-2}{-1}\right) = (-1,2)$.
Required transformed equation is $ax^2 + by^2 + \left(\frac{-g^2}{a} - \frac{f^2}{b} + c\right) = 0$
$\Rightarrow X^2 - Y^2 + (-1 + 4 + 0) = 0$
$\Rightarrow X^2 - Y^2 + 3 = 0$ is in the form of $mx^2 + ny^2 + k = 0$
$\Rightarrow m = 1, n = -1, k = 3$
35. a-s; b-s; c-q; d-p. Conceptual Formula $\theta = \frac{1}{2}\text{Tan}^{-1}\left(\frac{2h}{a-b}\right), (a \neq b)$

16. Binomial Theorem

WORKSHEET – 1

Level – I

1) B	2) B	3) C	4) A	5) C	6) A	7) C
8) B	9) B	10) D				

Level – II

1) B	2) C	3) A	4) D	5) A	6) A	7) C
8) B	9) A	10) A	11) D	12) C	13) B	14) D
15) C	16) B	17) C	18) B	19) A	20) A	21) A
22) A,C	23) B	24) A	25) C	26) a-r,b-p,c-s,d-q		27) 5
28) C,D	29) D	30) A	31) C	32) D	33) B	34) A
35) B						

1. General term $T_{r+1} = n_{c_r} \cdot x^{n-r} a^r$

$$T_4 = T_{3+1} = 7_{13} \cdot (2x)^{7-3} \cdot (3y)^3 = \frac{7 \times 6 \times 5}{6} \times 2^4 \cdot x^4 \cdot 3^3 \cdot y^3 = 35 \times 16 \times 27 x^4 y^3$$

$$= 15120 x^4 y^3$$

2. The coefficient of middle term in $(1-x)^{2n}$ is

$$t_{r+1} = n_{c_r} x^{n-r} \cdot y^r$$

Middle term $= T_{\frac{2n}{2}+1} = 2n_{c_n} \cdot (1)^{2n-n} \cdot (-x)^n = 2n_{c_n} \cdot (-1)^n \cdot x^n$

Coefficient of middle term $= 2n_{c_n} \cdot (-1)^n$

3. ${}^{n}C_r + n_{C_{r-1}} = {}^{n+1}C_r \Rightarrow {}^{18}C_5 + {}^{18}C_6 = {}^{19}C_6 = 19C_{13} \therefore n = 19.$

4. $\left(\mathrm{x} - \frac{1}{2\mathrm{x}}\right)^{10}$; middle term is $\mathrm{T}_{\frac{10}{2}+1} = \mathrm{T}_{5+1} = \mathrm{T} = 6$

$$\therefore \mathrm{T}_6 = \mathrm{T}_{5+1} = {}^{10}\mathrm{C}_5 \cdot (\mathrm{x})^{10-5} \cdot \left(\frac{-1}{2\mathrm{x}}\right)^5 = 10_{\mathrm{C}_5} \cdot \mathrm{x}^5 \cdot \frac{-1}{2^5 \cdot \mathrm{x}^5} = \frac{-63}{8}$$

5. $(2x-1)^{13}; 12$ th term $= T_{11+1} = {}^{13}C_{11}(2x)^{13-11}(-1)^{11} = {}^{-13}C_2(2x)^2.$

6. $\left(\frac{3x}{50} - \frac{5}{2x}\right) \Rightarrow p = 3, n = 8$

$T_{n-p+2} = T_{8-3+2} = T_7 = T_{6+1}$

n = 8 r = 6

$$T_{r+1} \Rightarrow T_{6+1} = 8_{c_6} \cdot \left(\frac{3x}{5}\right)^2 \cdot \left(\frac{-5}{2x}\right)^6$$

$$= 28 \cdot \frac{9 \cdot x^2}{25} \times \frac{5^6}{2^6} \cdot x^6 = \frac{39375}{16 \cdot x^4}$$

7. $10_{c_5} + 10_{c_4} + 10_{c_4} + 10_{c_3} = 11_{c_5} + 11_{c_4} = 12_{c_5}$

8. ${}^{15}C_{r+3} = {}^{15}C_{2r-3}$, then $15 = r + 3 + 2r - r \Rightarrow 3r = 15 = r = 5$

9. Middle term in the expansion $\left(2x - \frac{3}{x}\right)^{15}$ is

   If n = 15 (odd)

   $\left(\frac{n+1}{2}\right)^{\text{th}}$ term and $\left(\frac{4+3}{2}\right)^{\text{th}}$ terms are middle is

   $\left(\frac{15+1}{2}\right)$ and $\left(\frac{15+3}{2}\right)$ terms = 8 th and 9 th term

   $T_8 = T_{7+1} = {}^{15}C_7(2x)^8\left(\frac{-3}{x}\right)^7 = {}^{-15}C_7 2^8 3^7 \cdot x$

   $T_9 = T_{8+1} = {}^{15}C_8(2x)^7\left(\frac{-3}{x}\right)^8$

10. $2\mathrm{n}_{\mathrm{C}_{\mathrm{n}+1}} + 2 \cdot 2\mathrm{n}_{\mathrm{C}_{\mathrm{n}}} + 2\mathrm{n}_{\mathrm{C}_{\mathrm{n}-1}} = 2\mathrm{n}_{\mathrm{C}_{\mathrm{n}+1}} + 2\mathrm{n}_{\mathrm{C}_{\mathrm{n}}} + 2\mathrm{n}_{\mathrm{C}_{\mathrm{n}}} + 2\mathrm{n}_{\mathrm{C}_{\mathrm{n}-1}} = 2\mathrm{n} + 1_{\mathrm{C}_{\mathrm{n}+1}} + 2\mathrm{n} + 1_{\mathrm{C}_{\mathrm{n}}} = 2\mathrm{n} + 2_{\mathrm{C}_{\mathrm{n}+1}}$

11. $(1 + x)^{40} \Rightarrow n + 1 = 41$

12. $\left(\mathrm{x} + \frac{2}{3}\mathrm{x}^2\right)^7 \Rightarrow \mathrm{T}_3 = \mathrm{T}_{2+1} = {}^7\mathrm{C}_2 \cdot \mathrm{x}^{7-2} \cdot \left(\frac{2}{3}\mathrm{x}^2\right)^2 = {}^7\mathrm{C}_2 \cdot \mathrm{x}^5 \cdot \frac{4}{9} \cdot \mathrm{x}^4 = {}^7\mathrm{C}_2 \cdot \frac{4}{9} \cdot \mathrm{x}^9$

   $\mathrm{T}_4 = \mathrm{T}_{3+1} = {}^7\mathrm{C}_3 \cdot \mathrm{x}^{7-3} \cdot \left(\frac{2}{3}\mathrm{x}^2\right)^3 = {}^7\mathrm{C}_3 \cdot \mathrm{x}^4 \cdot \frac{8}{27} \cdot \mathrm{x}^6 = {}^7\mathrm{C}_3 \cdot \frac{8}{27} \cdot \mathrm{x}^{10}$

   $\frac{\mathrm{T}_3}{\mathrm{T}_4} = \frac{{}^7\mathrm{C}_2 \cdot \frac{4}{9} \cdot \mathrm{x}^9}{{}^7\mathrm{C}_3 \cdot \frac{8}{27} \cdot \mathrm{x}^{10}} = \frac{\frac{7!}{\mathrm{s!}\,2!} \times \frac{4}{9} \cdot \mathrm{x}^9}{\frac{7!}{4!\,3!} \times \frac{8}{27} \cdot \mathrm{x}^{10}} = \frac{4!\,3!}{5!\,2!} \times \frac{27}{8} \times \frac{4}{9} \times \frac{1}{\mathrm{x}} = \frac{3}{5} \cdot \frac{3}{2} \cdot \frac{1}{\mathrm{x}} = \frac{9}{10\mathrm{x}} = \frac{0.9}{\mathrm{x}}$

13. ${}^{28}C_r$: ${}^{24}C_{r-4} = 225:11$

   $\frac{\frac{28!}{(28-r)!\,r!}}{\frac{24!}{(24-r+4)!\,(r-4)!}} = \frac{225}{11}$

   $\frac{28 \times 27 \times 26 \times 25 \times 24!}{r(r-1)(r-2)(r-3)(r-4)!} \times \frac{(r-4)!}{24!} = \frac{225}{11}$

   $\therefore r = 14$

14. 5th term of $\left(2x^2 + \frac{3}{x}\right)^5 = 10 \Rightarrow {}^5C_4(2x^2)^{5-4}\left(\frac{3}{x}\right)^4 = 10 \Rightarrow 5 \cdot (2x^2)\frac{3^4}{x^4x^2} = 10; \mathrm{x} = 9$

15. $x^6 - 6_{c_1}x^5 + 6_{c_2}x^4 - 6_{c_3}x^3 + 6_{c_4} - x^2 - 6_{c_5}c + 6_{c_6}x^0$

   $(x-1)^6 = x^6 - 6x^5 + 15x^4 - 20x^3 + 15x^2 - 6x + 1$
   $(x+1)^6 = x^6 - 6x^5 + 15x^4 - 20x^3 + 15x^2 + 6x + 1$
   $(x-1)^6 + (x+1)^6 = 2[x^6 + 15x^4 + 15x^2 + 1]$
   $2x^6 + 30x^4 + 30x^2 + 2$

16. If ${}^nP_r = 840$, ${}^nC_r = 35$ then

$$r! = \frac{{}^{n}P_r}{{}^{n}C_r}\frac{840}{35} = 24 = 4 \times 3 \times 2 \times 1 = 4! \Rightarrow \frac{n!}{(n-4)!} = 840$$
$$\Rightarrow n(n-1)(n-2)(n-3) = 840 = 7 \times 6 \times 5 \times 4$$

17. $\left(\frac{q}{2x} - px\right)^8$

$$t_5 = 1120,\ t_{4+1} = 1120$$
$$t_{4+1} = 8_{c_4} \cdot \left(\frac{q}{2x}\right)^4 \cdot (-px)^4 = 1120$$
$$= \frac{8 \times 7 \times 6 \times 5}{1 \times 2 \times 3 \times 4} \cdot \frac{q^4}{2^4} \cdot p^4 \cdot x^4 = 1120$$
$$= 70 \times p^4 \cdot q^4 = 1120 \times 2$$
$$p^4q^4 = 2^4 \cdot 2^4 \Rightarrow (pq)^4 = 4 \Rightarrow \text{pq} = 4$$

clearly, p = 4 and q = 1(since p + q = 5) and $p > q > 0$,

18. $n_{C_r} = 1, n_{c_1} \cdot ax = 8x \Rightarrow na = 8$

and $n_{c_2} \cdot a^2x^2 = 24x^2 \Rightarrow \frac{n(n-1)}{2}a^2 = 24 \Rightarrow n(n-1)a^2 = 48$, on solving n = 4 and a = 2.

19. $\frac{7^{\text{th}}\text{ term from the beginning}}{7^{\text{th}}\text{ term from the end}} = \frac{1}{6} \Rightarrow \frac{n_{c_6} \cdot \left(2^{1/3}\right)^{n-6} \cdot \left(\frac{1}{3^{1/3}}\right)^6}{n_{c_{n-6}} \cdot \left(2^{1/3}\right)^5 \left(\frac{1}{3^{1/3}}\right)^{n-6}} = \frac{1}{6}$ $n = 9$ will satisfy the equation.

$$\text{n}_{\text{C}_\text{r}} + 4\left(\text{n}_{\text{C}_{\text{r}-1}}\right) + 6\left(\text{n}_{\text{C}_{\text{r}-2}}\right) + 4\left(\text{n}_{\text{C}_{\text{r}-3}}\right) + \text{n}_{\text{C}_{\text{r}-4}}$$
$$= \text{n}_{\text{C}_\text{r}} + \text{n}_{\text{C}_{\text{r}-1}} + 3 \cdot \text{n}_{\text{C}_{\text{r}-1}} + 3 \cdot \text{n}_{\text{C}_{\text{r}-2}} + 3 \cdot \text{n}_{\text{C}_{\text{r}-2}} + 3\text{n}_{\text{C}_{\text{r}-3}} + \text{n}_{\text{C}_{\text{r}-3}} + \text{n}_{\text{C}_{\text{r}-4}}$$
$$= (\text{n}+1)_{\text{C}_\text{r}} + 3(\text{n}+1)_{\text{C}_{\text{r}-1}} + 3(\text{n}+1)_{\text{C}_{\text{r}-2}} + (\text{n}+1)_{\text{C}_{\text{r}-3}}$$
$$= (\text{n}+1)_{\text{C}_\text{r}} + (\text{n}+1)_{\text{C}_{\text{r}-1}} + 2(\text{n}+1)_{\text{C}_{\text{r}-1}} + 2 \cdot (\text{n}+1)_{\text{C}_{\text{r}-2}} + (\text{n}+1)_{\text{C}_{\text{r}-2}} + (\text{n}+1)_{\text{C}_{\text{r}-3}}$$
$$= (\text{n}+2)_{\text{C}_\text{r}} + 2(\text{n}+2)_{\text{C}_{\text{r}-1}} + (\text{n}+2)_{\text{C}_{\text{r}-2}} = (\text{n}+2)_{\text{C}_\text{r}} + (\text{n}+2)_{\text{C}_{\text{r}-1}} + (\text{n}+2)_{\text{C}_{\text{r}-1}} + (\text{n}+2)_{\text{C}_{\text{r}-2}}$$
$$= (\text{n}+3)_{\text{C}_\text{r}} + (\text{n}+3)_{\text{C}_{\text{r}-1}} = (\text{n}+4)_{\text{C}_\text{r}}$$

21. If the 5 th term is for terms the 4 th term and 4 th term is 6 times the 3rd term is $(1 - ex)^n$

$$T_5 = 4T_4 \qquad T_4 = 6 \cdot T_3$$
$$n_{C_4} \cdot x^4 = 4 \cdot n_{C_3} \cdot x^3 \qquad n_{C_3} \cdot x^3 = 6 \cdot n_{C_2} \cdot x^2$$
$$\frac{n_{C_4}}{n_{C_3}}x = 4 \qquad \frac{n_{C_3}}{n_{C2}}x = 6$$
$$\frac{n-4+1}{4}x = 4 \qquad \frac{n-3+1}{3}x = 6$$
$$\Rightarrow \frac{(n-3)x}{(n-2)x} = \frac{16}{18}$$
$$\Rightarrow 18n - 54 = 16n - 32$$
$$\Rightarrow 2n = 22 \Rightarrow n = 11\& \qquad x = 2.$$

22. ${}^{n}P_r = 1320\ {}^{n}C_r = 220$

$$r! = \frac{{}^{n}P_r}{{}^{n}C_r} = \frac{1320}{220} = 6$$

$\frac{{}^{n}P_r}{{}^{n}C_r} = r! \Rightarrow r! = 6 = 3 \times 2 \times 1 = 3!$

r = 3 and $n_{p_3} = 1320 \Rightarrow n = 12$

23 [B]

24. [A]

25. [C]

Sol: $(m - n)^7$

The total number of terms in the expansion = 8

The 1st term = $m^{7;}$ The 2nd term = $-7\ m^6n$

The 3rd term = $\frac{7\times6}{2}m^5n^2 = 21m^5n^2$; The 4th term = $-\frac{21\times5}{3}m^4n^3 = -35m^4n^3$

26. a) Middle term $T_6 = -10_{c_5}\left(\frac{6}{7}\right)^5 x^5y^5$

b) Middle term $T_5 = 8_{c_4}(14)^4(pq)^4$

c) Middle terms T_aT_{10}

$T_{10} = 17_{c_4}4^85^9 \times 43$

d) Middle term $T_{11} = 20_{c_{10}}(15)^{10}a^{10}$

27. The number of Terms are, if n is odd $\left[\frac{n+1}{2}\right] = \frac{9+1}{2} = 5$

28. $2\ {}^{n}C_5 = {}^{n}C_4 + {}^{n}C_6 \Rightarrow 2 = \frac{{}^{n}C_4}{{}^{n}C_5} + \frac{{}^{n}C_6}{{}^{n}C_5}$

$$2 = \frac{5}{n-5+1} + \frac{n-6+1}{6}$$

$$2 = \frac{5}{n-4} + \frac{n-5}{6}$$

Solutions of the above equation are 7 or 14

29. $n = 100$ which is even

The number of terms = $\frac{n}{2} = 50$

30. $\left[\left(x+\frac{1}{x}\right)^2 - \left(x-\frac{1}{x}\right)^2\right]^{108} = 4^{108}$; Which is only one term

31. n = 49, which is odd

The number of terms = $\frac{n+1}{2} = \frac{50}{2} = 25$

32. $(1+ax)^n = 1 + {}^{n}C_1 \cdot ax + {}^{n}C_2 \cdot (ax)^2 +$

n.ax = 8x ⇒ na = 8

$$\frac{n(n-1)}{2}a^2x^2 = 24x^2 \Rightarrow \frac{na(n-1)a}{2} = 24$$

By solving n = 4 and a = 2

33.

$(x+a)^n = x^n + {}^nC_1x^{n-1}a + {}^nC_2x^{n-2} + {}^nC_3x^{n-3}a^3 + \cdots . + {}^nC_na^n$

$= (x^n + {}^nC_2x^{n-2}a^2 + \cdots + ({}^nC_1x^{n-1}a + {}^nC_3x^{n-3}a^3 + \cdots .) = P + Q$

and $(x-a)^n = x^n - {}^nC_1x^{n-1}a + {}^nC_2x^{n-2}a^2 - {}^nC_3x^{n-3}a^3 + \cdots \ldots . + (-1)^n\,{}^nC_na^n$

$= (x^n + {}^nC_2x^{n-2}a^2 + \cdots \ldots)({}^nC_1x^{n-1}a + {}^nC_3x^{n-3}a^3 + \cdots \ldots) = P - Q.$

$P^2 - Q^2 = (P+Q)(P-Q) = (x+a)^n(x-a)^n = (x^2-a^2)^n$

34. $PQ = (P+Q)^2 - (P-Q)^2 = (x+a)^{2n} - (x-a)^{2n}$

35. $p^2 + q^2 = (p+q)^2 - 2pq$

$p^2 + q^2 = (x+a)^{2n} - \left(\dfrac{(x+a)^{2n} - (x-a)^{2n}}{2}\right) = \dfrac{(x+a)^{2n} + (x-a)^{2n}}{2}$

WORKSHEET – 2

Level – I

1.	B	3.	C	5.	B	7.	B	9.	A
2.	C	4.	C	6.	B	8.	A	10.	D

Level – II

1) C	2) A	3) C	4) D	5) A	6) C	7) C
8) A	9) D	10) D	11) C	12) B	13)B	14) D
15) B	16) C	17) C	18) B	19) A	20) A	21) B
22) A,B	23) A	24) B	25) C	26) D	27) B,C,D	28) 7
29) D	30) A	31)A	32) C,D	33) A		

1. $r = \frac{(n+1)1\times 1}{1\times 1+1}$

$r = \frac{(15)\left|\frac{2x}{3}\right|}{\left|\frac{2x}{3}\right|+1}$ when $x = 4/5$

$r = \frac{(15)\left(\frac{8}{15}\right)}{\frac{8}{15}+1} = \frac{(15)(8)}{23} = \frac{120}{23} \Rightarrow T_{[r]+1} = T_{5+1} = T_6$ is N.G.T.

2. Use $r = \frac{np}{p+q}, n = 9, p = 2, q = 1$

3. We have, $r = \frac{np}{p+q} = \frac{24}{4} = 6$

$\therefore T_7 = 8_{c_6}(x^3)^2\left(\frac{-1}{x}\right)^6 = 8_{c_6} = 28.$

4. $\left(2x - \frac{3}{x}\right)^8$

$T_{r+1} = n_{c_r}(x)^{n-r}(y)^2 = 8_{c_r}(2x)^{8-r}\left(\dfrac{-3}{x}\right)^r$

$= 8_{c_r}(2x^{8-r})x^{8-2r}(-3)^r \cdot (x^{-r}) = 8_{c_r}(2x^{8-r})(-3)^r x^{8-2r}$

But, we require the coefficient of x^{-4}

$\therefore 8-2r=-4$

$\Rightarrow -2r=-12 \Rightarrow r=6$

(or) Use the formula, $r=\frac{np-k}{p+q}$. Where, $n=8, p=q=1 \& k=-4$.

$\therefore$ The coefficient of $x^{-4} = 8_{c_6}(2)^{8-6}(-3)^6$

5. In $(3+kx)^9$, $T_{r+1} = {}^9C_r \cdot 3^{9-r} \cdot (kx)^r = {}^9C_r 3^{9-r} k^r x^r$

x^2 coefficient $= {}^9C_2 3^7 k^2$

x^3 coefficient $= {}^9C_3 3^6 k^3$

Given ${}^9C_2 3^7 k^2 = {}^9C_3 3^6 k^3$

$k=\frac{9}{7}$

6. $(2+3x)^9 = (2)^9\left(1+\frac{3x}{2}\right)^9$

$r=6$

$T_7 = {}^9C_6(2)^3(3x)^6 = {}^9C_3(2)^3\left(3\cdot\frac{3}{2}\right)^6 = \frac{7.3^{13}}{2}$

7. $r=\frac{(n+1)|x|}{|x|+1}=\frac{(11+1)\left|\frac{3}{2},\frac{3}{4},\frac{3}{2}\right|}{\left|\frac{3}{2},\frac{3}{4},\frac{3}{2}\right|+1}=\frac{12\left(\frac{3}{4}\right)}{\left(\frac{7}{4}\right)}=\frac{36}{7}$; $r=5$

$T_6 = {}^{11}C_5(3x)^6(2y)^5 = {}^{11}C_5\left(3\cdot\frac{2}{3}\right)^6\left(2\cdot\frac{3}{4}\right)^5 = {}^{11}C_5 \times 486$

8. Largest Binomial Co-efficient is 5_{c_2} or 5_{c_3}

9. Use the formula $\frac{(n+1)|x|}{1+|x|}$

10. In the expansion of $\left(y+\frac{c^3}{y^2}\right)^{10}$, the rth term is given by

$T_r = {}^{10}C_{r-1}y^{10-r+1}\left(\frac{c^3}{y^2}\right)^{r-1} = {}^{10}C_{r-1}y^{11-r}\frac{c^{3r-3}}{y^{2r-2}} = {}^{10}C_{r-1}y^{11-r-2r+2}c^{3r-3}$

Thus, $T_r = {}^{10}C_{r-1}y^{13-3r}c^{3r-3}$

Let the rth term contains y^{-2} $\therefore 13-3r=-2 \Rightarrow r=\frac{15}{3}=5$.

Therefore, 5th term will contain y^{-2}

From (1), $T_5 = {}^{10}C_{5-1}c^{15-3}y^{-2} = {}^{10}C_4c^{12}y^{-2}$

$\therefore$ Coefficient of $y^{-2} = {}^{10}C_4c^{12} = \frac{10!c^{12}}{4!(10-4)!} = \frac{10!c^{12}}{4!6!} = \frac{10\times9\times8\times7}{4\times3\times2\times1}\times c^{12} = 210c^{12}$

11. In $\left(2+\frac{x}{3}\right)^n$, coefficient of x^7 = coefficient of x^8

${}^nC_7 2^{n-7}3^{-7} = {}^nC_8 2^{n-8}3^{-8} \Rightarrow \frac{{}^nC_8}{{}^nC_7}=6 \Rightarrow \frac{n-7}{8}=6 \Rightarrow n=55$

12. In $\left(\sqrt{\frac{x}{3}}+\frac{3}{2x^2}\right)^{10}$, $T_{r+1} = {}^{10}C_r\left(\frac{x}{3}\right)^{\frac{10-r}{2}}\left(\frac{3}{2x^2}\right)^r = T_{r+1} = {}^{10}C_r\left(\frac{1}{3}\right)^{\frac{10-r}{2}}\left(\frac{3}{2}\right)^r x^{\frac{10-2}{2}-2r}$

    To get the independent term $\frac{10-2}{2} - 2r = 0 \Rightarrow r = 2$

13. T_{r+1} is the term independent of x is $\left(ax+\frac{6}{x}\right)^{14}$

    Then $r = \frac{np}{p+q} = \frac{14.1}{1.1} = \frac{14}{2} = 7$

14. In $(1-x^2)^{10}, T_{r+1} = 10_{c_r}(-x^2)^r$. But, we need coefficient of x^{10}. so, r = 5.

    $\therefore$ Coefficient of $x^{10} = -{}^{10}C_5$.

    To get independent term in $(x-2/x)^{10}$,

    $r = \frac{np}{p+r} = \frac{10\cdot(1)}{1+1} = 5.$

    $\therefore T_6 = 10_{c_5}\cdot x^5\left(\frac{-2}{x}\right)^5 = -10_{c_5}2^5$ is the required independent term.

    Now, the required ratio is $-10_{c_5}: -10_{c_5}(2)^5 = 1:2^5 = 1:32.$

15. Co-efficient of $x^{17} = -(1+2+3+\cdots.+18) = -171$

16. $\sum_{m=0}^{100} {}^{100}C_m(x-3)^{100-m}\cdot 2^m = (1-x)^{100}$. Coefficient of x^{53} is ${}^{100}c_{53}(-1)^{53}$

17. In $\left(ax^2-\frac{1}{bx}\right)^{11}$, $T_{r+1} = {}^{11}C_r(ax^2)^{11-r}\left(\frac{-1}{bx}\right)^r$ to get the coefficient of x^7, $22-3r = 7$

    $\Rightarrow r = 5 \Rightarrow T_6 = {}^{11}C_5 a^6\left(-\frac{1}{b}\right)^5 x^7$

    In $\left(ax-\frac{1}{bx^2}\right)^{11}$, $T_{r+1} = {}^{11}C_r(ax)^{11-r}\left(\frac{-1}{bx^2}\right)^r$ to get the coefficient of $x^{-7}, 11-3r = -7$

    $\Rightarrow r = 6 \Rightarrow T_7 = {}^{11}C_6 a^5\left(-\frac{1}{b}\right)^6 x^{-7}$

    Given that $-{}^{11}C_5\frac{a^6}{b^5} = {}^{11}C_6\frac{a^5}{b^6} \Rightarrow ab = -1$

18. $\left(x^2+\frac{1}{x}\right)^{2n} \Rightarrow r = \frac{(2n\times 2)-p}{2+1} = \frac{4n-p}{3}$ $\therefore$ coefficient of x^p in ${}^{2n}C^3_{\underline{4n-p}}$

19. We have, $\left(\frac{x+1}{x^{2/3}-x^{1/3}+1}-\frac{x-1}{x-x^{1/2}}\right)^{10}$

    $= \left(\frac{\left(x^{1/3}\right)^3+1^3}{x^{2/3}-x^{1/3}+1}-\frac{\left(x^{1/2}\right)^2-1^2}{x^{1/2}(x^{1/2}-1)}\right)^{10} = \left[x^{\frac{1}{3}}+1-\left(1+x^{-\frac{1}{2}}\right)\right]^{10} = \left(x^{1/3}-x^{-1/2}\right)^{10}$

We have, $T_{r+1} = {}^{10}C_r\left(x^{1/3}\right)^{10-r}\left(-x^{-1/2}\right)^r \ldots..(1) = {}^{10}C_r x^{\frac{10-r}{3}-\frac{r}{2}}\cdot(-1)^r$

It will be independent of x, if $\frac{10-r}{3}-\frac{r}{2} = 0 \Rightarrow 20-2r-3r = 0 \Rightarrow r = 4$

Putting r = 4 in (1), we get $T_5 = {}^{10}C_4(-1)^4 = {}^{10}C_4 = 210$

20. Coefficient of x^5 in $(1+x)^{10}\left(1+\frac{1}{x}\right)^{20}$ = Coefficient of x^{25} in $(1+x)^{30}$

21. $\left(x+\frac{1}{x^2}\right)^{n-3}$

here the coefficient of x^{2r} is not zero $\frac{(n-3)-2r}{3} \neq 0 \Rightarrow \frac{n-2r}{3}-1 \neq 0 \Rightarrow \frac{n-2r}{3} \neq 1 \Rightarrow \frac{n-2r}{3}$ should be a positive integer.

22. $(3+2x)^{50} = 3^{50}\left(1+\frac{2}{3}x\right)^{50}$

$$m = \frac{(n+1)|x|}{|x|+1} = \frac{(51)\left|\frac{2}{15}\right|}{\left|\frac{2}{15}\right|+1} = \frac{51\times 2}{2+15} = 6$$

$\therefore 6^{th}$ and 7th terms are numerically greatest terms.

23. Largest binomial coefficient in $(1+x)^{19} = {}^{19}C_{\frac{19-1}{2}}$ or ${}^{19}C_{\frac{19+1}{2}} = {}^{19}C_9$ or ${}^{19}C_{10}$.

24. Largest binomial coefficient in $(1+x)^{24} = {}^{24}C_{12}$ as n = 24 (even).

25. Largest binomial coefficient in $(1+x)^{22}$ is ${}^{22}C_{11}$.

so r = 11 Now $13_{C_{11}} = \frac{13!}{(13-11)!11!} = \frac{13\times 12}{2} = 13\times 6 = 78$.

26. Ans: 4th term.

$$r = \frac{(n+1)|X|}{|X|+1} = \frac{6\left|\frac{2x}{3}\right|}{\left|\frac{2x}{3}\right|+1} = \frac{24}{7}$$

Integral part of r is 3 .

$\therefore$ Numerically greatest term is T_4.

27. Co-efficient of x^2 is ' 1 '

28. $T_{r+1} = {}^9C_r(3x^2)^{9-r}\left(\frac{-1}{3x}\right)^r = {}^9C_r(3)^{9-r}\left(\frac{-1}{3}\right)^r x^{18-2r-r}$

To get the independent term x coefficient is zero. i.e. $18-3r=0 \Rightarrow r=6$

29. Co-efficient of ' x ' in $\left(x^2+\frac{k}{x}\right)^5$ is $270\ {}^5C_3\cdot(1)^{5-3}\cdot(k)^3 = 270$

$$\Rightarrow \frac{5!}{(5-3)!\,31}\cdot k^3 = 270$$

$$\Rightarrow \frac{120}{2\times 6}\cdot k^3 = 270$$

$$\Rightarrow \frac{120}{12}k^3 = 270$$

$$\Rightarrow k^3 = \frac{270}{10}$$

$$\Rightarrow k^3 = 27 \Rightarrow k = 3$$

30. To get independent term of x in $\left(x^2-\frac{1}{x}\right)^9$,

$$r = \frac{np}{p+q} = \frac{9(2)}{2+1} = \frac{18}{3} = 6$$

Required term $= {}^{9}C_6(x^2)^{9-6} \cdot \left(\frac{-1}{x}\right)^6 = {}^{9}C_6 \cdot x^6 \cdot \frac{1}{x^6} = 84.$

31. In $(3 + kx)^9$

coefficient x^2 = coefficient of x^3

$${}^{9}C_2 \cdot 3^7 \cdot k^2 = {}^{9}C_3 \cdot 3^6 \cdot k^3 \Rightarrow {}^{9}C_2 \cdot 3 = {}^{9}C_3 \cdot k$$

$$\frac{9!}{7!\,2!} \cdot 3 = \frac{9!}{6!\,3!} \cdot k$$

$$\Rightarrow \frac{3}{7} = \frac{k}{3} \Rightarrow k = \frac{9}{7}$$

32. $\left(x^3 + 3 \cdot 2^{-\log_{\sqrt{2}}\sqrt{x^3}}\right)^{11} = \left(x^3 + 3 \cdot 2^{-\log_2 x^3}\right)^{11}$

$$= \left(x^3 + \frac{3}{2^{\log_2 x^3}}\right)^{11} = \left(x^3 + \frac{3}{x^3}\right)^{11}$$

$$= {}^{11}C_0(x^3)^{11} + {}^{11}C_1(x^3)^{10} \cdot \left(\frac{3}{x^3}\right) + {}^{11}C_2(x^3)^9 \cdot \left(\frac{3}{x^3}\right)^2 + {}^{11}C_3(x^3)^6 \cdot \left(\frac{3}{x^3}\right)^3 + \cdots + {}^{11}C_{11} \cdot \left(\frac{3}{x^3}\right)^{11}$$

$$= {}^{11}C_0 \cdot x^{33} + {}^{11}C_1 \cdot 3 \cdot x^{27} + {}^{11}C_2 x^2 \cdot x^{21} + {}^{11}C_3 \cdot 3^3 \cdot x^{15} + \cdots + {}^{11}C_{10} \cdot \frac{3^{10}}{x^{27}} + {}^{11}C_{11} \cdot \frac{3^{11}}{x^{33}}$$

33. Let T_r and T_{r+1} denote the rth and $(r + 1)$ th terms in the expansion of $(1 + x)^{10}$.

Then, $T_r = {}^{10}C_{r-1}x^{r-1}$ and $T_{r+1} = {}^{10}C_r x^r$

$$\therefore \frac{T_{r+1}}{T_r} = \frac{{}^{10}C_r x^r}{{}^{10}C_{r-1}x^{r-1}} = \frac{{}^{10}C_r}{{}^{10}C_{r-1}}$$

$$\therefore \frac{T_{r+1}}{T_r} = \frac{11-r}{r}x$$

$$\Rightarrow \frac{T_{r+1}}{T_r} = \left(\frac{11-r}{r}\right) \times \frac{2}{3} [\because x = 2/3]$$

Now, $\therefore \frac{T_{r+1}}{T_r} > 1$

$$\left(\frac{11-r}{r}\right) \times \frac{2}{3} > 1 \Rightarrow 22 > 5r \Rightarrow r < 4\frac{2}{5}$$

$\therefore (4 + 1)$ th i.e 5th term is the greatest term.

Putting $r = 4$ in T_{r+1}, we get $T_5 = {}^{10}C_5x^4[\because x = 2/3] \Rightarrow T_5 = 210\left(\frac{2}{3}\right)^4$

17. Partial Fractions

WORKSHEET – 1

Level – I

1)B	2)C	3)C	4)D	5)A	6)D	7)C
8)A	9)B	10)C				

Level – II

1) C	2) D	3) A	4) B	5) B	6) B	7) C
8) B	9) C	10) B	11) B	12) C	13) A	14) C
15) C	16) B	17) B	18) C	19) A	20) B	21) B
22) A,B	23) A	24) B	25) A	26) *a - q b - s c - p d -r*		27) A,C,D
28) C	29) D	30) B	31) B,C,D	32) 5		

1. $x = 3(x-2) + A(x-3)$; Put $x = 2, 2 = A(-1) \Rightarrow A = -2$.
2. $1 = A(x-1) + B(x)$

   Put $x = 1 \Rightarrow B = 1$

   Put $x = 0 \Rightarrow A(-1) = 1 \Rightarrow A = -1$.
3. Option (A)
4. $ax = 2(x-1) + 1(x+2) + 2a = +6 \Rightarrow a = 3$.
5. $3x + 7 = A(x-2) + B(x-1)$

   Put $x = 2,\ 13 = B(1) \Rightarrow B = 13$

   $2\ B = 2 \times 13 = 26$.
6. $5x - 7 = A(x-4) + B(x+2)$

   Put $x = 4 \Rightarrow B = \frac{13}{6}$

   Put $x = -2 \Rightarrow 6A = 17 \Rightarrow A = \frac{17}{6}$.

   $\therefore A - B = \frac{17}{6} - \frac{13}{6} = \frac{2}{3}$.
7. $x^2 + x + 1 = (x+2)^2 - 3(x+2) + k$

   $\Rightarrow x^2 + x + 1 = x^2 + 4 + 4x - 3x - 6 + k$

   $\Rightarrow k = 3$.
8. $x - 4 = A(x-3) - 1(x-2)$

   Put $x = 2 \Rightarrow -2 = A(-1) \Rightarrow A = 2$.
9. $ax + b = (3x+4) - 3$

   $ax + b = 3x + 1$

   $a = 3, b = -1$.

10. $x + 4 = A(x + 2)(x + 1) + B(x - 2)(x + 1) + C(x - 2)(x + 2)$

    Put $x = -2$

    $\Rightarrow 2 = B(-4)(-1) \Rightarrow B = \frac{1}{2}$

    Put $x = 2 \Rightarrow A(4)(3) = 6 \Rightarrow A = \frac{1}{2}$

    Put $x = 0 \Rightarrow 0 + 4 = A(2) + B(-2) + B(-2)(1)$

    $4 = 2\ A - 2\ B - 4C$

    $-4C = 4 \Rightarrow C = -1$

    $\therefore A + B + C = \frac{1}{2} + \frac{1}{2} - 1 = 0.$

11. $3x - 2 = A(x + 2) + B(x - 1)$

    Put $x = 1,\ \ A = \frac{1}{3}$

    Put $x = -2,\ \ + 8 = B(-3) \Rightarrow B = \frac{8}{3}$

    $(B, A) = \left(\frac{8}{3}, \frac{1}{3}\right)$

12. $3x = 2(x + a) + (x - 6)$

    Put $x = 6,\ a = 3.$

13. $1 = \frac{1}{k}[x + 5 - x + 5] \Rightarrow k = 10.$

14. $\frac{x^2+5x+7}{(x+1)^3} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3}$

    $x^2 + 5x + 7 = A(x + 1)^2 + B(x + 1) + C$

    $x^2 + 5x + 7 = Ax^2 + A + 2Ax + Bx + B + C$

    $A = 1$

    $2\ A + B = 5$

    $A + B + C = 7$

    $2 + B = 5$

    $C = 7 - 4$

    $B = 3$

    $C = 3$

    $= \frac{1}{x+1} + \frac{3}{(x+1)^2} + \frac{3}{(x+1)^3}.$

15. $1 - 2x = A(3 - x) + B(1 + x)$

    $A = \frac{3}{4}, B = \frac{-5}{4}.$

16. $3x^3 - 8x^2 + 10 = A(x - 1)^3 + B(x - 1)^2 + C(x - 1) + D$

    $3x^3 - 8x^2 + 10 = Ax^3 - A - 3Ax^2 + 3Ax + Bx^2 + B - 2Bx + Cx - C + D$

$-3\ A + B = -83\ A - 2\ B + C = 0$

$B = 1 \quad C = -7$

$D = 5$

$A + B + C + D = 3 + 1 - 7 + 5 = 9 - 7 = 2.$

17. Put $x - 1 = p; 2x^2 + 3x + 5 = 2(1 + p)^2 + 3(1 + p) + 5$

$\frac{2}{p} + \frac{7}{p^2} + \frac{10}{p^3}$

18. put $x = 2$ in $\frac{x^2+1}{(x-1)^8}, \Rightarrow A_1 = 5$

19. Let $e^x = y, \frac{y+2}{(y-1)(2y-3)} = \frac{3}{y-1} + \frac{B}{2y-3}$; Put $y = \frac{3}{2} \Rightarrow B = 7.$

20. $\frac{\log(x+1)}{(2\log x-3)(\log x-2)} = \frac{\alpha}{2\log|x|-3} + \frac{\beta}{2\log|x|-2}$

$\log(x + 1) = \alpha(\log x - 2) + \beta(2\log x - 3)$

Put $x = 1, \ -2\alpha - 3\beta = 1$

Put $x = e, \ -\alpha - \beta = 2$

$\alpha = -5, \beta = 3 \Rightarrow \alpha^2 - \beta^2 = 16.$

21. $x^3 + 1 = A(x^2 - x) + B(x^2 + x) + C(x^2 - 1)$

$\therefore A + B + C = 0$

$-A + B = 0$

$-C = 1$

$A = \frac{1}{2} \quad A = B \quad C = -1$

$\left(\frac{A}{B}\right) = \frac{\frac{1}{2}}{\frac{1}{2}} = 1 = \tan\frac{\pi}{4}$

22. $1 = A(1 + 3x) + B(1 - 2x)$

$A + B = 1$

$3\ A - 2\ B = 0$

$\frac{2}{2}\ B + B = 1$

$A = \frac{2}{3}\ B$

$5\ B = 3$

$A = \frac{2}{5}$

$B = \frac{3}{5}$

$A: B = 2: 3.$

$3\ \mathrm{A} - 2\ \mathrm{B} = 3\left(\frac{2}{5}\right) - 2\left(\frac{3}{5}\right) = \frac{6}{5} - \frac{6}{5} = 0.$

23. $1 = \mathrm{A}(1-2\mathrm{x})(1-3\mathrm{x}) + \mathrm{B}(1-\mathrm{x})(1-3\mathrm{x}) + \mathrm{C}(1-\mathrm{x})(1-2\mathrm{x})$

    Put $\mathrm{x} = 1,\ \ \mathrm{A} = \frac{1}{2}$

    Put $\mathrm{x} = -1,$

    $12\ \mathrm{A} + 8\ \mathrm{B} + 6\mathrm{C} = 1$

    $8B + 6c = -5$

    $B = -4$

    $\mathrm{A} + \mathrm{C} - \mathrm{B} = 9\mathrm{C} = \frac{9}{2}$

    $2\ \mathrm{A} + 2\ \mathrm{B} = -11.$

24. Option (2)

25. Option (1)

26. $a \to q\ b \to s\ c \to p d \to r$

    $3x - 18 = A(x^2)(x+3) + Bx(x+3) + C(x+3) + D(x^3)$

    Put $x = -3 \Rightarrow D = 1$

    put $x = 0 \Rightarrow c = -6$

    Comparing Co-efficients of x^3

    $A + D = 0 \Rightarrow A = -1$

    Comparing Co-efficients of x

    $3B + C = 3 \Rightarrow B = 3$

27. $\mathrm{x}^2 + 3\mathrm{x} + 1 = \mathrm{A}\big(\mathrm{x}(\mathrm{x}+1) + \mathrm{B}(\mathrm{x}+1) + \mathrm{C}(\mathrm{x}^2)\big)$

    Put $\mathrm{x} = -1,\ \mathrm{C} = -1$

    Put $\mathrm{x} = 0,\ \ \mathrm{B} = 1$

    Put $\mathrm{x} = 1,\ 2\ \mathrm{A} + 2\ \mathrm{B} + \mathrm{C} = 5$

    $\mathrm{A} = 2$

    $\mathrm{B} + \mathrm{C} = 0.$

28. -3 (Conceptual)

29. 4 (Conceptual)

30. $\frac{x}{(x-1)^2(x-2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x-2}$

    $\Rightarrow x = A(x-1)(x-2) + B(x-2) + C(x-1)^2$

    put $x = 2 \Rightarrow B = -1$

    put $x = 2 \Rightarrow c = 2$

    Comparing the co-efficients of x^2

$A + C = 0$

$A = -2$

31. $1 = Ax^2(x-a) + Bx(x-a) + C(x-a) + Dx^3$

put x = 0 ⇒ $c = \frac{-1}{a}$; put $x = a \Rightarrow D = \frac{-1}{a^3}$

Comparing the co-efficients of x^3; $A + D = 0 \Rightarrow A = \frac{1}{a^3}$

Comparing the co-efficients of x^2

$B - aA = 0 \Rightarrow B = \frac{-1}{a^2}$

32. $x^2 + 1 = A(x+1)^2 + B(x+1) + C(1)$

Put $x = -0,1 = A + B + C \Rightarrow A + B = -1$

Put $x = -1,2 = A(0) + B(0) + C \Rightarrow C = 2$

Put $x = 1,2 = 4\ A + 2\ B + C$

$4\ A + 2\ B = 0$

$A + B = -1$

$2\ A + B = 0$

$1 + B = -1$

$A + B = -1$

$B = -2$

$A = 1$

$A - B + C = 1 + 2 + 2 \Rightarrow 5.$

WORKSHEET – 2

Level – I

1) 3	2) 2	3) 1	4) 2	5) 1	6) 1	7) 1
8) 1	9) 1	10) 1				

Level – II

1) C	2) B	3) B	4) C	5) D	6) A	7) D
8) C	9) A	10) B	11) C	12) A	13) C	14) A
15) C	16) C	17) D	18) A	19) B	20) C	21) A,C,D
22) B	23) D	24) A	25) a -s ,b - p ,c - q ,d -r		26) A,B,D	27) C
28) D	29) A	30) A,B,D				

1. $x^2 + 10 = x^2 + 2 + k \Rightarrow k = 8.$

2. $2x + 1 = A(x^2 + 1) + (Bx + C)(x - 1)$

$2x + 1 = Ax^2 + A + Bx^2 + Cx \cdot Bx - C$

$A = 0\ \ B = 0\ C = 0$

$A + B = 0.$

3. $x^2 = k[a^2x^2 + a^2b^2 - b^2x^2 - b^2a^2]$

   $k = \frac{1}{a^2 - b^2}.$

4. $(x - 1)^2 = A(x^2 + 1) + Bx^2 + Cx$

   $A = 1,\ B = 0, C = -2$

   $(A, B, C) = (1,0,-2).$

5. $x^2 + 3x + 5 = A(x^2 + 2) + (Bx + C)(x + 2)$

   $x^2 + 3x + 5 = Ax^2 + 2\ A + Bx^2 + 2Bx + Cx + 2C$

   Put $x = -2, 4 - 6 + 5 = 6\ A \Rightarrow A = \frac{1}{2}$

   $A + B + C = \frac{1}{2} + \frac{1}{2} + 2\ \ B = \frac{1}{2}\ C = 2$

   $A + B + C = 3.$

6. $x^2 - 5x + 6)x^3 - 6x^2 + 10x - 2(x - 1$

   $$\frac{x^3 + 5x^2 + 6x}{x^2 + 4x - 2} \Rightarrow f(x) = x - 1$$

7. $2x^2 + 3x + 4 = A(x^2 + 2) + (Bx + C)(x - 1)$

   $A = 3,\ \ B = -1,\ C = 2$

   $(A, B, C) = (3, -1, 2).$

8. $x^2 + 2x + 1 = A(x^2 + 1)(Bx + C)(x)$

   $x^2 + 2x + 1 = Ax^2 + A + Bx^2 + Cx$

   $A + B = 1\ \ B + C = 2$

   Put $x = 0,$

   $1 = A\ C = 2$

   $B = 0$

   $A + B + C = 1 + 0 + 2 = 3$

9. $4x^2 + 7x + 11 = A(x^2 + 1) + (Bx + C)(x - 1)$

   Put $x = 1,\ \ A = 11,\ \ B = -7,\ C = 0.$

10. $x^2 + 2 = (Ax + B)(x^2 + 3) + (Cx + D)(x^2 + 1)$

    $x^2 + 2 = Ax^3 + 3Ax + Bx^2 + 3\ B + Cx^3 + Cx + Dx^2 + D$

    $A = 0,\ B + D = 1 \Rightarrow 3\ B + D = 2.$

11. $x + 2 = (Ax + B)(x^2 + 3) + (Cx + D)(x^2 + 1)$

    $A = \frac{1}{2}, C = \frac{-1}{2}.$

12. $1 = A(x^2 + x + 1) + (Bx + C)(x + 1)$

    $1 = (A + B)x^2 + (-A + B + C)x + A + C$

$A = \frac{1}{3}$, $B = \frac{-1}{3}, C = \frac{2}{3}$.

13. $x^2 = (Ax + B)(x^2 + 1)^2 + (Cx + D)(x^2 + 1) + Ex + F$

$A = 0, B = 0, C = 0, D = 1, E = 0, F = -1$

14. $2x + 1 = A(x^2 + 1) + (Bx + c)x - 1$

Put $x = 0$,

$A - C = 1$,

$A = -B$

Put $x = 1$,

$A = \frac{3}{2}$, $C = \frac{1}{2}$

15. $3x^4 + 2 =$

$Ax^2(x - 1)(x - 2) + Bx(x^2 - 3x + 2) + C(x^2 - 3x + 2) + D(x - 2) + E(x - 1)$

$x = 1$,

$D = -5$

$x = 2$,

$E = 50$

Comparing the x^4 coefficients,

$Ax^4 = 3x^4 \Rightarrow A = 3$

$\therefore A + D + E = 3 - 5 + 50 = 48.$

16. $x^2 + 2x + 1 = x^4 + 2x^2 + 1 - x^2$

$(x^2 + 1)^2 - x^2 \Rightarrow (x^2 + x + 1)(x^2 - x + 1)$

17. $x^2 + x - 2)x^4\ (x^2 - x + 3$

$$\frac{x^4 + x^3 - 2x^2}{-x^3 + 2x^2}$$
$$\frac{-x^3 - x^2 + 2x}{3x^2 - 2x}$$
$$\frac{3x^2 + 3x - 6}{-5x + 6}$$

18. $x^2 + 1 = (-Ax + B)(x^2 - x + 1) + (Cx + D)(x^2 + x + 1)$

$B + D = 1,\ -A + C = 0\ \ A - B + C + D = 0$

$A = -C.$

19. $\frac{X^4+3x-1}{(x+1)^2(x-1)} = (x - 1) + \frac{C}{x+1} + \frac{D}{(x+1)^2} + \frac{E}{x-1}$

$A = 1, D = 1/2$

$\because A + D = 3/2$

20. $x^4 + 24x^2 + 28 = (Ax + B)(x^2 + 1)^2 + (Cx + D)(x^2 + 1) + Ex + F$

$$A = 0,\ C = 0,\ 2A + C = 0$$

21. $x^2 + x + 1 = A(x^2 + 2) + (Bx + C)(x + 1)$

Put $x = 0$,

$2A + C = 1 \quad A = \frac{1}{3}$

$A + B = 1$

$2A + B \Rightarrow B = \frac{2}{3}\ C = \frac{1}{3} \quad \therefore A = C$

24. $3x - 1 = A(1 - x + x^2) + (Bx + C)(2 + x)$

put $x = -2 \Rightarrow -7 = 7A \Rightarrow A = -1$

Comparing Co-efficient of x, x^2, constant terms

$2B + C - A = 3,\ A + B = 0,\ A + 2C = -1$

25. $x^3 = k(2x - 1)(x + 2)(x - 3) + A(x + 2)(x - 3) + B(2x - 1)(2x - 1)(x + 2)$

$K = \frac{Co - \text{ efficient of } x^3}{Co - \text{ efficient of } x^3} = \frac{1}{2}$

put $x = \frac{1}{2} \Rightarrow A\left(\frac{1}{2} + 2\right)\left(\frac{1}{2} - 3\right) = \frac{1}{8} \Rightarrow A(-25) = \frac{1}{2} \Rightarrow A = \frac{-1}{50}$

put $x = -2 \Rightarrow B(-5)(-5) = -8 \Rightarrow B = \frac{8}{25}$

put $x = 3 \Rightarrow C(5)(5) = 9 \Rightarrow c = \frac{9}{25}$

26. $(x + 2)^2 = Ax^2 + 2Ax + 3A + Bx^2 + Cx$

$A + B = 1,\ 2A + C = 4$

$B = \frac{1}{3}, C = \frac{4}{3}$

$A + B = 1$

27. $x^3 + x^2 + 1 = Ax^3 + 3Ax + Bx^2 + 3B + Cx^3 + 2Cx + Dx^2 + 2D.$

$A + C = 1,\ D = 2,\ \ B = -1,\ C = 5,\ \ A = -4$

$3A + 2C = -2.$

28. $3B + 2D - A = 5.$

29. $\frac{B+D}{A+C} = 1.$

30. $x^2 + 2x + 1 = x^4 + 2x^2 + 1 - x^2$

$(x^2 + 1)^2 - x^2 \Rightarrow (x^2 + x + 1)(x^2 - x + 1)$

$3x^3 - 2x^2 - 1 = (Ax + B)(x^2 - x = 1) + (Cx + D)(x^2 + x + 1)$

$\Rightarrow A + C = 3, B + D = -1,$

$-A + B + C + D = -2, A - B + C + D = 0$

Solving $A = 2, B = 1,\ C = 1\ D = -2$